VCU220

VCU220 SYS.3 SYSTEM ARCHITECTURAL DESIGN

Detailed HW/SW Allocation, Safety Requirements, DVP & Validation Handoff

Document ID	VCU-EV220-SYS3-ARCH-REQ-001
Revision	Rev A2 - Final Technical Hardening
Technical Freeze / Authority	JAN-2026 Program Baseline / Rev A2 hardening review 14-Aug-2026
Vehicle	220 kW peak / 110 kW continuous EV platform
Architecture	Central ECU with independent safety MCU + high-performance cockpit/application SoC
Process Scope	Automotive SPICE 4.0 SYS.1 / SYS.2 / SYS.3 with HWE.1 / SWE.1 / SYS.4 / SYS.5 / VAL.1 handoff
Safety / Cybersecurity	ISO 26262 vehicle safety lifecycle; ISO/SAE 21434 TARA; final ASIL from vehicle HARA/FSC/TSC
Evidence Status	Engineering baseline. Planned requirements/tests are not executed release or compliance evidence.

Independent consultant-derived systems engineering work product. Supplier/OEM source rights remain with their respective owners.

1. SYS.3 Architectural Intent and Design Rules

SYS.3 defines the internal architecture and allocates system requirements to hardware, safety MCU software, HPA/hypervisor services, cockpit software, networks and external ECU contracts. The architecture is intentionally fail-bounded: a central ECU may consolidate packaging and compute, but no single QM/HPA fault is allowed to create valid torque/HV authority.

Safety authority: Safety MCU output, vehicle state, E2E/SecOC validity and local inverter/ESC/BMS permissives collectively bound the actuation path. Android/POSIX requests remain non-authoritative.

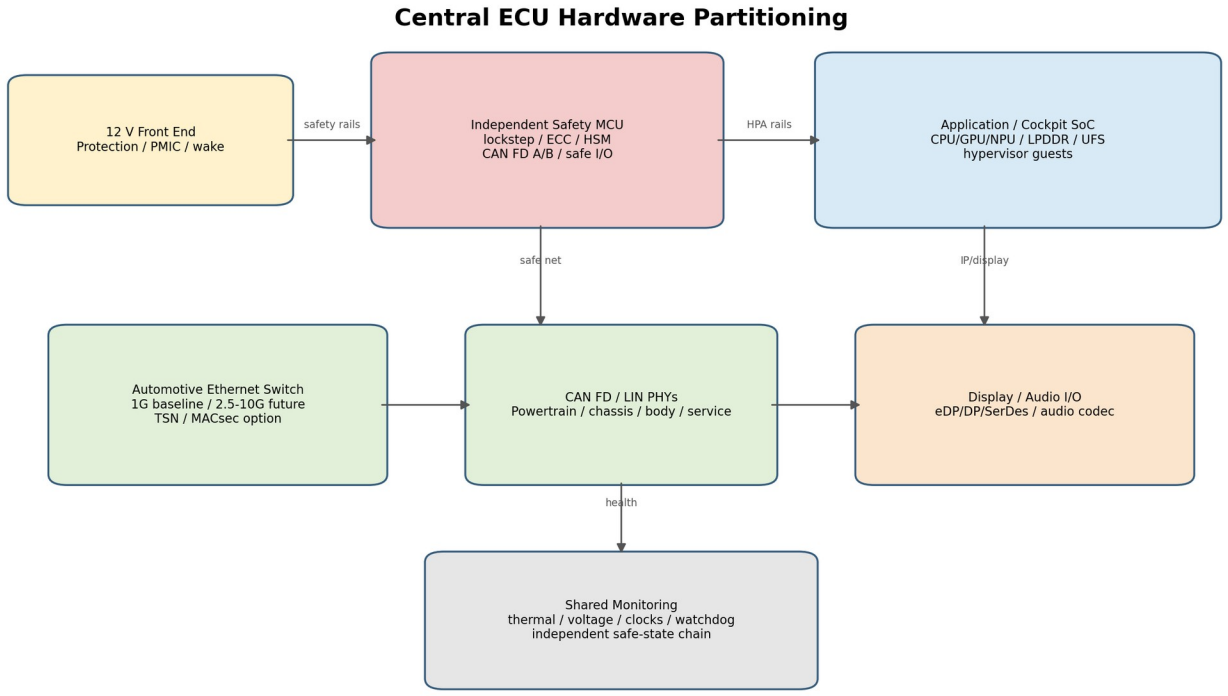


Figure 1 - Central ECU hardware partition

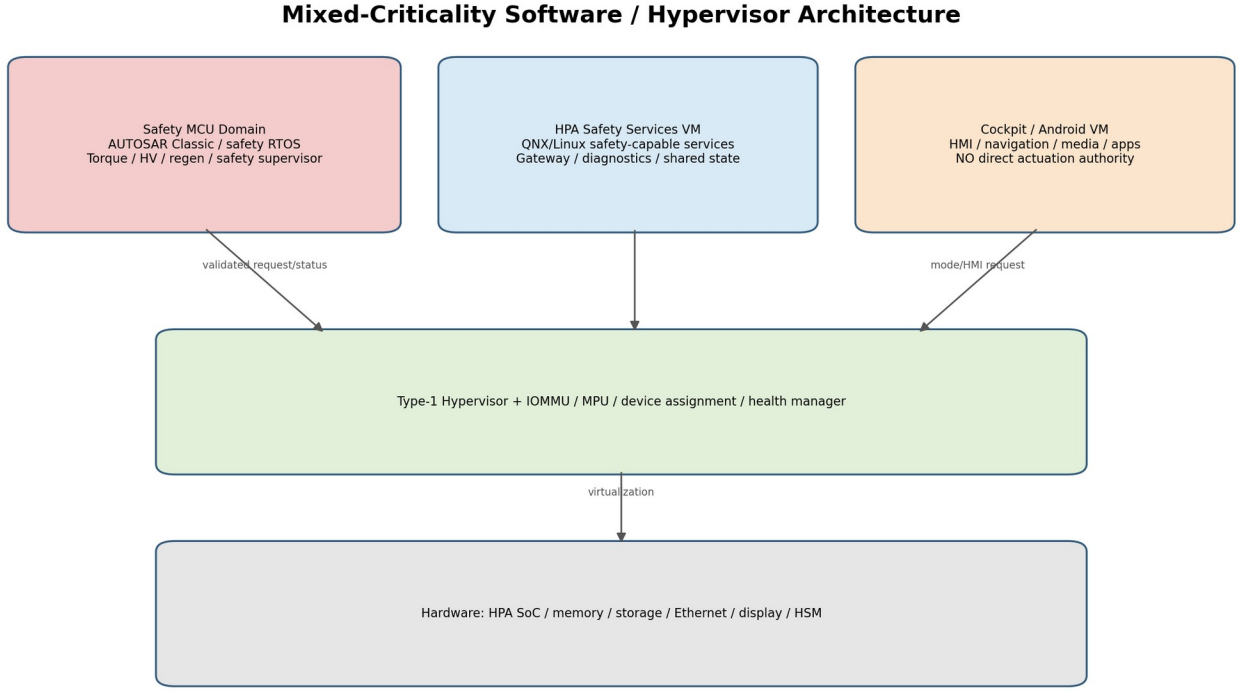


Figure 2 - Hypervisor and software partitioning

2. System Block Architecture B01-B20

B01 - LV Power Input, PMIC, Wake/Sleep & Power-State Control

Technical overview. The low-voltage front end is the first containment boundary for central-compute common-cause faults. It converts the vehicle 12 V network into independent safety and HPA rails, supervises undervoltage/overvoltage/reverse-polarity conditions and implements controlled wake/sleep sequencing.

Design justification. The safety MCU rail must not depend on successful Android/HPA boot. Brownout handling shall remove authority before corrupted execution is possible and shall preserve diagnostic reset reason.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B01-01	12V input envelope	The LV input shall operate across the released EV low-voltage supply envelope and shall never create an unsafe actuation output during undervoltage, overvoltage, reverse polarity or power interruption.	Electrical DVP; safe outputs through all supply states.	HW/SW	Safety-related
VCU220-SYS3-B01-02	Power consumption	Central ECU electronics shall remain <=45 W nominal and <=75 W peak for the released workload, excluding external displays/amplifiers/antennas.	Power measurements at worst CPU/GPU/network load.	HW/SW	Safety-related
VCU220-SYS3-B01-03	Safety power domain	Safety MCU/PMIC/CAN path shall use an independently supervised rail path from the HPA application SoC where practicable.	Main SoC reset/rail fault does not remove safety communication.	HW/SW	Safety-related
VCU220-SYS3-B01-04	Brownout handling	Power manager shall predict/qualify brownout and enter non-permissive torque/HV states before safety MCU execution becomes invalid.	No uncontrolled output during LV ramps/dropouts.	SW/System	Safety-related
VCU220-SYS3-B01-05	Hold-up	Safety domain shall retain sufficient energy to publish safe-state/degradation and store critical event context after primary rail loss.	>=50 ms design reference or FSC-derived value.	SW/System	Safety-related
VCU220-SYS3-B01-06	Wake sources	Wake shall support ignition/network/charger/service sources with authorization and debounce.	No wake storm; source identifiable.	SW/System	Safety-related
VCU220-SYS3-B01-07	Sleep sequencing	Safety domain shall verify torque=0/HV state/network quiescence before deep sleep.	No sleep while motion authority is active.	SW/System	Safety-related
VCU220-SYS3-B01-08	Standby current	Low-power mode current shall meet vehicle quiescent-current allocation and support TC10/CAN wake architecture.	Vehicle parasitic current budget passes.	SW/System	Safety-related
VCU220-SYS3-B01-09	PMIC watchdog	Independent watchdog/window/function monitoring shall supervise safety MCU execution and rails.	Watchdog fault forces configured safe-state outputs.	HW/SW	Safety-related
VCU220-SYS3-B01-10	Power diagnostics	Rail voltage/current/reset-cause/PMIC faults shall be timestamped and exposed to diagnostics.	Fault cause retrievable after reset.	HW/SW	Safety-related

B02 - Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway

Technical overview. The independent safety MCU is the authoritative deterministic motion-control domain. It owns torque authorization, critical plausibility, HV-state permissives, watchdog supervision and the independent safety communications path.

Design justification. Selecting a lockstep/ECC/HSM MCU outside the cockpit SoC prevents a single hypervisor, GPU, Android or HPA power fault from becoming a direct torque/HV common-cause failure.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety

VCU220-SYS3-B02-01	Safety MCU capability	Safety MCU shall provide lockstep/ECC/MPU/HSM/watchdog/safety management features sufficient for the final ISO 26262 allocation.	Supplier safety manual and FMEDA assumptions close.	HW/SW	Safety-related
VCU220-SYS3-B02-02	Torque authority	Safety MCU shall own the authoritative propulsion torque output and final clamps.	No HPA/Android direct torque path.	SW/System	Safety-related
VCU220-SYS3-B02-03	Safe outputs	Torque enable/HV permissive/safety CAN outputs shall default non-permissive during reset/unpowered/fault.	Bench fault injection passes.	SW/System	Safety-related
VCU220-SYS3-B02-04	Cycle time	Critical motion arbitration shall execute at <=1 ms internal base rate or a WCET-proven rate supporting the 10 ms safety reaction target.	WCET margin >=30% reference.	SW/System	Safety-related
VCU220-SYS3-B02-05	Diverse supervision	Independent monitor shall compare requested torque against pedal/brake/gear/speed/BMS/inverter context.	Injected corrupt request detected within budget.	HW/SW	Safety-related
VCU220-SYS3-B02-06	Actuation gateway	Safety MCU shall mediate safety CAN motion/HV commands and verify acknowledgement/feedback.	Missing ack transitions to degrade/fault.	SW/System	Safety-related
VCU220-SYS3-B02-07	Clock supervision	Independent clock monitors shall detect out-of-range or stalled safety clocks.	Clock fault enters safe state.	HW/SW	Safety-related
VCU220-SYS3-B02-08	Memory integrity	ECC/CRC/MPU shall protect code, calibration and state used by torque/HV control.	Fault injection detects/contains memory errors.	HW/SW	Safety-related
VCU220-SYS3-B02-09	Freedom from interference	Shared HPA resources shall not block safety MCU peripherals, reset, clock or power path.	Common-cause analysis and fault injection pass.	SW/System	Safety-related
VCU220-SYS3-B02-10	Safety logs	Safety MCU shall retain compact freeze-frame of safety transitions independent of rich OS availability.	Minimum 30 s pre/post event references or event-based buffer.	SW/System	Safety-related

B03 - Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization

Technical overview. The HPA SoC consolidates cockpit, gateway and high-level services under a Type-1 hypervisor. CPU/GPU/NPU, LPDDR and UFS are sized from measured workloads and thermal envelopes rather than nominal benchmark claims.

Design justification. The SoC may host safety-supporting services, but authoritative vehicle motion remains bounded by the safety MCU unless a future decomposition is independently justified.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B03-01	Application SoC headroom	HPA SoC shall support cockpit, gateway, service and non-authoritative vehicle services with >=30% CPU/memory/network reserve at release.	Target benchmark and stress test pass.	HW/SW	Safety-related
VCU220-SYS3-B03-02	Virtualization	Type-1 or equivalent hardware-assisted virtualization shall isolate Android/Linux/QNX guests and devices.	No cross-partition unauthorized MMIO/DMA.	SW/System	Safety-related
VCU220-SYS3-B03-03	Safety island use	If SoC includes safety cores, they may support monitored cockpit/gateway functions but shall not replace the independent torque safety MCU in the reference design.	Architecture review passes.	SW/System	Safety-related
VCU220-SYS3-B03-04	Memory bandwidth	LPDDR subsystem shall meet worst mixed IVI/gateway workload without starving safety-relevant HPA services.	QoS stress test; >=30% reserve.	HW/SW	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B03-05	Storage	UFS/NVMe/eMMC storage shall provide A/B update partitions, logging endurance and power-loss recovery appropriate to lifecycle.	Power-cut/update endurance test passes.	HW/SW	Safety-related
VCU220-SYS3-B03-06	IOMMU	DMA-capable guest devices shall be constrained by IOMMU/SMMU mappings.	Negative DMA tests cannot access safety regions.	SW/System	Safety-related
VCU220-SYS3-B03-07	GPU isolation	GPU/display resources serving QM guests shall not prevent safety telltale/cluster fallback path.	GPU hang fault test passes.	SW/System	Safety-related
VCU220-SYS3-B03-08	Boot chain	SoC boot shall authenticate hypervisor and guest images before privileged execution.	Tampered image rejected.	SW/System	Safety-related
VCU220-SYS3-B03-09	Thermal throttling	SoC thermal manager shall shed noncritical IVI workloads before affecting required safety/cockpit warning functions.	Thermal chamber stress passes.	HW/SW	Safety-related
VCU220-SYS3-B03-10	Crash containment	Guest kernel panic/reboot shall remain within guest/partition and preserve safety MCU operation.	Fault injection passes.	SW/System	Safety-related

B04 - Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs

Technical overview. Vehicle I/O converts physical driver and vehicle inputs into qualified signals. Accelerator, brake, gear, ignition, crash and local controls require range, plausibility, timing and diagnostic qualification before use.

Design justification. Sensor or input faults are represented as quality states, not silently substituted with last-known values where doing so can increase vehicle authority.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B04-01	Pedal sensing	Accelerator channels shall be range/plausibility/correlation checked before use in torque demand.	Open/short/stuck/correlation faults detected.	HW/SW	Safety-related
VCU220-SYS3-B04-02	Brake input	Brake switch/pressure/ESC brake status shall be qualified and capable of revoking positive torque.	Brake override timing passes.	HW/SW	Safety-related
VCU220-SYS3-B04-03	Gear/state input	Drive/reverse/park/neutral inputs shall be plausibility checked against vehicle speed and actuator state.	Illegal combinations non-permissive.	HW/SW	Safety-related
VCU220-SYS3-B04-04	Ignition/start	Start/ready request shall be debounced, authorized and state-qualified.	No unexpected READY transition.	SW/System	Safety-related
VCU220-SYS3-B04-05	Crash/HVIL	Crash/HVIL/emergency signals shall be available to safety MCU without dependence on Android.	Direct/independent path verified.	SW/System	Safety-related
VCU220-SYS3-B04-06	Steering input	Steering/wheel-angle data used for mode/chassis coordination shall carry quality/age status.	Stale data not used to increase authority.	HW/SW	Safety-related
VCU220-SYS3-B04-07	Vehicle speed	Speed source shall use qualified wheel/inverter/ESC data with plausibility and fallback.	Faulted source isolated.	SW/System	Safety-related
VCU220-SYS3-B04-08	Analog diagnostics	Safety analog inputs shall support open/short-to-battery/ground diagnostics as applicable.	DVP faults detected.	SW/System	Safety-related
VCU220-SYS3-B04-09	Debounce timing	Input debounce/filter shall not mask faults beyond FTTI or add excessive control latency.	Timing budget met.	SW/System	Safety-related
VCU220-SYS3-B04-10	Input calibration	Sensor scaling/offset/range calibration shall be signed, variant-bound and range constrained.	Wrong calibration rejected.	HW/SW	Safety-related

B05 - Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager

Technical overview. Torque arbitration combines driver demand, validated mode shaping, cruise/ADAS requests, ESC overrides and the minimum available capability from BMS, inverter/motor and thermal supervisors.

Design justification. The output is a bounded authoritative torque request with reason codes. The independent safety monitor must be able to revoke positive torque without waiting for the HPA/cockpit domain.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B05-01	Torque equation	Net torque request shall combine driver demand, cruise/ADAS request where present, traction/brake overrides and all available limits using an explicitly ordered arbitration.	Traceable algorithm and SIL/HIL boundary tests.	SW/System	Safety-related
VCU220-SYS3-B05-02	Pedal map	Pedal-to-torque mapping shall be monotonic for road modes and constrained by mode calibration.	No non-monotonic unsafe region.	SW/System	Safety-related
VCU220-SYS3-B05-03	Torque clamp	Final request shall be clamped by BMS discharge SOP, inverter/motor torque, thermal, speed, gear, traction and regulatory/valet caps.	Minimum-limit selection verified.	SW/System	Safety-related
VCU220-SYS3-B05-04	Brake override	Qualified brake application shall reduce/revoke positive torque according to calibration and safety concept.	<=10 ms internal response reference.	SW/System	Safety-related
VCU220-SYS3-B05-05	Direction safety	Positive/negative torque sign shall be consistent with selected gear and vehicle motion state.	No unintended sign reversal.	SW/System	Safety-related
VCU220-SYS3-B05-06	Rate limiter	Torque slew/jerk shall be bounded; emergency torque cut may bypass comfort slew where safety requires.	Normal jerk <=2.0 m/s3 vehicle target; fault cut per FSC.	SW/System	Safety-related
VCU220-SYS3-B05-07	Feedback monitor	Requested torque shall be compared to inverter delivered/estimated torque and acceleration response.	Deviation fault detection within calibrated window.	SW/System	Safety-related
VCU220-SYS3-B05-08	Degraded drive	Single noncritical sensor/network faults shall lead to a documented limp/degraded torque cap where safe.	Degraded-state HIL/vehicle passes.	SW/System	Safety-related
VCU220-SYS3-B05-09	Latency	Pedal acquisition to valid torque PDU transmit shall meet <=6 ms internal target and <=15 ms E2E vehicle target.	Timing trace passes.	SW/System	Safety-related
VCU220-SYS3-B05-10	Torque audit	Every limit contributing to final torque shall expose source/reason for diagnostics and calibration.	Reason-code trace available.	SW/System	Safety-related

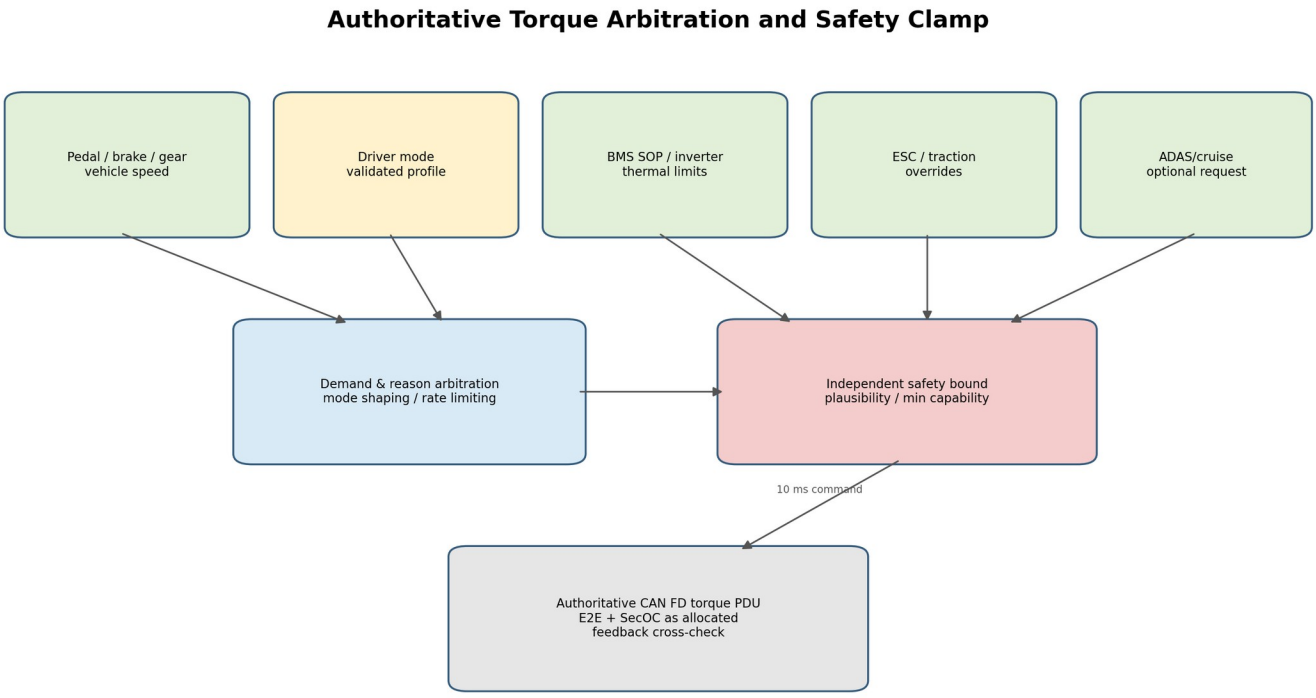


Figure 3 - Authoritative torque arbitration flow

B06 - Regenerative Braking, Deceleration Coordination & Brake-Energy Manager

Technical overview. Regenerative braking management computes desired motor regen but does not own final wheel-slip or hydraulic-friction safety. ESC/brake ECU remains the local brake authority.

Design justification. This allocation avoids duplicating ABS/ESC safety logic in the VCU and permits regen to fall back smoothly when battery charge SOP, inverter limits or wheel stability reduce regenerative capability.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B06-01	Decel request	VCU shall compute regenerative braking availability/request but ESC/brake ECU remains final brake-blend authority.	Interface arbitration verified.	SW/System	Safety-related
VCU220-SYS3-B06-02	Charge SOP	Regen shall never exceed BMS charge power/current/SOC/temperature constraints.	Boundary sweep passes.	SW/System	Safety-related
VCU220-SYS3-B06-03	Wheel slip	ABS/ESC active state shall revoke or reduce regen immediately as defined by brake ICD.	Brake HIL passes.	SW/System	Safety-related
VCU220-SYS3-B06-04	Low speed fade	Regen shall fade at low motor speed according to inverter capability and friction-brake coordination.	No decel discontinuity beyond target.	SW/System	Safety-related
VCU220-SYS3-B06-05	One-pedal	One-pedal calibration shall be mode/market configurable and preserve brake-light/deceleration legal behavior.	Vehicle/homologation validation.	SW/System	Safety-related
VCU220-SYS3-B06-06	Brake pedal blend	Pedal deceleration demand shall be split into regen/friction without increasing stopping distance or causing wheel lock.	Brake dyno/HIL/vehicle.	SW/System	Safety-related
VCU220-SYS3-B06-07	Energy accounting	Recovered energy and regen limit reasons shall be measured/logged for diagnostics and efficiency analysis.	Consistency check passes.	SW/System	Safety-related
VCU220-SYS3-B06-08	Regen fault	Loss/stale brake or BMS data shall reduce regen authority rather than increase it.	Fault injection passes.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B06-09	Jerk	Regen transitions shall be jerk-limited except where ABS/ESC safety intervention requires faster action.	Vehicle target <=2.0 m/s3 normal transition.	SW/System	Safety-related
VCU220-SYS3-B06-10	Mode influence	Driver mode may change regen feel only inside BMS/brake/traction hard limits.	Mode tests pass.	SW/System	Safety-related

B07 - Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control

Technical overview. Driver Mode Manager translates Eco/Normal/Sport/Track/Valet requests into bounded calibration profiles. The HMI request is untrusted until validated in the safety domain.

Design justification. A 50 ms acknowledgement does not mean an instantaneous torque-map change. Torque, regen and chassis requests are blended over a calibrated jerk-bounded window; Track mode cannot disable local ABS/ESC hard protection.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B07-01	Mode set	Support Eco, Normal, Sport, Track and Valet/Custom with signed variant calibration.	All modes have versioned calibration sets.	SW/System	Safety-related
VCU220-SYS3-B07-02	Request validation	HMI/physical-switch request shall be validated against speed, HV, thermal, BMS SOP, ESC permission, market and diagnostics.	Rejected state returns reason.	SW/System	Safety-related
VCU220-SYS3-B07-03	Ack timing	Accepted/rejected mode shall be published to HMI within <=50 ms reference.	HIL timing passes.	SW/System	Safety-related
VCU220-SYS3-B07-04	Blend timing	Torque/regen/chassis calibration shall cross-fade over jerk-bounded 200-500 ms reference or longer if required by torque delta.	No step torque from mode switch.	SW/System	Safety-related
VCU220-SYS3-B07-05	Eco cap	Eco power cap shall default to 110 kW reference but remain calibration/vehicle-SOP limited.	Cap test passes.	SW/System	Safety-related
VCU220-SYS3-B07-06	Valet cap	Valet shall default <=80 kW plus optional speed/geofence restrictions; authorization required to relax.	Cyber/HIL passes.	SW/System	Safety-related
VCU220-SYS3-B07-07	Track gating	Track mode requires explicit market/vehicle permissions and ESC acceptance; it shall not defeat ESC/ABS hard protections.	Closed-course/market config tests.	SW/System	Safety-related
VCU220-SYS3-B07-08	Thermal precondition	Sport/Track can request pre-chill but transition is rejected/degraded if thermal system cannot support it.	Thermal HIL passes.	HW/SW	Safety-related
VCU220-SYS3-B07-09	Persistent preference	User preference may persist but vehicle shall always start in a safe market-approved default or revalidate stored mode before activation.	Power-cycle tests pass.	HW/SW	Safety-related
VCU220-SYS3-B07-10	Calibration safety	Mode calibration bounds and dependencies shall be independently reviewed and protected from unsigned updates.	Calibration audit passes.	SW/System	Safety-related

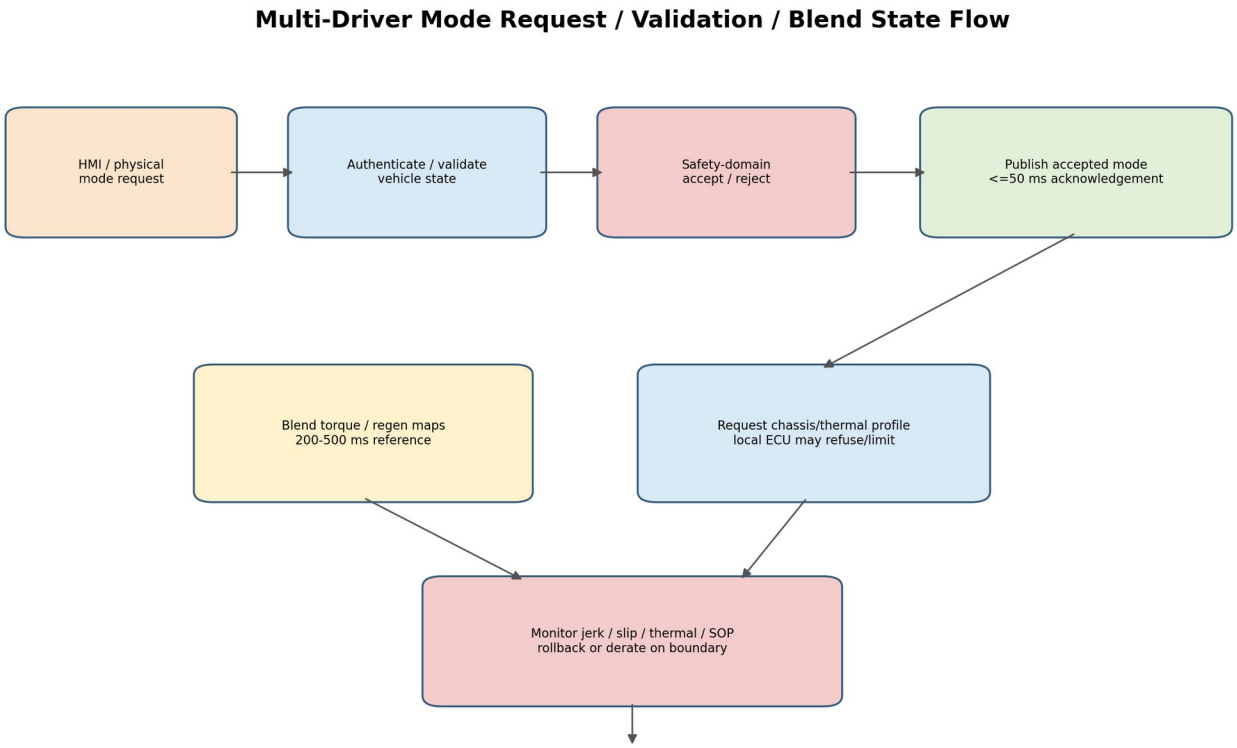


Figure 4 - Driver-mode request, validation and blending

B08 - Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager

Technical overview. Chassis coordination exchanges steering, yaw/slip, suspension and stability states with EPS/ESC/chassis controllers. The VCU may request performance profiles but local ECUs retain actuator-level safety authority.

Design justification. Loss of chassis data reduces capability; it must never increase allowed torque, slip or track-mode authority.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B08-01	ESC authority	ESC/ABS intervention shall override VCU propulsion/regen requests according to chassis ICD.	HIL proves authority.	SW/System	Safety-related
VCU220-SYS3-B08-02	EPS mode request	VCU may request steering assist mode but EPS retains steering safety limits and fault handling.	ICD/vehicle tests.	SW/System	Safety-related
VCU220-SYS3-B08-03	Suspension request	VCU may request damping/ride-height mode only within suspension ECU safe envelope.	Mode tests.	HW/SW	Safety-related
VCU220-SYS3-B08-04	Yaw/slip state	VCU shall consume quality/age-qualified ESC yaw/slip/traction state for torque arbitration.	Stale state handled non-permissively.	SW/System	Safety-related
VCU220-SYS3-B08-05	Launch control	Any launch/track feature shall require approved chassis state and vehicle conditions; no generic unrestricted wheel-slip target.	Feature safety review.	SW/System	Safety-related
VCU220-SYS3-B08-06	Hill hold	Drive torque shall coordinate with brake hold/release to prevent rollback or torque fight.	Grade vehicle tests.	SW/System	Safety-related
VCU220-SYS3-B08-07	Reverse	Reverse torque limits and mode availability shall be separately calibrated and constrained.	Reverse tests.	SW/System	Safety-related
VCU220-SYS3-B08-08	Chassis comm loss	Loss of ESC/EPS/suspension state shall transition to conservative torque/mode limits.	Fault injection.	HW/SW	Safety-related
VCU220-SYS3-B08-09	State consistency	Chassis mode	HMI/HIL passes.	HW/SW	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		acknowledgement shall be checked before reporting the driver mode fully active.			
VCU220-SYS3-B08-10	Dynamics logging	ESC overrides, traction events and mode state shall be event logged for calibration/field analysis.	Logging trace.	SW/System	Safety-related

B09 - HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks

Technical overview. HV orchestration coordinates vehicle READY, precharge request, inverter enable and shutdown. In the reference allocation, the BMS owns pack contactor and precharge actuation/protection.

Design justification. This prevents two independent controllers from fighting over contactors. Direct VCU contactor control is a separate architecture variant requiring explicit HARA/FSC/TSC and hardware safety allocation.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B09-01	HV state machine	VCU shall implement OFF, ACCESSORY, HV_REQUEST, PRECHARGE, READY, DRIVE, CHARGE, SHUTDOWN and FAULT coordination states.	HIL state coverage 100% planned.	SW/System	Safety-related
VCU220-SYS3-B09-02	BMS authority	BMS shall own pack contactor actuation/protection in reference architecture; VCU sends HV state request and validates feedback.	No direct VCU contactor drive unless variant safety case.	SW/System	Safety-related
VCU220-SYS3-B09-03	Precharge	READY/propulsion enable requires BMS-reported precharge complete and inverter DC-link ready.	No early drive enable.	SW/System	Safety-related
VCU220-SYS3-B09-04	Timeout	Precharge/HV state transitions shall have bounded timeout and retry/cooldown policy from BMS/inverter specifications.	Fault injection passes.	SW/System	Safety-related
VCU220-SYS3-B09-05	Crash	Crash/HVIL severe fault shall revoke propulsion and request BMS safe HV response.	Crash HIL.	SW/System	Safety-related
VCU220-SYS3-B09-06	Charge interlock	Drive torque shall be inhibited when charge connector/charging state requires stationary safe behavior.	Charge-drive conflict tests.	SW/System	Safety-related
VCU220-SYS3-B09-07	Weld detection	VCU shall consume BMS contactor weld/stuck status and prevent unsafe restart.	BMS fault tests.	SW/System	Safety-related
VCU220-SYS3-B09-08	Shutdown	Key-off/shutdown sequence shall coordinate torque zero, inverter disable, HV request off, data save and low-power transition.	Power-cycle tests.	SW/System	Safety-related
VCU220-SYS3-B09-09	800V variant	HV state logic shall be voltage-class independent using interface thresholds/configuration rather than duplicated unsafe branches.	Variant regression.	SW/System	Safety-related
VCU220-SYS3-B09-10	HV event record	HV state transition reasons/timeouts/feedback mismatch shall be logged.	Diagnostics trace.	SW/System	Safety-related

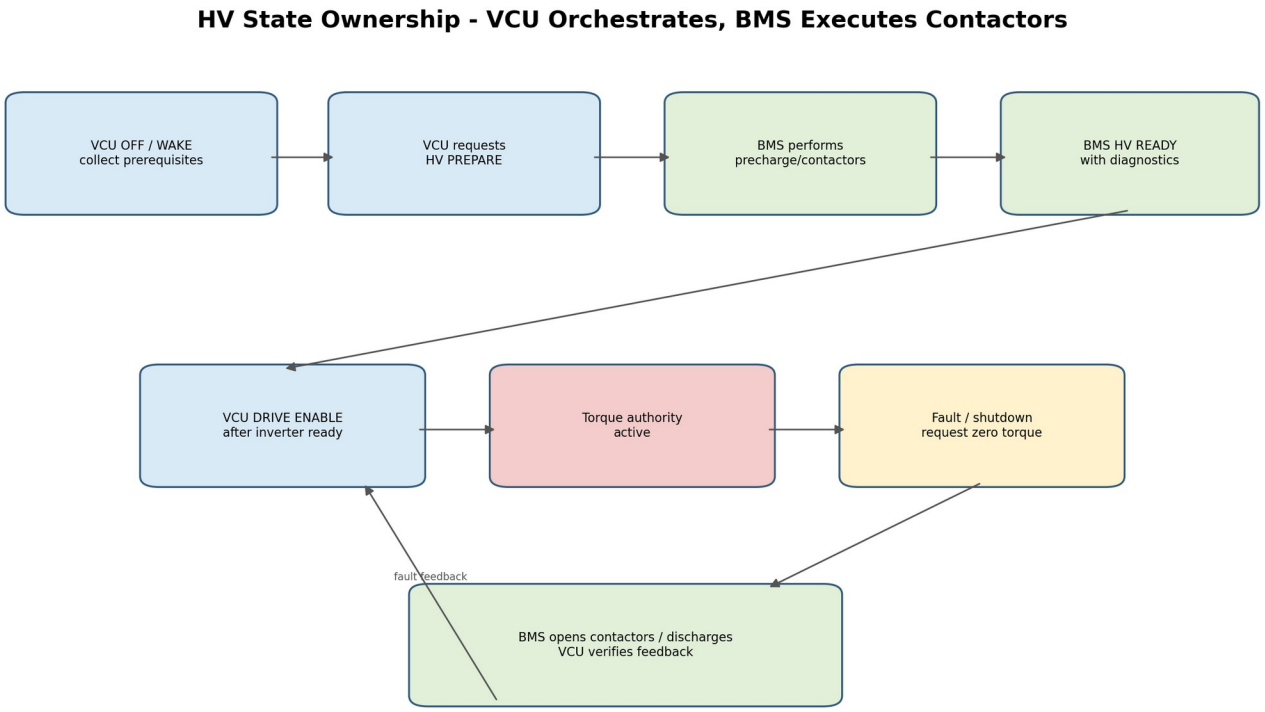


Figure 5 - HV-state sequence and BMS ownership

B10 - Energy, Thermal, Performance Limit & Derating Supervisor

Technical overview. Thermal and energy supervision aggregates BMS SOP, inverter/motor thermal limits, coolant/chiller status and vehicle energy strategy to derive monotonic capability limits.

Design justification. No universal 65 C cutoff is frozen. Supplier-normalized capability and calibrated thresholds control derating so the architecture remains reusable across battery/inverter variants.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B10-01	Limit arbitration	Available propulsion/regen shall select minimum valid limit from BMS, inverter, motor, thermal, chassis and mode policies.	Boundary sweep.	SW/System	Safety-related
VCU220-SYS3-B10-02	No universal 65C	VCU shall not use a single hard-coded battery/inverter temperature limit; supplier normalized capability limits are authoritative.	Calibration review.	SW/System	Safety-related
VCU220-SYS3-B10-03	Derate hysteresis	Thermal/power derates shall have hysteresis and controlled recovery to avoid oscillation.	Thermal HIL.	HW/SW	Safety-related
VCU220-SYS3-B10-04	Peak timer	220 kW peak shall be time/energy/temperature limited; continuous target 110 kW shall not be assumed at all ambient/SOC.	Dyno/HIL.	SW/System	Safety-related
VCU220-SYS3-B10-05	Coolant prioritization	Thermal supervisor shall arbitrate battery, inverter, motor and cabin requests based on mode and component risk.	Thermal bench.	SW/System	Safety-related
VCU220-SYS3-B10-06	Preconditioning	Track/Sport preconditioning shall not starve safety-critical thermal protection.	HIL.	SW/System	Safety-related
VCU220-SYS3-B10-07	Power reserve	Driver-mode power cap shall only reduce available capability; it shall not raise supplier SOP.	Boundary tests.	HW/SW	Safety-related
VCU220-SYS3-B10-08	Sensor invalid	Invalid thermal/SOP data shall reduce capability using a documented fallback.	Fault injection.	SW/System	Safety-related
VCU220-SYS3-B10-09	Recovery slew	Power recovery after	Vehicle tests.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		cooling/SOC improvement shall be rate-limited to avoid torque jumps.			
VCU220-SYS3-B10-10	Reason codes	Derate source and active limit shall be published to HMI/diagnostics.	Traceability.	SW/System	Safety-related

B11 - CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management

Technical overview. Network architecture separates safety CAN FD paths from vehicle IP/cockpit Ethernet. CAN scheduling is deadline-driven; Ethernet uses VLAN/QoS/time-sync and optional MACsec according to the cybersecurity concept.

Design justification. 1GBASE-T1 is the baseline for IP/service traffic. Dedicated display links carry pixel streams; 2.5/10G is reserved for future central/zonal consolidation rather than overloading the baseline requirement.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B11-01	CAN architecture	At least two independent CAN FD paths or equivalent safety network separation shall support powertrain/chassis/HV functions as allocated.	Single-bus fault retains safe response.	HW/SW	Safety-related
VCU220-SYS3-B11-02	CAN rates	500 kbps arbitration / 2 Mbps data is baseline; higher data phase may be used only after transceiver/network qualification.	Network conformance.	HW/SW	Safety-related
VCU220-SYS3-B11-03	Bus load	Nominal sustained safety CAN load <=40% and fault burst <=60% reference.	Worst-case traffic analysis.	SW/System	Safety-related
VCU220-SYS3-B11-04	Ethernet baseline	1000BASE-T1 supports IP diagnostics/service/cockpit; design reserves 2.5/10G uplink option.	Traffic matrix.	HW/SW	Safety-related
VCU220-SYS3-B11-05	Switch QoS	Ethernet switch shall support VLAN/QoS/TSN features needed to isolate critical traffic from OTA/IVI bursts.	Network stress.	SW/System	Safety-related
VCU220-SYS3-B11-06	E2E	Safety PDUs shall use AUTOSAR E2E profile selected by payload/stack; Profile 4 is preferred for suitable long safety PDUs.	Negative sequence/CRC tests.	SW/System	Safety-related
VCU220-SYS3-B11-07	SecOC	Where TARA allocates authenticity, SecOC shall add freshness/authenticator independent of E2E.	Replay/spoof tests.	SW/System	Safety-related
VCU220-SYS3-B11-08	Gateway filtering	Routing between IVI/telematics and safety CAN shall be allow-list controlled with direction/rate/service restrictions.	Pen test.	SW/System	Safety-related
VCU220-SYS3-B11-09	Sleep/wake	CAN partial networking and Ethernet TC10-class wake shall support low-power state without false wake storms.	Sleep/wake tests.	SW/System	Safety-related
VCU220-SYS3-B11-10	Network diagnostics	Bus-off, link state, timeouts, errors and routing drops shall be logged with counters.	Fault injection.	SW/System	Safety-related

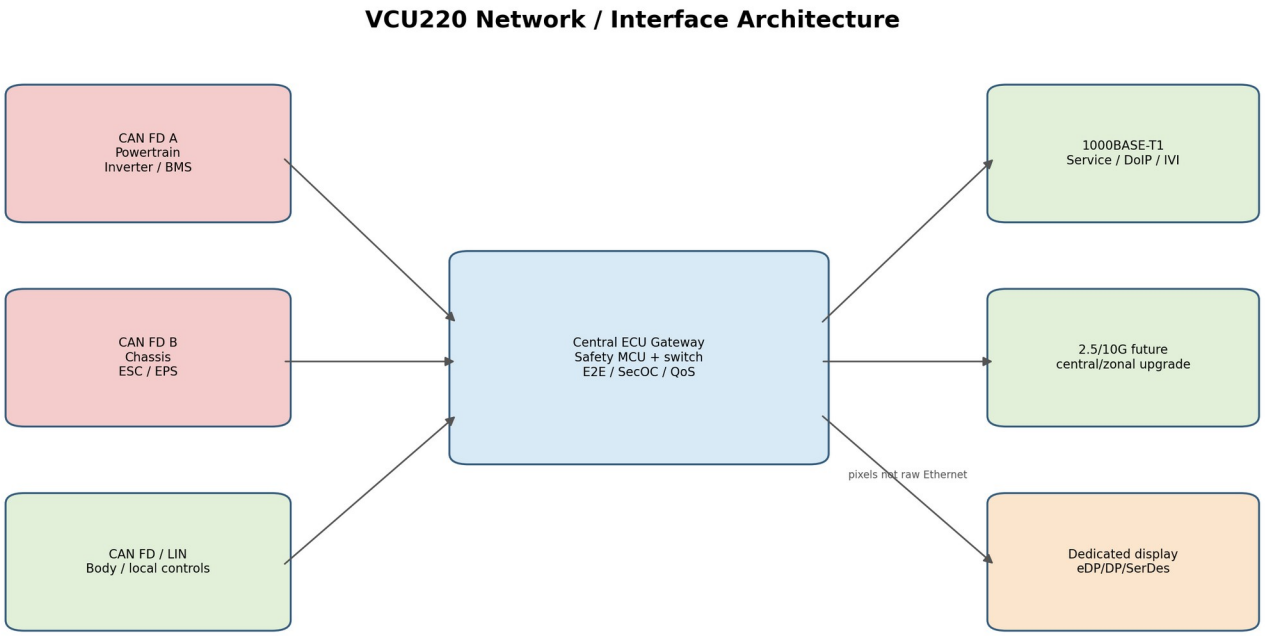


Figure 6 - Network and transport architecture

B12 - Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness

Technical overview. Time synchronization and middleware maintain signal age, monotonic timestamps, bounded queues and typed service contracts between safety and HPA partitions.

Design justification. Shared memory is treated as an interface requiring ownership, freshness and bounded capacity. QM services cannot overwrite safety-owned command memory or actuation peripherals.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B12-01	Global time	Central ECU shall maintain synchronized monotonic vehicle time for cross-domain logs and service data.	Time jump/holdover tests.	SW/System	Safety-related
VCU220-SYS3-B12-02	PTP	Ethernet time synchronization shall use 802.1AS/gPTP where distributed time-critical services require it.	Offset budget verified.	SW/System	Safety-related
VCU220-SYS3-B12-03	Safety time	Safety MCU shall not depend solely on HPA/Android wall-clock for control deadlines.	Independent timer verified.	SW/System	Safety-related
VCU220-SYS3-B12-04	SOME/IP	Service-oriented vehicle APIs may use SOME/IP/ara::com or OEM equivalent with versioned contracts.	Interface compatibility test.	SW/System	Safety-related
VCU220-SYS3-B12-05	IPC	Cross-partition IPC shall be typed, bounded, versioned and access-controlled.	Fuzz/negative test.	SW/System	Safety-related
VCU220-SYS3-B12-06	Data age	Every safety-relevant input shall carry age/quality and be rejected when stale.	Timeout tests.	SW/System	Safety-related
VCU220-SYS3-B12-07	Memory pools	Safety-relevant HPA services shall use bounded/preallocated communication queues to avoid unbounded runtime allocation.	Stress/OOM test.	HW/SW	Safety-related
VCU220-SYS3-B12-08	Service startup	Service discovery/startup shall not block safety MCU readiness.	Boot test.	SW/System	Safety-related
VCU220-SYS3-B12-09	Clock fault	Clock/source mismatch shall degrade timestamp-	Fault injection.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		dependent services without increasing torque authority.			
VCU220-SYS3-B12-10	Schema control	Signal/service definitions shall be configuration controlled with compatibility rules across OTA versions.	Update integration test.	SW/System	Safety-related

B13 - Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary

Technical overview. Cockpit/HMI presents driver mode, safety telltales, warnings, range and vehicle state. Rich Android rendering is not the only path for safety-critical indication.

Design justification. Critical telltales require monitored source data and a fallback/independent indication strategy appropriate to the safety analysis.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B13-01	Android boundary	Android/IVI shall not directly own torque/HV actuation interfaces.	Architecture/penetration test.	SW/System	Functional/QM
VCU220-SYS3-B13-02	Mode request	HMI sends driver-mode request; safety MCU returns authoritative accepted state/reason.	HIL.	SW/System	Functional/QM
VCU220-SYS3-B13-03	Safety telltale	Critical telltales shall have monitored fallback/overlay independent of rich compositor.	GPU/Android hang test.	SW/System	Safety-related
VCU220-SYS3-B13-04	Display latency	Accepted mode/safety state shall be visible within <=100 ms reference after authoritative state change.	HMI timing.	HW/SW	Functional/QM
VCU220-SYS3-B13-05	Cluster watchdog	Safety-relevant display path shall detect frozen/incorrect rendering where allocated.	Display fault injection.	HW/SW	Functional/QM
VCU220-SYS3-B13-06	Audio/haptic	Critical warning escalation shall use redundant audio/haptic channel where safety concept requires.	HMI test.	SW/System	Functional/QM
VCU220-SYS3-B13-07	User profile	User personalization cannot modify signed safety calibration outside allowed range.	Cyber test.	SW/System	Functional/QM
VCU220-SYS3-B13-08	Track UX	Track mode shall clearly indicate altered response and any road-use restriction.	Human-factors/market review.	SW/System	Functional/QM
VCU220-SYS3-B13-09	Display transport	Multi-display video shall use DP/eDP/SerDes; Ethernet remains service/control backbone.	SI/display test.	HW/SW	Functional/QM
VCU220-SYS3-B13-10	Privacy	Driver/user data handling shall follow OEM privacy/security requirements and be separated from safety logging.	Security/privacy review.	SW/System	Functional/QM

B14 - Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging

Technical overview. Diagnostics provide UDS/DoIP, DTCs, freeze frames, event logs and service routines while preventing service commands from bypassing vehicle-state and cybersecurity permissions.

Design justification. Production diagnostics must preserve traceability and avoid creating hidden torque/HV authority during service or plant modes.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B14-01	UDS services	VCU shall support project-approved UDS services and DIDs/RIDs/DTCs with session/security/state constraints.	Diagnostic conformance.	SW/System	Functional/QM
VCU220-SYS3-B14-02	DoIP	Ethernet diagnostics shall support DoIP and vehicle-gateway routing constraints.	DoIP conformance.	SW/System	Functional/QM
VCU220-SYS3-B14-03	DTC ownership	DTCs shall distinguish	Fault mapping coverage.	SW/System	Functional/QM

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		safety MCU, HPA partition, network, mode, HV, thermal and external ECU interface faults.			
VCU220-SYS3-B14-04	Freeze frame	Safety DTCs shall record mode, torque demand/final, limits, brake/gear/HV state, network age and reset reason.	HIL inspection.	SW/System	Functional/QM
VCU220-SYS3-B14-05	Flight recorder	Circular recorder shall capture pre/post critical torque/HV/mode/cyber events with time sync.	Capacity/trigger test.	SW/System	Functional/QM
VCU220-SYS3-B14-06	Service safety	Actuator/service routines require park/torque-zero/HV-safe preconditions as appropriate.	Negative service tests.	SW/System	Functional/QM
VCU220-SYS3-B14-07	Calibration readback	Service tool shall read active calibration/variant/software IDs and signatures.	Audit test.	SW/System	Functional/QM
VCU220-SYS3-B14-08	Remote diagnostics	Cloud/fleet remote diagnostic access shall be read-only by default and separately authorized for active routines.	Cyber test.	SW/System	Functional/QM
VCU220-SYS3-B14-09	Log wear	Storage logging shall use wear-managed bounded write rates.	Endurance analysis.	SW/System	Functional/QM
VCU220-SYS3-B14-10	Privacy retention	Retention/export policies shall separate safety evidence, warranty and personal data.	Data governance review.	SW/System	Functional/QM

B15 - Boot, OTA Software Update, Configuration, Variant & Calibration Management

Technical overview. Boot, update and configuration management control dependencies across safety MCU, HPA SoC, hypervisor, Android/Linux/QNX guests and calibrations.

Design justification. A/B or rollback-capable activation, compatibility manifests and safe activation states are required so an interrupted or mixed-version update cannot create an unsafe vehicle-control combination.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B15-01	Secure boot	Safety MCU, hypervisor and privileged software shall authenticate before execution.	Tamper boot tests.	SW/System	Safety-related
VCU220-SYS3-B15-02	A/B update	Update architecture shall use A/B or equivalent rollback-safe atomic activation for HPA software.	Power-cut update test.	SW/System	Safety-related
VCU220-SYS3-B15-03	Safety MCU update	Safety MCU update shall use controlled bootloader, vehicle-safe state and recovery path.	Interrupted update test.	HW/SW	Safety-related
VCU220-SYS3-B15-04	Dependency manifest	Vehicle update manifest shall bind compatible MCU/SoC/guest/calibration/variant versions.	Mismatch rejected.	SW/System	Safety-related
VCU220-SYS3-B15-05	Vehicle state	OTA activation shall require safe parked/power/HV conditions and adequate LV energy.	State negative tests.	SW/System	Safety-related
VCU220-SYS3-B15-06	Rollback	Rollback to known vulnerable/unsafe version shall be prevented outside approved recovery policy.	Cyber test.	SW/System	Safety-related
VCU220-SYS3-B15-07	Calibration update	Safety calibration is independently signed/range checked and may have separate approval workflow.	Negative calibration tests.	SW/System	Safety-related
VCU220-SYS3-B15-08	Variant coding	Region/powertrain/track/valet/cockpit variants shall be signed and topology compatible.	EOL/update test.	SW/System	Safety-related
VCU220-SYS3-B15-09	SUMS trace	Update status, software IDs, RXSWIN-equivalent identifiers where applicable and evidence shall support	Audit.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		ISO 24089/UNECE R156 obligations.			
VCU220-SYS3-B15-10	Recovery	Failed guest update shall not disable safety MCU torque/HV safe-state capability.	Fault injection.	SW/System	Safety-related

B16 - Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services

Technical overview. Cybersecurity establishes roots of trust, secure boot, HSM-backed keys, authenticated diagnostics/update, gateway filtering, SecOC freshness/authentication and security monitoring.

Design justification. AUTOSAR E2E and SecOC are separate: E2E addresses communication safety integrity while SecOC provides authenticity/freshness. Security monitoring must not block deterministic safety reaction.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B16-01	HSM	Safety MCU/SoC shall use HSM/secure enclave for keys, secure boot and critical authentication operations.	Key extraction negative test.	HW/SW	Safety-related
VCU220-SYS3-B16-02	SecOC	Safety PDU authenticator/freshness parameters shall be TARA- and bus-budget-derived.	Replay/spoof test.	SW/System	Safety-related
VCU220-SYS3-B16-03	MACsec	Ethernet MACsec may protect sensitive backbone flows where switch/PHY and latency budget support it.	Line-rate/latency test.	SW/System	Safety-related
VCU220-SYS3-B16-04	IDS	Gateway shall monitor anomalous CAN/Ethernet diagnostics/routing/security events without becoming a safety single point of failure.	Pen/stress test.	SW/System	Safety-related
VCU220-SYS3-B16-05	Debug	Production JTAG/SWD/debug shall be disabled or strongly authenticated and physically controlled.	EOL negative test.	SW/System	Safety-related
VCU220-SYS3-B16-06	Key lifecycle	Keys/certificates shall support production provisioning, rotation, revocation and service replacement.	Lifecycle audit.	SW/System	Safety-related
VCU220-SYS3-B16-07	Privilege	Android/Linux apps shall use least privilege; no direct raw CAN device access for normal apps.	OS security test.	SW/System	Safety-related
VCU220-SYS3-B16-08	Firewall	Gateway firewall shall enforce source/destination/service/rate policy between domains.	Pen test.	SW/System	Safety-related
VCU220-SYS3-B16-09	TARA update	Security monitoring/vulnerability changes shall update TARA and controls through lifecycle.	CSMS audit.	SW/System	Safety-related
VCU220-SYS3-B16-10	Safety-security coanalysis	Security control failure that can affect safety shall have explicit safety impact and degraded response.	Joint review.	HW/SW	Safety-related

B17 - Functional Safety Supervisor, Freedom-from-Interference & Health Management

Technical overview. Safety supervision coordinates freedom-from-interference, watchdogs, deadline monitoring, cross-checks and safe-state escalation across the centralized ECU.

Design justification. The safety case treats the HPA/cockpit domain as potentially faulty or compromised and demonstrates that shared resources cannot violate safety MCU timing, memory, network or power assumptions.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B17-01	Safety supervisor	Independent safety supervisor shall monitor torque, mode, HV, network, power, clock and MCU health.	Fault injection.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B17-02	FFI	QM HPA partitions shall not corrupt safety MCU memory/I/O/timing/power.	Common-cause/FFI evidence.	SW/System	Safety-related
VCU220-SYS3-B17-03	Reaction budget	Critical monitor DTI + reaction shall support <=10 ms internal motion-inhibit capability reference.	WCET/HIL timing.	SW/System	Safety-related
VCU220-SYS3-B17-04	Safe-state selection	Fault handling shall choose torque inhibit, degraded drive, controlled stop or HV isolation based on hazard/context.	Scenario HIL.	SW/System	Safety-related
VCU220-SYS3-B17-05	Watchdog diversity	Application task watchdog plus independent PMIC/external supervision shall detect stuck/overrun control.	Watchdog tests.	HW/SW	Safety-related
VCU220-SYS3-B17-06	Calibration monitor	Safety supervisor shall range/check CRC/signature of safety calibration at startup and update.	Corrupt calibration test.	SW/System	Safety-related
VCU220-SYS3-B17-07	Network monitor	Counter/age/E2E/SecOC failures shall not increase authority.	Network fault injection.	SW/System	Safety-related
VCU220-SYS3-B17-08	External feedback	Inverter/BMS/brake acknowledgements and delivered-state feedback shall be cross-checked.	Actuator mismatch tests.	SW/System	Safety-related
VCU220-SYS3-B17-09	Latent diagnostics	Safety mechanisms shall be proof-tested at startup/runtime intervals allocated by safety analysis.	Diagnostic coverage evidence.	SW/System	Safety-related
VCU220-SYS3-B17-10	Safety case evidence	FMEDA/FMEA/FTA/WCET/ test evidence shall trace to safety goals and supplier assumptions.	Safety-case audit.	SW/System	Safety-related

B18 - Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design

Technical overview. Thermal, EMC, SI/PI and mechanical design close the physical implementation of a high-power central ECU. Power integrity and memory/network signal integrity are validated at worst workload and temperature.

Design justification. The 70 W nominal / 130 W sustained peak ECU budget excludes external display backlights, amplifiers and powered peripherals unless explicitly reallocated. Thermal resistance figures are early enclosure design starts, not semiconductor junction guarantees.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B18-01	Thermal resistance	Central ECU cooling shall maintain silicon/case junction margins at 130 W sustained peak and released ambient; <=0.35 C/W case-to-ambient assembly reference is the Rev A2 50C ambient / 95C case design start at 130 W; final supplier junction/package model controls.	Thermal CFD/chamber.	HW/SW	Safety-related
VCU220-SYS3-B18-02	Power integrity	PDN shall meet SoC/DDR/MCU transient droop/noise limits under CPU/GPU/network load steps.	PI simulation + scope.	HW/SW	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B18-03	DDR SI	LPDDR routing/timing/eye margin shall close supplier design guide and PCB stack-up.	SI simulation + margining.	HW/SW	Safety-related
VCU220-SYS3-B18-04	PCIe/storage SI	PCIe/UFS/NVMe interfaces shall close channel insertion/return-loss/jitter targets.	SI/BER.	HW/SW	Safety-related
VCU220-SYS3-B18-05	Ethernet SI	100/1000/2.5/10G T1 channels shall meet selected PHY/OPEN Alliance channel masks.	S-parameter/BER.	HW/SW	Safety-related
VCU220-SYS3-B18-06	EMC emissions	Worst rich-HMI/GPU/network activity shall meet CISPR 25/OEM limits.	EMC chamber.	HW/SW	Safety-related
VCU220-SYS3-B18-07	EMC immunity	Safety outputs shall remain defined through ISO 11452/OEM immunity.	EMC immunity.	HW/SW	Safety-related
VCU220-SYS3-B18-08	Mechanical	Connector/PCB/heatsink shall meet mounting vibration/shock/environment qualification.	ISO 16750-3/4 or OEM.	HW/SW	Safety-related
VCU220-SYS3-B18-09	Condensation/corrosion	Enclosure/coating/venting shall suit cabin/console mounting and lifetime environment.	Environmental DVP.	HW/SW	Safety-related
VCU220-SYS3-B18-10	Serviceability	Thermal interface, connectors and labels shall support production/service without damaging safety configuration.	DFA/service review.	SW/System	Safety-related

B19 - Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning

Technical overview. Manufacturing/EOL provisions hardware identity, firmware, keys, calibration and variant coding and verifies network, I/O, PMIC, memory and safety-chain function before release.

Design justification. A vehicle cannot leave EOL with an untrusted topology or mismatched safety calibration. Golden references and station MSA are controlled production evidence.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B19-01	Secure provisioning	EOL shall provision keys/certificates/device identity/variant using authenticated tooling.	Duplicate/unsigned identity rejected.	HW/SW	Safety-related
VCU220-SYS3-B19-02	MCU test	EOL shall verify safety MCU power, watchdog, memory tests, analog/discrete I/O and CAN channels.	100% functional test.	HW/SW	Safety-related
VCU220-SYS3-B19-03	SoC test	EOL shall verify boot chain, memory/storage, Ethernet, display link and guest launch.	100% functional test.	HW/SW	Safety-related
VCU220-SYS3-B19-04	Power test	EOL shall verify standby/nominal/peak current, rail monitoring and controlled reset.	Limits pass.	HW/SW	Safety-related
VCU220-SYS3-B19-05	Network identity	CAN/Ethernet node IDs, MACs, VLAN/config and DoIP identity shall be variant-bound.	Topology scan.	SW/System	Safety-related
VCU220-SYS3-B19-06	Calibration	Pedal/analog scaling and vehicle-specific calibration checksum/version shall be verified.	Golden reference compare.	SW/System	Safety-related
VCU220-SYS3-B19-07	Safety path	Fault injection shall verify PMIC watchdog, safe-state, torque gateway inhibit and critical telltale fallback.	EOL/FAT evidence.	SW/System	Safety-related
VCU220-SYS3-B19-08	Configuration binding	Vehicle VIN/variant/powertrain/inverter/BMS/cockpit configuration shall be cryptographically/checksum bound.	Mismatch blocks READY.	SW/System	Safety-related
VCU220-SYS3-B19-09	Station quality	Test-station master references, fixture IDs,	MSA/audit.	SW/System	Safety-related

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
		calibration status and drift shall be controlled.			
VCU220-SYS3-B19-10	Genealogy	PCB/ECU serial, silicon IDs, software/calibration, key provisioning and test results shall be retained.	Traceability complete.	SW/System	Safety-related

B20 - Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS

Technical overview. Lifecycle engineering covers FRACAS, field logging, supplier PCN/EOL, cyber/update support, spare strategy and long-term calibration/configuration compatibility.

Design justification. Centralization increases common dependency on the ECU, so long-term availability, obsolescence and update support are architectural requirements rather than after-SOP procurement tasks.

Requirement ID	Name	Requirement Statement	Acceptance / Boundary	Allocation	Safety
VCU220-SYS3-B20-01	Mission life	Critical HW/SW components shall have a 15-year vehicle support/lifecycle plan or vehicle-program requirement with replacement/obsolescence strategy.	Lifecycle plan approved.	HW/SW	Functional/QM
VCU220-SYS3-B20-02	PCN/EOL	Critical semiconductor/software suppliers shall provide PCN/EOL/change notification and requalification support.	Supplier agreement.	SW/System	Functional/QM
VCU220-SYS3-B20-03	FRACAS	Field faults/nuisance trips/cyber incidents shall feed closed-loop problem solving and DFMEA/TARA updates.	Monthly/quarterly review.	SW/System	Functional/QM
VCU220-SYS3-B20-04	Telemetry	Non-personal health counters shall track resets, thermal throttling, network errors, storage wear and safety interventions.	Fleet analytics.	SW/System	Functional/QM
VCU220-SYS3-B20-05	Storage endurance	Log/update storage wear shall be monitored and sized for lifecycle write workload.	Endurance model.	HW/SW	Functional/QM
VCU220-SYS3-B20-06	Thermal aging	SoC/PMIC/capacitor thermal-hours and fan/pump if any shall be tracked or conservatively modeled.	Reliability model.	HW/SW	Functional/QM
VCU220-SYS3-B20-07	Calibration governance	Mode/torque/thermal calibration changes shall have review, simulation/HIL regression and release approval.	Audit.	SW/System	Functional/QM
VCU220-SYS3-B20-08	Software compatibility	Long-term update strategy shall maintain interface compatibility or provide migration for BMS/inverter/ESC variants.	OTA regression.	SW/System	Functional/QM
VCU220-SYS3-B20-09	Spares/tool support	Programming/service tools and critical spare ECUs shall be supported through service period.	Service plan.	SW/System	Functional/QM
VCU220-SYS3-B20-10	Decommissioning	Retired ECU shall sanitize credentials and preserve legally required safety/diagnostic records.	Decommissioning procedure.	HW/SW	Functional/QM

3. Hardware Safety Requirements (HSR)

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
HSR-VCU-001	VCU220-SYS3-B01-01	SG-01	Hardware shall implement and diagnose the physical safety mechanism required for 12v input envelope, including independent fault	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			containment where credited.			
HSR-VCU-002	VCU220-SYS3-B01-04	SG-02	Hardware shall implement and diagnose the physical safety mechanism required for brownout handling, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-003	VCU220-SYS3-B01-07	SG-03	Hardware shall implement and diagnose the physical safety mechanism required for sleep sequencing, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-004	VCU220-SYS3-B01-10	SG-04	Hardware shall implement and diagnose the physical safety mechanism required for power diagnostics, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-005	VCU220-SYS3-B02-03	SG-05	Hardware shall implement and diagnose the physical safety mechanism required for safe outputs, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-006	VCU220-SYS3-B02-06	SG-06	Hardware shall implement and diagnose the physical safety mechanism required for actuation gateway, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-007	VCU220-SYS3-B02-09	SG-07	Hardware shall implement and diagnose the physical safety mechanism required for freedom from interference, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-008	VCU220-SYS3-B03-02	SG-08	Hardware shall implement and diagnose the physical safety mechanism required for virtualization, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-009	VCU220-SYS3-B03-05	SG-09	Hardware shall implement and diagnose the physical safety mechanism required for storage, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-010	VCU220-SYS3-B03-08	SG-10	Hardware shall implement and diagnose the physical safety mechanism required for boot chain, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-011	VCU220-SYS3-B04-01	SG-11	Hardware shall implement and diagnose the physical safety mechanism	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			required for pedal sensing, including independent fault containment where credited.			
HSR-VCU-012	VCU220-SYS3-B04-04	SG-12	Hardware shall implement and diagnose the physical safety mechanism required for ignition/start, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-013	VCU220-SYS3-B04-07	SG-01	Hardware shall implement and diagnose the physical safety mechanism required for vehicle speed, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-014	VCU220-SYS3-B04-10	SG-02	Hardware shall implement and diagnose the physical safety mechanism required for input calibration, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-015	VCU220-SYS3-B05-03	SG-03	Hardware shall implement and diagnose the physical safety mechanism required for torque clamp, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-016	VCU220-SYS3-B05-06	SG-04	Hardware shall implement and diagnose the physical safety mechanism required for rate limiter, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-017	VCU220-SYS3-B05-09	SG-05	Hardware shall implement and diagnose the physical safety mechanism required for latency, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-018	VCU220-SYS3-B06-02	SG-06	Hardware shall implement and diagnose the physical safety mechanism required for charge sop, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-019	VCU220-SYS3-B06-05	SG-07	Hardware shall implement and diagnose the physical safety mechanism required for one-pedal, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-020	VCU220-SYS3-B06-08	SG-08	Hardware shall implement and diagnose the physical safety mechanism required for regen fault, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-021	VCU220-SYS3-B07-01	SG-09	Hardware shall implement and diagnose the physical safety mechanism	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			required for mode set, including independent fault containment where credited.			
HSR-VCU-022	VCU220-SYS3-B07-04	SG-10	Hardware shall implement and diagnose the physical safety mechanism required for blend timing, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-023	VCU220-SYS3-B07-07	SG-11	Hardware shall implement and diagnose the physical safety mechanism required for track gating, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-024	VCU220-SYS3-B07-10	SG-12	Hardware shall implement and diagnose the physical safety mechanism required for calibration safety, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-025	VCU220-SYS3-B08-03	SG-01	Hardware shall implement and diagnose the physical safety mechanism required for suspension request, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-026	VCU220-SYS3-B08-06	SG-02	Hardware shall implement and diagnose the physical safety mechanism required for hill hold, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-027	VCU220-SYS3-B08-09	SG-03	Hardware shall implement and diagnose the physical safety mechanism required for state consistency, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-028	VCU220-SYS3-B09-02	SG-04	Hardware shall implement and diagnose the physical safety mechanism required for bms authority, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-029	VCU220-SYS3-B09-05	SG-05	Hardware shall implement and diagnose the physical safety mechanism required for crash, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-030	VCU220-SYS3-B09-08	SG-06	Hardware shall implement and diagnose the physical safety mechanism required for shutdown, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-031	VCU220-SYS3-B10-01	SG-07	Hardware shall implement and diagnose the physical safety mechanism	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			required for limit arbitration, including independent fault containment where credited.			
HSR-VCU-032	VCU220-SYS3-B10-04	SG-08	Hardware shall implement and diagnose the physical safety mechanism required for peak timer, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-033	VCU220-SYS3-B10-07	SG-09	Hardware shall implement and diagnose the physical safety mechanism required for power reserve, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-034	VCU220-SYS3-B10-10	SG-10	Hardware shall implement and diagnose the physical safety mechanism required for reason codes, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-035	VCU220-SYS3-B11-03	SG-11	Hardware shall implement and diagnose the physical safety mechanism required for bus load, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-036	VCU220-SYS3-B11-06	SG-12	Hardware shall implement and diagnose the physical safety mechanism required for e2e, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-037	VCU220-SYS3-B11-09	SG-01	Hardware shall implement and diagnose the physical safety mechanism required for sleep/wake, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-038	VCU220-SYS3-B12-02	SG-02	Hardware shall implement and diagnose the physical safety mechanism required for ptp, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-039	VCU220-SYS3-B12-05	SG-03	Hardware shall implement and diagnose the physical safety mechanism required for ipc, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-040	VCU220-SYS3-B12-08	SG-04	Hardware shall implement and diagnose the physical safety mechanism required for service startup, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-041	VCU220-SYS3-B13-01	SG-05	Hardware shall implement and diagnose the physical safety mechanism required for android	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			boundary, including independent fault containment where credited.			
HSR-VCU-042	VCU220-SYS3-B13-04	SG-06	Hardware shall implement and diagnose the physical safety mechanism required for display latency, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-043	VCU220-SYS3-B13-07	SG-07	Hardware shall implement and diagnose the physical safety mechanism required for user profile, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-044	VCU220-SYS3-B13-10	SG-08	Hardware shall implement and diagnose the physical safety mechanism required for privacy, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-045	VCU220-SYS3-B14-03	SG-09	Hardware shall implement and diagnose the physical safety mechanism required for dtc ownership, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-046	VCU220-SYS3-B14-06	SG-10	Hardware shall implement and diagnose the physical safety mechanism required for service safety, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-047	VCU220-SYS3-B14-09	SG-11	Hardware shall implement and diagnose the physical safety mechanism required for log wear, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-048	VCU220-SYS3-B15-02	SG-12	Hardware shall implement and diagnose the physical safety mechanism required for a/b update, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-049	VCU220-SYS3-B15-05	SG-01	Hardware shall implement and diagnose the physical safety mechanism required for vehicle state, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-050	VCU220-SYS3-B15-08	SG-02	Hardware shall implement and diagnose the physical safety mechanism required for variant coding, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-051	VCU220-SYS3-B16-01	SG-03	Hardware shall implement and diagnose the physical safety mechanism	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

HSR	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
			required for hsm, including independent fault containment where credited.			
HSR-VCU-052	VCU220-SYS3-B16-04	SG-04	Hardware shall implement and diagnose the physical safety mechanism required for ids, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-053	VCU220-SYS3-B16-07	SG-05	Hardware shall implement and diagnose the physical safety mechanism required for privilege, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-054	VCU220-SYS3-B16-10	SG-06	Hardware shall implement and diagnose the physical safety mechanism required for safety-security coanalysis, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-055	VCU220-SYS3-B17-03	SG-07	Hardware shall implement and diagnose the physical safety mechanism required for reaction budget, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-056	VCU220-SYS3-B17-06	SG-08	Hardware shall implement and diagnose the physical safety mechanism required for calibration monitor, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-057	VCU220-SYS3-B17-09	SG-09	Hardware shall implement and diagnose the physical safety mechanism required for latent diagnostics, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-058	VCU220-SYS3-B18-02	SG-10	Hardware shall implement and diagnose the physical safety mechanism required for power integrity, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-059	VCU220-SYS3-B18-05	SG-11	Hardware shall implement and diagnose the physical safety mechanism required for ethernet si, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP
HSR-VCU-060	VCU220-SYS3-B18-08	SG-12	Hardware shall implement and diagnose the physical safety mechanism required for mechanical, including independent fault containment where credited.	Block/FSC timing	Central ECU HW / safety MCU / transceiver / PMIC	WCCA + bench fault injection + environmental DVP

4. Software Safety Requirements

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
SWSR-VCU-001	VCU220-SYS3-B01-02	SG-01	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for power consumption with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-002	VCU220-SYS3-B01-05	SG-02	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for hold-up with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-003	VCU220-SYS3-B01-08	SG-03	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for standby current with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-004	VCU220-SYS3-B02-01	SG-04	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety mcu capability with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-005	VCU220-SYS3-B02-04	SG-05	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for cycle time with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-006	VCU220-SYS3-B02-07	SG-06	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for clock supervision with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-007	VCU220-SYS3-B02-10	SG-07	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety logs with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-008	VCU220-SYS3-B03-03	SG-08	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety island use with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-009	VCU220-SYS3-B03-06	SG-09	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for iommu with	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			bounded execution and non-permissive stale-data behavior.			
SWSR-VCU-010	VCU220-SYS3-B03-09	SG-10	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for thermal throttling with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-011	VCU220-SYS3-B04-02	SG-11	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for brake input with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-012	VCU220-SYS3-B04-05	SG-12	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for crash/hvil with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-013	VCU220-SYS3-B04-08	SG-01	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for analog diagnostics with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-014	VCU220-SYS3-B05-01	SG-02	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for torque equation with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-015	VCU220-SYS3-B05-04	SG-03	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for brake override with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-016	VCU220-SYS3-B05-07	SG-04	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for feedback monitor with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-017	VCU220-SYS3-B05-10	SG-05	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for torque audit with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-018	VCU220-SYS3-B06-03	SG-06	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			reaction for wheel slip with bounded execution and non-permissive stale-data behavior.			
SWSR-VCU-019	VCU220-SYS3-B06-06	SG-07	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for brake pedal blend with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-020	VCU220-SYS3-B06-09	SG-08	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for jerk with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-021	VCU220-SYS3-B07-02	SG-09	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for request validation with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-022	VCU220-SYS3-B07-05	SG-10	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for eco cap with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-023	VCU220-SYS3-B07-08	SG-11	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for thermal precondition with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-024	VCU220-SYS3-B08-01	SG-12	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for esc authority with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-025	VCU220-SYS3-B08-04	SG-01	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for yaw/slip state with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-026	VCU220-SYS3-B08-07	SG-02	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for reverse with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-027	VCU220-SYS3-B08-10	SG-03	Safety software shall implement deterministic validation, state handling, diagnostics	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			and safe/degraded reaction for dynamics logging with bounded execution and non-permissive stale-data behavior.			
SWSR-VCU-028	VCU220-SYS3-B09-03	SG-04	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for precharge with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-029	VCU220-SYS3-B09-06	SG-05	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for charge interlock with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-030	VCU220-SYS3-B09-09	SG-06	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for 800v variant with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-031	VCU220-SYS3-B10-02	SG-07	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for no universal 65c with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-032	VCU220-SYS3-B10-05	SG-08	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for coolant prioritization with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-033	VCU220-SYS3-B10-08	SG-09	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for sensor invalid with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-034	VCU220-SYS3-B11-01	SG-10	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for can architecture with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-035	VCU220-SYS3-B11-04	SG-11	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for ethernet baseline with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-036	VCU220-SYS3-B11-07	SG-12	Safety software shall	Block/FSC timing	Safety MCU / safety	SIL/HIL fault injection +

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			implement deterministic validation, state handling, diagnostics and safe/degraded reaction for secoc with bounded execution and non-permissive stale-data behavior.		services	WCET + trace review
SWSR-VCU-037	VCU220-SYS3-B11-10	SG-01	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for network diagnostics with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-038	VCU220-SYS3-B12-03	SG-02	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety time with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-039	VCU220-SYS3-B12-06	SG-03	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for data age with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-040	VCU220-SYS3-B12-09	SG-04	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for clock fault with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-041	VCU220-SYS3-B13-02	SG-05	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for mode request with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-042	VCU220-SYS3-B13-05	SG-06	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for cluster watchdog with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-043	VCU220-SYS3-B13-08	SG-07	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for track ux with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-044	VCU220-SYS3-B14-01	SG-08	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for uds services with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-045	VCU220-SYS3-B14-04	SG-09	Safety software shall implement deterministic	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			validation, state handling, diagnostics and safe/degraded reaction for freeze frame with bounded execution and non-permissive stale-data behavior.			
SWSR-VCU-046	VCU220-SYS3-B14-07	SG-10	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for calibration readback with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-047	VCU220-SYS3-B14-10	SG-11	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for privacy retention with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-048	VCU220-SYS3-B15-03	SG-12	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety mcu update with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-049	VCU220-SYS3-B15-06	SG-01	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for rollback with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-050	VCU220-SYS3-B15-09	SG-02	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for sums trace with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-051	VCU220-SYS3-B16-02	SG-03	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for secoc with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-052	VCU220-SYS3-B16-05	SG-04	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for debug with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-053	VCU220-SYS3-B16-08	SG-05	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for firewall with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-054	VCU220-SYS3-B17-01	SG-06	Safety software shall implement deterministic	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

SWSR	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			validation, state handling, diagnostics and safe/degraded reaction for safety supervisor with bounded execution and non-permissive stale-data behavior.			
SWSR-VCU-055	VCU220-SYS3-B17-04	SG-07	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safe-state selection with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-056	VCU220-SYS3-B17-07	SG-08	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for network monitor with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-057	VCU220-SYS3-B17-10	SG-09	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for safety case evidence with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-058	VCU220-SYS3-B18-03	SG-10	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for ddr si with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-059	VCU220-SYS3-B18-06	SG-11	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for emc emissions with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review
SWSR-VCU-060	VCU220-SYS3-B18-09	SG-12	Safety software shall implement deterministic validation, state handling, diagnostics and safe/degraded reaction for condensation/corrosion with bounded execution and non-permissive stale-data behavior.	Block/FSC timing	Safety MCU / safety services	SIL/HIL fault injection + WCET + trace review

5. Hardware Functional Requirement Handoff

HWFR	Block	Hardware Functional Requirement	Verification
HWFR-VCU-001	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-002	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway hardware implementation shall allocate	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	
HWFR-VCU-003	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-004	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-005	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-006	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-007	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-008	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-009	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-010	B10	Energy, Thermal, Performance Limit & Derating Supervisor hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-011	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-012	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-013	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary hardware implementation shall allocate interfaces,	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	
HWFR-VCU-014	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-015	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-016	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-017	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-018	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-019	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-020	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 1, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-021	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-022	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-023	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-024	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs hardware implementation shall allocate interfaces, power, diagnostics and margin for	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		functional requirement set 2, with selected-device WCCA and DVP evidence.	
HWFR-VCU-025	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-026	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-027	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-028	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-029	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-030	B10	Energy, Thermal, Performance Limit & Derating Supervisor hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-031	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-032	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-033	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-034	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-035	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management hardware implementation shall allocate interfaces, power,	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	
HWFR-VCU-036	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-037	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-038	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-039	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-040	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 2, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-041	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-042	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-043	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-044	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-045	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-046	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3,	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		with selected-device WCCA and DVP evidence.	
HWFR-VCU-047	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-048	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-049	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-050	B10	Energy, Thermal, Performance Limit & Derating Supervisor hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-051	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-052	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-053	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-054	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-055	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-056	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-057	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3,	Schematic/PCB review + bench/DVP

HWFR	Block	Hardware Functional Requirement	Verification
		with selected-device WCCA and DVP evidence.	
HWFR-VCU-058	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-059	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP
HWFR-VCU-060	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS hardware implementation shall allocate interfaces, power, diagnostics and margin for functional requirement set 3, with selected-device WCCA and DVP evidence.	Schematic/PCB review + bench/DVP

6. Software Functional Requirement Handoff

SWFR	Block	Software Functional Requirement	Verification
SWFR-VCU-001	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-002	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-003	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-004	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-005	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-006	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-007	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-008	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager software implementation shall provide the	SIL/unit/HIL + integration

SWFR	Block	Software Functional Requirement	Verification
		required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	
SWFR-VCU-009	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-010	B10	Energy, Thermal, Performance Limit & Derating Supervisor software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-011	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-012	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-013	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-014	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-015	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-016	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-017	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-018	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-019	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration
SWFR-VCU-020	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 1.	SIL/unit/HIL + integration

SWFR	Block	Software Functional Requirement	Verification
SWFR-VCU-021	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-022	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-023	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-024	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-025	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-026	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-027	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-028	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-029	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-030	B10	Energy, Thermal, Performance Limit & Derating Supervisor software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-031	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-032	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-033	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary software implementation shall provide the	SIL/unit/HIL + integration

SWFR	Block	Software Functional Requirement	Verification
		required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	
SWFR-VCU-034	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-035	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-036	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-037	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-038	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-039	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-040	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 2.	SIL/unit/HIL + integration
SWFR-VCU-041	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-042	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-043	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-044	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-045	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement	SIL/unit/HIL + integration

SWFR	Block	Software Functional Requirement	Verification
		set 3.	
SWFR-VCU-046	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-047	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-048	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-049	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-050	B10	Energy, Thermal, Performance Limit & Derating Supervisor software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-051	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-052	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-053	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-054	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-055	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-056	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-057	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-058	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	SIL/unit/HIL + integration

SWFR	Block	Software Functional Requirement	Verification
		software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	
SWFR-VCU-059	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration
SWFR-VCU-060	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS software implementation shall provide the required state machine, calibration, timeout, diagnostics and observability for functional requirement set 3.	SIL/unit/HIL + integration

7. Control Algorithms and State Machines

Block	Algorithm / Function	Logic / State Sequence	Inputs / Outputs
B05	Torque Arbitration	Acquire/qualify pedal+brake+gear+speed -> compute mode-shaped driver demand -> merge cruise/ADAS if applicable -> apply brake/ESC overrides -> clamp by BMS/inverter/motor/thermal/mode limits -> rate limit -> independent safety check -> transmit torque PDU -> compare feedback.	Pedal/brake/gear/speed/BMS/inverter/ESC/mode -> authoritative torque/reason
B06	Regen Blend	Driver decel/one-pedal -> calculate desired decel -> calculate BMS/inverter/motor regen capability -> send regen capability/request to brake ECU -> brake ECU owns friction blend and wheel-slip -> feedback actual decel/regen -> energy log.	Brake/ESC/BMS/inverter/motor -> regen request/capability
B07	Mode Transition	Request -> authenticate source/profile -> validate state/market/thermal/chassis -> accept/reject -> cross-fade pedal/regen/chassis/thermal calibration using jerk/slew limits -> wait external ECU acknowledgements -> declare ACTIVE.	HMI/physical switch/vehicle state -> accepted mode/reason
B09	HV State	OFF -> ACCESSORY -> HV_REQUEST -> BMS precharge -> verify HV_READY + inverter ready -> READY -> DRIVE; any severe fault -> torque zero -> inverter disable -> BMS HV-off request -> SHUTDOWN/FAULT.	Ignition/BMS/inverter/charge/crash -> HV/drive permissive
B10	Limit Manager	Collect BMS SOP, inverter/motor torque, thermal/chassis limits -> qualify age -> minimum valid positive/negative torque capability -> apply mode caps only as reductions -> hysteresis/rate-limited recovery -> publish reason.	Capability inputs -> propulsion/regen limits
B17	Safety Supervisor	1 ms monitor loop -> plausibility of authoritative torque vs inputs/limits -> monitor timing/rails/network/calibration -> qualify fault -> select safe/degraded state -> revoke/inhibit authority -> record freeze-frame -> controlled recovery.	Safety inputs -> safe-state authority

8. Timing and Network Design Budgets

8.1 Pedal-to-torque latency

Stage	Budget ms
Pedal/input acquisition + qualification	2.0
Torque arbitration / limits / mode mapping	2.0
Safety monitor / authorization	1.0

Stage	Budget ms
CAN scheduling/transmit to inverter	2.0
Inverter reception/local plausibility	3.0
Vehicle-level reserve	5.0

8.2 Internal safety-reaction capability

Stage	Budget ms
Safety monitor cycle	1.0
Fault qualification	2.0
Safe-state decision	1.0
CAN/inhibit transmit	2.0
Inverter local inhibit/ack reference	4.0

CAN FD planning uses 500 kbit/s arbitration and 2 Mbit/s data. The workbook contains an approximate load model only; final acceptance requires bit-accurate network scheduling including arbitration, stuffing, retries, gateways and fault traffic. Safety message deadlines are verified on the final DBC/ARXML configuration.

9. DVP Strategy and Complete DVP Index

The master workbook contains the complete 400-row DVP with scenario, observables and acceptance criteria. This document includes the complete controlled DVP index plus the detailed system validation set to keep the review document navigable while preserving full traceability in the packaged workbook.

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-001-A	VCU220-SYS3-B01-01	B01	Power bench / LV transient / brownout	Nominal/boundary verification of 12V input envelope	DV	Open
DVP-VCU-001-B	VCU220-SYS3-B01-01	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for 12V input envelope	DV	Open
DVP-VCU-002-A	VCU220-SYS3-B01-02	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Power consumption	DV	Open
DVP-VCU-002-B	VCU220-SYS3-B01-02	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Power consumption	DV	Open
DVP-VCU-003-A	VCU220-SYS3-B01-03	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Safety power domain	DV	Open
DVP-VCU-003-B	VCU220-SYS3-B01-03	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Safety power domain	DV	Open
DVP-VCU-004-A	VCU220-SYS3-B01-04	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Brownout handling	DV	Open
DVP-VCU-004-B	VCU220-SYS3-B01-04	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Brownout handling	DV	Open
DVP-VCU-005-A	VCU220-SYS3-B01-05	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Hold-up	DV	Open
DVP-VCU-005-B	VCU220-SYS3-B01-05	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Hold-up	DV	Open
DVP-VCU-006-A	VCU220-SYS3-B01-06	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Wake sources	DV	Open
DVP-VCU-006-B	VCU220-SYS3-B01-06	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Wake sources	DV	Open
DVP-VCU-007-A	VCU220-SYS3-B01-07	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Sleep sequencing	DV	Open
DVP-VCU-007-B	VCU220-SYS3-B01-07	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Sleep sequencing	DV	Open
DVP-VCU-008-A	VCU220-SYS3-B01-08	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Standby current	DV	Open
DVP-VCU-008-B	VCU220-SYS3-B01-08	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Standby current	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-009-A	VCU220-SYS3-B01-09	B01	Power bench / LV transient / brownout	Nominal/boundary verification of PMIC watchdog	DV	Open
DVP-VCU-009-B	VCU220-SYS3-B01-09	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for PMIC watchdog	DV	Open
DVP-VCU-010-A	VCU220-SYS3-B01-10	B01	Power bench / LV transient / brownout	Nominal/boundary verification of Power diagnostics	DV	Open
DVP-VCU-010-B	VCU220-SYS3-B01-10	B01	Power bench / LV transient / brownout	Fault/degraded-case injection for Power diagnostics	DV	Open
DVP-VCU-011-A	VCU220-SYS3-B02-01	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Safety MCU capability	DV	Open
DVP-VCU-011-B	VCU220-SYS3-B02-01	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Safety MCU capability	DV	Open
DVP-VCU-012-A	VCU220-SYS3-B02-02	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Torque authority	DV	Open
DVP-VCU-012-B	VCU220-SYS3-B02-02	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Torque authority	DV	Open
DVP-VCU-013-A	VCU220-SYS3-B02-03	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Safe outputs	DV	Open
DVP-VCU-013-B	VCU220-SYS3-B02-03	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Safe outputs	DV	Open
DVP-VCU-014-A	VCU220-SYS3-B02-04	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Cycle time	DV	Open
DVP-VCU-014-B	VCU220-SYS3-B02-04	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Cycle time	DV	Open
DVP-VCU-015-A	VCU220-SYS3-B02-05	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Diverse supervision	DV	Open
DVP-VCU-015-B	VCU220-SYS3-B02-05	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Diverse supervision	DV	Open
DVP-VCU-016-A	VCU220-SYS3-B02-06	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Actuation gateway	DV	Open
DVP-VCU-016-B	VCU220-SYS3-B02-06	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Actuation gateway	DV	Open
DVP-VCU-017-A	VCU220-SYS3-B02-07	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Clock supervision	DV	Open
DVP-VCU-017-B	VCU220-SYS3-B02-07	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Clock supervision	DV	Open
DVP-VCU-018-A	VCU220-SYS3-B02-08	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Memory integrity	DV	Open
DVP-VCU-018-B	VCU220-SYS3-B02-08	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Memory integrity	DV	Open
DVP-VCU-019-A	VCU220-SYS3-B02-09	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Freedom from interference	DV	Open
DVP-VCU-019-B	VCU220-SYS3-B02-09	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Freedom from interference	DV	Open
DVP-VCU-020-A	VCU220-SYS3-B02-10	B02	Safety MCU bench/HIL fault injection	Nominal/boundary verification of Safety logs	DV	Open
DVP-VCU-020-B	VCU220-SYS3-B02-10	B02	Safety MCU bench/HIL fault injection	Fault/degraded-case injection for Safety logs	DV	Open
DVP-VCU-021-A	VCU220-SYS3-B03-01	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Application SoC headroom	DV	Open
DVP-VCU-021-B	VCU220-SYS3-B03-01	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Application SoC headroom	DV	Open
DVP-VCU-022-A	VCU220-SYS3-B03-02	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Virtualization	DV	Open
DVP-VCU-022-B	VCU220-SYS3-B03-02	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Virtualization	DV	Open
DVP-VCU-023-A	VCU220-SYS3-B03-03	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Safety island use	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-023-B	VCU220-SYS3-B03-03	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Safety island use	DV	Open
DVP-VCU-024-A	VCU220-SYS3-B03-04	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Memory bandwidth	DV	Open
DVP-VCU-024-B	VCU220-SYS3-B03-04	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Memory bandwidth	DV	Open
DVP-VCU-025-A	VCU220-SYS3-B03-05	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Storage	DV	Open
DVP-VCU-025-B	VCU220-SYS3-B03-05	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Storage	DV	Open
DVP-VCU-026-A	VCU220-SYS3-B03-06	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of IOMMU	DV	Open
DVP-VCU-026-B	VCU220-SYS3-B03-06	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for IOMMU	DV	Open
DVP-VCU-027-A	VCU220-SYS3-B03-07	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of GPU isolation	DV	Open
DVP-VCU-027-B	VCU220-SYS3-B03-07	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for GPU isolation	DV	Open
DVP-VCU-028-A	VCU220-SYS3-B03-08	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Boot chain	DV	Open
DVP-VCU-028-B	VCU220-SYS3-B03-08	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Boot chain	DV	Open
DVP-VCU-029-A	VCU220-SYS3-B03-09	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Thermal throttling	DV	Open
DVP-VCU-029-B	VCU220-SYS3-B03-09	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Thermal throttling	DV	Open
DVP-VCU-030-A	VCU220-SYS3-B03-10	B03	SoC benchmark / hypervisor fault injection	Nominal/boundary verification of Crash containment	DV	Open
DVP-VCU-030-B	VCU220-SYS3-B03-10	B03	SoC benchmark / hypervisor fault injection	Fault/degraded-case injection for Crash containment	DV	Open
DVP-VCU-031-A	VCU220-SYS3-B04-01	B04	I/O fault injection / HIL	Nominal/boundary verification of Pedal sensing	DV	Open
DVP-VCU-031-B	VCU220-SYS3-B04-01	B04	I/O fault injection / HIL	Fault/degraded-case injection for Pedal sensing	DV	Open
DVP-VCU-032-A	VCU220-SYS3-B04-02	B04	I/O fault injection / HIL	Nominal/boundary verification of Brake input	DV	Open
DVP-VCU-032-B	VCU220-SYS3-B04-02	B04	I/O fault injection / HIL	Fault/degraded-case injection for Brake input	DV	Open
DVP-VCU-033-A	VCU220-SYS3-B04-03	B04	I/O fault injection / HIL	Nominal/boundary verification of Gear/state input	DV	Open
DVP-VCU-033-B	VCU220-SYS3-B04-03	B04	I/O fault injection / HIL	Fault/degraded-case injection for Gear/state input	DV	Open
DVP-VCU-034-A	VCU220-SYS3-B04-04	B04	I/O fault injection / HIL	Nominal/boundary verification of Ignition/start	DV	Open
DVP-VCU-034-B	VCU220-SYS3-B04-04	B04	I/O fault injection / HIL	Fault/degraded-case injection for Ignition/start	DV	Open
DVP-VCU-035-A	VCU220-SYS3-B04-05	B04	I/O fault injection / HIL	Nominal/boundary verification of Crash/HVIL	DV	Open
DVP-VCU-035-B	VCU220-SYS3-B04-05	B04	I/O fault injection / HIL	Fault/degraded-case injection for Crash/HVIL	DV	Open
DVP-VCU-036-A	VCU220-SYS3-B04-06	B04	I/O fault injection / HIL	Nominal/boundary verification of Steering input	DV	Open
DVP-VCU-036-B	VCU220-SYS3-B04-06	B04	I/O fault injection / HIL	Fault/degraded-case injection for Steering input	DV	Open
DVP-VCU-037-A	VCU220-SYS3-B04-07	B04	I/O fault injection / HIL	Nominal/boundary verification of Vehicle speed	DV	Open
DVP-VCU-037-B	VCU220-SYS3-B04-07	B04	I/O fault injection / HIL	Fault/degraded-case injection for Vehicle speed	DV	Open
DVP-VCU-038-A	VCU220-SYS3-B04-08	B04	I/O fault injection / HIL	Nominal/boundary verification of Analog diagnostics	DV	Open
DVP-VCU-038-B	VCU220-SYS3-B04-08	B04	I/O fault injection / HIL	Fault/degraded-case injection for Analog diagnostics	DV	Open
DVP-VCU-039-A	VCU220-SYS3-B04-09	B04	I/O fault injection / HIL	Nominal/boundary verification of Debounce	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
				timing		
DVP-VCU-039-B	VCU220-SYS3-B04-09	B04	I/O fault injection / HIL	Fault/degraded-case injection for Debounce timing	DV	Open
DVP-VCU-040-A	VCU220-SYS3-B04-10	B04	I/O fault injection / HIL	Nominal/boundary verification of Input calibration	DV	Open
DVP-VCU-040-B	VCU220-SYS3-B04-10	B04	I/O fault injection / HIL	Fault/degraded-case injection for Input calibration	DV	Open
DVP-VCU-041-A	VCU220-SYS3-B05-01	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Torque equation	DV	Open
DVP-VCU-041-B	VCU220-SYS3-B05-01	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Torque equation	DV	Open
DVP-VCU-042-A	VCU220-SYS3-B05-02	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Pedal map	DV	Open
DVP-VCU-042-B	VCU220-SYS3-B05-02	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Pedal map	DV	Open
DVP-VCU-043-A	VCU220-SYS3-B05-03	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Torque clamp	DV	Open
DVP-VCU-043-B	VCU220-SYS3-B05-03	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Torque clamp	DV	Open
DVP-VCU-044-A	VCU220-SYS3-B05-04	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Brake override	DV	Open
DVP-VCU-044-B	VCU220-SYS3-B05-04	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Brake override	DV	Open
DVP-VCU-045-A	VCU220-SYS3-B05-05	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Direction safety	DV	Open
DVP-VCU-045-B	VCU220-SYS3-B05-05	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Direction safety	DV	Open
DVP-VCU-046-A	VCU220-SYS3-B05-06	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Rate limiter	DV	Open
DVP-VCU-046-B	VCU220-SYS3-B05-06	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Rate limiter	DV	Open
DVP-VCU-047-A	VCU220-SYS3-B05-07	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Feedback monitor	DV	Open
DVP-VCU-047-B	VCU220-SYS3-B05-07	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Feedback monitor	DV	Open
DVP-VCU-048-A	VCU220-SYS3-B05-08	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Degraded drive	DV	Open
DVP-VCU-048-B	VCU220-SYS3-B05-08	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Degraded drive	DV	Open
DVP-VCU-049-A	VCU220-SYS3-B05-09	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Latency	DV	Open
DVP-VCU-049-B	VCU220-SYS3-B05-09	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Latency	DV	Open
DVP-VCU-050-A	VCU220-SYS3-B05-10	B05	SIL/HIL/dyno/vehicle	Nominal/boundary verification of Torque audit	DV	Open
DVP-VCU-050-B	VCU220-SYS3-B05-10	B05	SIL/HIL/dyno/vehicle	Fault/degraded-case injection for Torque audit	DV	Open
DVP-VCU-051-A	VCU220-SYS3-B06-01	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Decel request	DV	Open
DVP-VCU-051-B	VCU220-SYS3-B06-01	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Decel request	DV	Open
DVP-VCU-052-A	VCU220-SYS3-B06-02	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Charge SOP	DV	Open
DVP-VCU-052-B	VCU220-SYS3-B06-02	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Charge SOP	DV	Open
DVP-VCU-053-A	VCU220-SYS3-B06-03	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Wheel slip	DV	Open
DVP-VCU-053-B	VCU220-SYS3-B06-03	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Wheel slip	DV	Open
DVP-VCU-054-A	VCU220-SYS3-B06-04	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Low speed fade	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-054-B	VCU220-SYS3-B06-04	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Low speed fade	DV	Open
DVP-VCU-055-A	VCU220-SYS3-B06-05	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of One-pedal	DV	Open
DVP-VCU-055-B	VCU220-SYS3-B06-05	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for One-pedal	DV	Open
DVP-VCU-056-A	VCU220-SYS3-B06-06	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Brake pedal blend	DV	Open
DVP-VCU-056-B	VCU220-SYS3-B06-06	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Brake pedal blend	DV	Open
DVP-VCU-057-A	VCU220-SYS3-B06-07	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Energy accounting	DV	Open
DVP-VCU-057-B	VCU220-SYS3-B06-07	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Energy accounting	DV	Open
DVP-VCU-058-A	VCU220-SYS3-B06-08	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Regen fault	DV	Open
DVP-VCU-058-B	VCU220-SYS3-B06-08	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Regen fault	DV	Open
DVP-VCU-059-A	VCU220-SYS3-B06-09	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Jerk	DV	Open
DVP-VCU-059-B	VCU220-SYS3-B06-09	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Jerk	DV	Open
DVP-VCU-060-A	VCU220-SYS3-B06-10	B06	Brake HIL/dyno/vehicle	Nominal/boundary verification of Mode influence	DV	Open
DVP-VCU-060-B	VCU220-SYS3-B06-10	B06	Brake HIL/dyno/vehicle	Fault/degraded-case injection for Mode influence	DV	Open
DVP-VCU-061-A	VCU220-SYS3-B07-01	B07	Mode HIL/vehicle	Nominal/boundary verification of Mode set	DV	Open
DVP-VCU-061-B	VCU220-SYS3-B07-01	B07	Mode HIL/vehicle	Fault/degraded-case injection for Mode set	DV	Open
DVP-VCU-062-A	VCU220-SYS3-B07-02	B07	Mode HIL/vehicle	Nominal/boundary verification of Request validation	DV	Open
DVP-VCU-062-B	VCU220-SYS3-B07-02	B07	Mode HIL/vehicle	Fault/degraded-case injection for Request validation	DV	Open
DVP-VCU-063-A	VCU220-SYS3-B07-03	B07	Mode HIL/vehicle	Nominal/boundary verification of Ack timing	DV	Open
DVP-VCU-063-B	VCU220-SYS3-B07-03	B07	Mode HIL/vehicle	Fault/degraded-case injection for Ack timing	DV	Open
DVP-VCU-064-A	VCU220-SYS3-B07-04	B07	Mode HIL/vehicle	Nominal/boundary verification of Blend timing	DV	Open
DVP-VCU-064-B	VCU220-SYS3-B07-04	B07	Mode HIL/vehicle	Fault/degraded-case injection for Blend timing	DV	Open
DVP-VCU-065-A	VCU220-SYS3-B07-05	B07	Mode HIL/vehicle	Nominal/boundary verification of Eco cap	DV	Open
DVP-VCU-065-B	VCU220-SYS3-B07-05	B07	Mode HIL/vehicle	Fault/degraded-case injection for Eco cap	DV	Open
DVP-VCU-066-A	VCU220-SYS3-B07-06	B07	Mode HIL/vehicle	Nominal/boundary verification of Valet cap	DV	Open
DVP-VCU-066-B	VCU220-SYS3-B07-06	B07	Mode HIL/vehicle	Fault/degraded-case injection for Valet cap	DV	Open
DVP-VCU-067-A	VCU220-SYS3-B07-07	B07	Mode HIL/vehicle	Nominal/boundary verification of Track gating	DV	Open
DVP-VCU-067-B	VCU220-SYS3-B07-07	B07	Mode HIL/vehicle	Fault/degraded-case injection for Track gating	DV	Open
DVP-VCU-068-A	VCU220-SYS3-B07-08	B07	Mode HIL/vehicle	Nominal/boundary verification of Thermal precondition	DV	Open
DVP-VCU-068-B	VCU220-SYS3-B07-08	B07	Mode HIL/vehicle	Fault/degraded-case injection for Thermal precondition	DV	Open
DVP-VCU-069-A	VCU220-SYS3-B07-09	B07	Mode HIL/vehicle	Nominal/boundary verification of Persistent preference	DV	Open
DVP-VCU-069-B	VCU220-SYS3-B07-09	B07	Mode HIL/vehicle	Fault/degraded-case injection for Persistent preference	DV	Open
DVP-VCU-070-A	VCU220-SYS3-B07-10	B07	Mode HIL/vehicle	Nominal/boundary verification of Calibration safety	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-070-B	VCU220-SYS3-B07-10	B07	Mode HIL/vehicle	Fault/degraded-case injection for Calibration safety	DV	Open
DVP-VCU-071-A	VCU220-SYS3-B08-01	B08	Chassis HIL/vehicle	Nominal/boundary verification of ESC authority	DV	Open
DVP-VCU-071-B	VCU220-SYS3-B08-01	B08	Chassis HIL/vehicle	Fault/degraded-case injection for ESC authority	DV	Open
DVP-VCU-072-A	VCU220-SYS3-B08-02	B08	Chassis HIL/vehicle	Nominal/boundary verification of EPS mode request	DV	Open
DVP-VCU-072-B	VCU220-SYS3-B08-02	B08	Chassis HIL/vehicle	Fault/degraded-case injection for EPS mode request	DV	Open
DVP-VCU-073-A	VCU220-SYS3-B08-03	B08	Chassis HIL/vehicle	Nominal/boundary verification of Suspension request	DV	Open
DVP-VCU-073-B	VCU220-SYS3-B08-03	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Suspension request	DV	Open
DVP-VCU-074-A	VCU220-SYS3-B08-04	B08	Chassis HIL/vehicle	Nominal/boundary verification of Yaw/slip state	DV	Open
DVP-VCU-074-B	VCU220-SYS3-B08-04	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Yaw/slip state	DV	Open
DVP-VCU-075-A	VCU220-SYS3-B08-05	B08	Chassis HIL/vehicle	Nominal/boundary verification of Launch control	DV	Open
DVP-VCU-075-B	VCU220-SYS3-B08-05	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Launch control	DV	Open
DVP-VCU-076-A	VCU220-SYS3-B08-06	B08	Chassis HIL/vehicle	Nominal/boundary verification of Hill hold	DV	Open
DVP-VCU-076-B	VCU220-SYS3-B08-06	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Hill hold	DV	Open
DVP-VCU-077-A	VCU220-SYS3-B08-07	B08	Chassis HIL/vehicle	Nominal/boundary verification of Reverse	DV	Open
DVP-VCU-077-B	VCU220-SYS3-B08-07	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Reverse	DV	Open
DVP-VCU-078-A	VCU220-SYS3-B08-08	B08	Chassis HIL/vehicle	Nominal/boundary verification of Chassis comm loss	DV	Open
DVP-VCU-078-B	VCU220-SYS3-B08-08	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Chassis comm loss	DV	Open
DVP-VCU-079-A	VCU220-SYS3-B08-09	B08	Chassis HIL/vehicle	Nominal/boundary verification of State consistency	DV	Open
DVP-VCU-079-B	VCU220-SYS3-B08-09	B08	Chassis HIL/vehicle	Fault/degraded-case injection for State consistency	DV	Open
DVP-VCU-080-A	VCU220-SYS3-B08-10	B08	Chassis HIL/vehicle	Nominal/boundary verification of Dynamics logging	DV	Open
DVP-VCU-080-B	VCU220-SYS3-B08-10	B08	Chassis HIL/vehicle	Fault/degraded-case injection for Dynamics logging	DV	Open
DVP-VCU-081-A	VCU220-SYS3-B09-01	B09	HV HIL/contactor model	Nominal/boundary verification of HV state machine	DV	Open
DVP-VCU-081-B	VCU220-SYS3-B09-01	B09	HV HIL/contactor model	Fault/degraded-case injection for HV state machine	DV	Open
DVP-VCU-082-A	VCU220-SYS3-B09-02	B09	HV HIL/contactor model	Nominal/boundary verification of BMS authority	DV	Open
DVP-VCU-082-B	VCU220-SYS3-B09-02	B09	HV HIL/contactor model	Fault/degraded-case injection for BMS authority	DV	Open
DVP-VCU-083-A	VCU220-SYS3-B09-03	B09	HV HIL/contactor model	Nominal/boundary verification of Precharge	DV	Open
DVP-VCU-083-B	VCU220-SYS3-B09-03	B09	HV HIL/contactor model	Fault/degraded-case injection for Precharge	DV	Open
DVP-VCU-084-A	VCU220-SYS3-B09-04	B09	HV HIL/contactor model	Nominal/boundary verification of Timeout	DV	Open
DVP-VCU-084-B	VCU220-SYS3-B09-04	B09	HV HIL/contactor model	Fault/degraded-case injection for Timeout	DV	Open
DVP-VCU-085-A	VCU220-SYS3-B09-05	B09	HV HIL/contactor model	Nominal/boundary verification of Crash	DV	Open
DVP-VCU-085-B	VCU220-SYS3-B09-05	B09	HV HIL/contactor model	Fault/degraded-case	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
				injection for Crash		
DVP-VCU-086-A	VCU220-SYS3-B09-06	B09	HV HIL/contactor model	Nominal/boundary verification of Charge interlock	DV	Open
DVP-VCU-086-B	VCU220-SYS3-B09-06	B09	HV HIL/contactor model	Fault/degraded-case injection for Charge interlock	DV	Open
DVP-VCU-087-A	VCU220-SYS3-B09-07	B09	HV HIL/contactor model	Nominal/boundary verification of Weld detection	DV	Open
DVP-VCU-087-B	VCU220-SYS3-B09-07	B09	HV HIL/contactor model	Fault/degraded-case injection for Weld detection	DV	Open
DVP-VCU-088-A	VCU220-SYS3-B09-08	B09	HV HIL/contactor model	Nominal/boundary verification of Shutdown	DV	Open
DVP-VCU-088-B	VCU220-SYS3-B09-08	B09	HV HIL/contactor model	Fault/degraded-case injection for Shutdown	DV	Open
DVP-VCU-089-A	VCU220-SYS3-B09-09	B09	HV HIL/contactor model	Nominal/boundary verification of 800V variant	DV	Open
DVP-VCU-089-B	VCU220-SYS3-B09-09	B09	HV HIL/contactor model	Fault/degraded-case injection for 800V variant	DV	Open
DVP-VCU-090-A	VCU220-SYS3-B09-10	B09	HV HIL/contactor model	Nominal/boundary verification of HV event record	DV	Open
DVP-VCU-090-B	VCU220-SYS3-B09-10	B09	HV HIL/contactor model	Fault/degraded-case injection for HV event record	DV	Open
DVP-VCU-091-A	VCU220-SYS3-B10-01	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Limit arbitration	DV	Open
DVP-VCU-091-B	VCU220-SYS3-B10-01	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Limit arbitration	DV	Open
DVP-VCU-092-A	VCU220-SYS3-B10-02	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of No universal 65C	DV	Open
DVP-VCU-092-B	VCU220-SYS3-B10-02	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for No universal 65C	DV	Open
DVP-VCU-093-A	VCU220-SYS3-B10-03	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Derate hysteresis	DV	Open
DVP-VCU-093-B	VCU220-SYS3-B10-03	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Derate hysteresis	DV	Open
DVP-VCU-094-A	VCU220-SYS3-B10-04	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Peak timer	DV	Open
DVP-VCU-094-B	VCU220-SYS3-B10-04	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Peak timer	DV	Open
DVP-VCU-095-A	VCU220-SYS3-B10-05	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Coolant prioritization	DV	Open
DVP-VCU-095-B	VCU220-SYS3-B10-05	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Coolant prioritization	DV	Open
DVP-VCU-096-A	VCU220-SYS3-B10-06	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Preconditioning	DV	Open
DVP-VCU-096-B	VCU220-SYS3-B10-06	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Preconditioning	DV	Open
DVP-VCU-097-A	VCU220-SYS3-B10-07	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Power reserve	DV	Open
DVP-VCU-097-B	VCU220-SYS3-B10-07	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Power reserve	DV	Open
DVP-VCU-098-A	VCU220-SYS3-B10-08	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Sensor invalid	DV	Open
DVP-VCU-098-B	VCU220-SYS3-B10-08	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Sensor invalid	DV	Open
DVP-VCU-099-A	VCU220-SYS3-B10-09	B10	Thermal/energy HIL/dyno	Nominal/boundary verification of Recovery slew	DV	Open
DVP-VCU-099-B	VCU220-SYS3-B10-09	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Recovery slew	DV	Open
DVP-VCU-100-A	VCU220-SYS3-B10-10	B10	Thermal/energy HIL/dyno	Nominal/boundary	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
				verification of Reason codes		
DVP-VCU-100-B	VCU220-SYS3-B10-10	B10	Thermal/energy HIL/dyno	Fault/degraded-case injection for Reason codes	DV	Open
DVP-VCU-101-A	VCU220-SYS3-B11-01	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of CAN architecture	DV	Open
DVP-VCU-101-B	VCU220-SYS3-B11-01	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for CAN architecture	DV	Open
DVP-VCU-102-A	VCU220-SYS3-B11-02	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of CAN rates	DV	Open
DVP-VCU-102-B	VCU220-SYS3-B11-02	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for CAN rates	DV	Open
DVP-VCU-103-A	VCU220-SYS3-B11-03	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Bus load	DV	Open
DVP-VCU-103-B	VCU220-SYS3-B11-03	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Bus load	DV	Open
DVP-VCU-104-A	VCU220-SYS3-B11-04	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Ethernet baseline	DV	Open
DVP-VCU-104-B	VCU220-SYS3-B11-04	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Ethernet baseline	DV	Open
DVP-VCU-105-A	VCU220-SYS3-B11-05	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Switch QoS	DV	Open
DVP-VCU-105-B	VCU220-SYS3-B11-05	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Switch QoS	DV	Open
DVP-VCU-106-A	VCU220-SYS3-B11-06	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of E2E	DV	Open
DVP-VCU-106-B	VCU220-SYS3-B11-06	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for E2E	DV	Open
DVP-VCU-107-A	VCU220-SYS3-B11-07	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of SecOC	DV	Open
DVP-VCU-107-B	VCU220-SYS3-B11-07	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for SecOC	DV	Open
DVP-VCU-108-A	VCU220-SYS3-B11-08	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Gateway filtering	DV	Open
DVP-VCU-108-B	VCU220-SYS3-B11-08	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Gateway filtering	DV	Open
DVP-VCU-109-A	VCU220-SYS3-B11-09	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Sleep/wake	DV	Open
DVP-VCU-109-B	VCU220-SYS3-B11-09	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Sleep/wake	DV	Open
DVP-VCU-110-A	VCU220-SYS3-B11-10	B11	CAN/Ethernet traffic + network fault injection	Nominal/boundary verification of Network diagnostics	DV	Open
DVP-VCU-110-B	VCU220-SYS3-B11-10	B11	CAN/Ethernet traffic + network fault injection	Fault/degraded-case injection for Network diagnostics	DV	Open
DVP-VCU-111-A	VCU220-SYS3-B12-01	B12	Middleware/time/IPC stress	Nominal/boundary verification of Global time	DV	Open
DVP-VCU-111-B	VCU220-SYS3-B12-01	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Global time	DV	Open
DVP-VCU-112-A	VCU220-SYS3-B12-02	B12	Middleware/time/IPC stress	Nominal/boundary verification of PTP	DV	Open
DVP-VCU-112-B	VCU220-SYS3-B12-02	B12	Middleware/time/IPC stress	Fault/degraded-case injection for PTP	DV	Open
DVP-VCU-113-A	VCU220-SYS3-B12-03	B12	Middleware/time/IPC stress	Nominal/boundary verification of Safety time	DV	Open
DVP-VCU-113-B	VCU220-SYS3-B12-03	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Safety time	DV	Open
DVP-VCU-114-A	VCU220-SYS3-B12-04	B12	Middleware/time/IPC stress	Nominal/boundary verification of SOME/IP	DV	Open
DVP-VCU-114-B	VCU220-SYS3-B12-04	B12	Middleware/time/IPC stress	Fault/degraded-case injection for SOME/IP	DV	Open
DVP-VCU-115-A	VCU220-SYS3-B12-05	B12	Middleware/time/IPC stress	Nominal/boundary verification of IPC	DV	Open
DVP-VCU-115-B	VCU220-SYS3-B12-05	B12	Middleware/time/IPC stress	Fault/degraded-case injection for IPC	DV	Open
DVP-VCU-116-A	VCU220-SYS3-B12-06	B12	Middleware/time/IPC stress	Nominal/boundary verification of Data age	DV	Open
DVP-VCU-116-B	VCU220-SYS3-B12-06	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Data age	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-117-A	VCU220-SYS3-B12-07	B12	Middleware/time/IPC stress	Nominal/boundary verification of Memory pools	DV	Open
DVP-VCU-117-B	VCU220-SYS3-B12-07	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Memory pools	DV	Open
DVP-VCU-118-A	VCU220-SYS3-B12-08	B12	Middleware/time/IPC stress	Nominal/boundary verification of Service startup	DV	Open
DVP-VCU-118-B	VCU220-SYS3-B12-08	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Service startup	DV	Open
DVP-VCU-119-A	VCU220-SYS3-B12-09	B12	Middleware/time/IPC stress	Nominal/boundary verification of Clock fault	DV	Open
DVP-VCU-119-B	VCU220-SYS3-B12-09	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Clock fault	DV	Open
DVP-VCU-120-A	VCU220-SYS3-B12-10	B12	Middleware/time/IPC stress	Nominal/boundary verification of Schema control	DV	Open
DVP-VCU-120-B	VCU220-SYS3-B12-10	B12	Middleware/time/IPC stress	Fault/degraded-case injection for Schema control	DV	Open
DVP-VCU-121-A	VCU220-SYS3-B13-01	B13	HMI/cluster fault injection	Nominal/boundary verification of Android boundary	DV	Open
DVP-VCU-121-B	VCU220-SYS3-B13-01	B13	HMI/cluster fault injection	Fault/degraded-case injection for Android boundary	DV	Open
DVP-VCU-122-A	VCU220-SYS3-B13-02	B13	HMI/cluster fault injection	Nominal/boundary verification of Mode request	DV	Open
DVP-VCU-122-B	VCU220-SYS3-B13-02	B13	HMI/cluster fault injection	Fault/degraded-case injection for Mode request	DV	Open
DVP-VCU-123-A	VCU220-SYS3-B13-03	B13	HMI/cluster fault injection	Nominal/boundary verification of Safety telltale	DV	Open
DVP-VCU-123-B	VCU220-SYS3-B13-03	B13	HMI/cluster fault injection	Fault/degraded-case injection for Safety telltale	DV	Open
DVP-VCU-124-A	VCU220-SYS3-B13-04	B13	HMI/cluster fault injection	Nominal/boundary verification of Display latency	DV	Open
DVP-VCU-124-B	VCU220-SYS3-B13-04	B13	HMI/cluster fault injection	Fault/degraded-case injection for Display latency	DV	Open
DVP-VCU-125-A	VCU220-SYS3-B13-05	B13	HMI/cluster fault injection	Nominal/boundary verification of Cluster watchdog	DV	Open
DVP-VCU-125-B	VCU220-SYS3-B13-05	B13	HMI/cluster fault injection	Fault/degraded-case injection for Cluster watchdog	DV	Open
DVP-VCU-126-A	VCU220-SYS3-B13-06	B13	HMI/cluster fault injection	Nominal/boundary verification of Audio/haptic	DV	Open
DVP-VCU-126-B	VCU220-SYS3-B13-06	B13	HMI/cluster fault injection	Fault/degraded-case injection for Audio/haptic	DV	Open
DVP-VCU-127-A	VCU220-SYS3-B13-07	B13	HMI/cluster fault injection	Nominal/boundary verification of User profile	DV	Open
DVP-VCU-127-B	VCU220-SYS3-B13-07	B13	HMI/cluster fault injection	Fault/degraded-case injection for User profile	DV	Open
DVP-VCU-128-A	VCU220-SYS3-B13-08	B13	HMI/cluster fault injection	Nominal/boundary verification of Track UX	DV	Open
DVP-VCU-128-B	VCU220-SYS3-B13-08	B13	HMI/cluster fault injection	Fault/degraded-case injection for Track UX	DV	Open
DVP-VCU-129-A	VCU220-SYS3-B13-09	B13	HMI/cluster fault injection	Nominal/boundary verification of Display transport	DV	Open
DVP-VCU-129-B	VCU220-SYS3-B13-09	B13	HMI/cluster fault injection	Fault/degraded-case injection for Display transport	DV	Open
DVP-VCU-130-A	VCU220-SYS3-B13-10	B13	HMI/cluster fault injection	Nominal/boundary verification of Privacy	DV	Open
DVP-VCU-130-B	VCU220-SYS3-B13-10	B13	HMI/cluster fault injection	Fault/degraded-case injection for Privacy	DV	Open
DVP-VCU-131-A	VCU220-SYS3-B14-01	B14	UDS/DoIP/service test	Nominal/boundary verification of UDS services	DV	Open
DVP-VCU-131-B	VCU220-SYS3-B14-01	B14	UDS/DoIP/service test	Fault/degraded-case injection for UDS services	DV	Open
DVP-VCU-132-A	VCU220-SYS3-B14-02	B14	UDS/DoIP/service test	Nominal/boundary verification of DoIP	DV	Open
DVP-VCU-132-B	VCU220-SYS3-B14-02	B14	UDS/DoIP/service test	Fault/degraded-case	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
				injection for DoIP		
DVP-VCU-133-A	VCU220-SYS3-B14-03	B14	UDS/DoIP/service test	Nominal/boundary verification of DTC ownership	DV	Open
DVP-VCU-133-B	VCU220-SYS3-B14-03	B14	UDS/DoIP/service test	Fault/degraded-case injection for DTC ownership	DV	Open
DVP-VCU-134-A	VCU220-SYS3-B14-04	B14	UDS/DoIP/service test	Nominal/boundary verification of Freeze frame	DV	Open
DVP-VCU-134-B	VCU220-SYS3-B14-04	B14	UDS/DoIP/service test	Fault/degraded-case injection for Freeze frame	DV	Open
DVP-VCU-135-A	VCU220-SYS3-B14-05	B14	UDS/DoIP/service test	Nominal/boundary verification of Flight recorder	DV	Open
DVP-VCU-135-B	VCU220-SYS3-B14-05	B14	UDS/DoIP/service test	Fault/degraded-case injection for Flight recorder	DV	Open
DVP-VCU-136-A	VCU220-SYS3-B14-06	B14	UDS/DoIP/service test	Nominal/boundary verification of Service safety	DV	Open
DVP-VCU-136-B	VCU220-SYS3-B14-06	B14	UDS/DoIP/service test	Fault/degraded-case injection for Service safety	DV	Open
DVP-VCU-137-A	VCU220-SYS3-B14-07	B14	UDS/DoIP/service test	Nominal/boundary verification of Calibration readback	DV	Open
DVP-VCU-137-B	VCU220-SYS3-B14-07	B14	UDS/DoIP/service test	Fault/degraded-case injection for Calibration readback	DV	Open
DVP-VCU-138-A	VCU220-SYS3-B14-08	B14	UDS/DoIP/service test	Nominal/boundary verification of Remote diagnostics	DV	Open
DVP-VCU-138-B	VCU220-SYS3-B14-08	B14	UDS/DoIP/service test	Fault/degraded-case injection for Remote diagnostics	DV	Open
DVP-VCU-139-A	VCU220-SYS3-B14-09	B14	UDS/DoIP/service test	Nominal/boundary verification of Log wear	DV	Open
DVP-VCU-139-B	VCU220-SYS3-B14-09	B14	UDS/DoIP/service test	Fault/degraded-case injection for Log wear	DV	Open
DVP-VCU-140-A	VCU220-SYS3-B14-10	B14	UDS/DoIP/service test	Nominal/boundary verification of Privacy retention	DV	Open
DVP-VCU-140-B	VCU220-SYS3-B14-10	B14	UDS/DoIP/service test	Fault/degraded-case injection for Privacy retention	DV	Open
DVP-VCU-141-A	VCU220-SYS3-B15-01	B15	OTA/update/power-cut test	Nominal/boundary verification of Secure boot	DV	Open
DVP-VCU-141-B	VCU220-SYS3-B15-01	B15	OTA/update/power-cut test	Fault/degraded-case injection for Secure boot	DV	Open
DVP-VCU-142-A	VCU220-SYS3-B15-02	B15	OTA/update/power-cut test	Nominal/boundary verification of A/B update	DV	Open
DVP-VCU-142-B	VCU220-SYS3-B15-02	B15	OTA/update/power-cut test	Fault/degraded-case injection for A/B update	DV	Open
DVP-VCU-143-A	VCU220-SYS3-B15-03	B15	OTA/update/power-cut test	Nominal/boundary verification of Safety MCU update	DV	Open
DVP-VCU-143-B	VCU220-SYS3-B15-03	B15	OTA/update/power-cut test	Fault/degraded-case injection for Safety MCU update	DV	Open
DVP-VCU-144-A	VCU220-SYS3-B15-04	B15	OTA/update/power-cut test	Nominal/boundary verification of Dependency manifest	DV	Open
DVP-VCU-144-B	VCU220-SYS3-B15-04	B15	OTA/update/power-cut test	Fault/degraded-case injection for Dependency manifest	DV	Open
DVP-VCU-145-A	VCU220-SYS3-B15-05	B15	OTA/update/power-cut test	Nominal/boundary verification of Vehicle state	DV	Open
DVP-VCU-145-B	VCU220-SYS3-B15-05	B15	OTA/update/power-cut test	Fault/degraded-case injection for Vehicle state	DV	Open
DVP-VCU-146-A	VCU220-SYS3-B15-06	B15	OTA/update/power-cut test	Nominal/boundary verification of Rollback	DV	Open
DVP-VCU-146-B	VCU220-SYS3-B15-06	B15	OTA/update/power-cut test	Fault/degraded-case injection for Rollback	DV	Open
DVP-VCU-147-A	VCU220-SYS3-B15-07	B15	OTA/update/power-cut test	Nominal/boundary verification of Calibration update	DV	Open
DVP-VCU-147-B	VCU220-SYS3-B15-07	B15	OTA/update/power-cut test	Fault/degraded-case injection for Calibration	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
				update		
DVP-VCU-148-A	VCU220-SYS3-B15-08	B15	OTA/update/power-cut test	Nominal/boundary verification of Variant coding	DV	Open
DVP-VCU-148-B	VCU220-SYS3-B15-08	B15	OTA/update/power-cut test	Fault/degraded-case injection for Variant coding	DV	Open
DVP-VCU-149-A	VCU220-SYS3-B15-09	B15	OTA/update/power-cut test	Nominal/boundary verification of SUMS trace	DV	Open
DVP-VCU-149-B	VCU220-SYS3-B15-09	B15	OTA/update/power-cut test	Fault/degraded-case injection for SUMS trace	DV	Open
DVP-VCU-150-A	VCU220-SYS3-B15-10	B15	OTA/update/power-cut test	Nominal/boundary verification of Recovery	DV	Open
DVP-VCU-150-B	VCU220-SYS3-B15-10	B15	OTA/update/power-cut test	Fault/degraded-case injection for Recovery	DV	Open
DVP-VCU-151-A	VCU220-SYS3-B16-01	B16	Penetration/cyber HIL	Nominal/boundary verification of HSM	DV	Open
DVP-VCU-151-B	VCU220-SYS3-B16-01	B16	Penetration/cyber HIL	Fault/degraded-case injection for HSM	DV	Open
DVP-VCU-152-A	VCU220-SYS3-B16-02	B16	Penetration/cyber HIL	Nominal/boundary verification of SecOC	DV	Open
DVP-VCU-152-B	VCU220-SYS3-B16-02	B16	Penetration/cyber HIL	Fault/degraded-case injection for SecOC	DV	Open
DVP-VCU-153-A	VCU220-SYS3-B16-03	B16	Penetration/cyber HIL	Nominal/boundary verification of MACsec	DV	Open
DVP-VCU-153-B	VCU220-SYS3-B16-03	B16	Penetration/cyber HIL	Fault/degraded-case injection for MACsec	DV	Open
DVP-VCU-154-A	VCU220-SYS3-B16-04	B16	Penetration/cyber HIL	Nominal/boundary verification of IDS	DV	Open
DVP-VCU-154-B	VCU220-SYS3-B16-04	B16	Penetration/cyber HIL	Fault/degraded-case injection for IDS	DV	Open
DVP-VCU-155-A	VCU220-SYS3-B16-05	B16	Penetration/cyber HIL	Nominal/boundary verification of Debug	DV	Open
DVP-VCU-155-B	VCU220-SYS3-B16-05	B16	Penetration/cyber HIL	Fault/degraded-case injection for Debug	DV	Open
DVP-VCU-156-A	VCU220-SYS3-B16-06	B16	Penetration/cyber HIL	Nominal/boundary verification of Key lifecycle	DV	Open
DVP-VCU-156-B	VCU220-SYS3-B16-06	B16	Penetration/cyber HIL	Fault/degraded-case injection for Key lifecycle	DV	Open
DVP-VCU-157-A	VCU220-SYS3-B16-07	B16	Penetration/cyber HIL	Nominal/boundary verification of Privilege	DV	Open
DVP-VCU-157-B	VCU220-SYS3-B16-07	B16	Penetration/cyber HIL	Fault/degraded-case injection for Privilege	DV	Open
DVP-VCU-158-A	VCU220-SYS3-B16-08	B16	Penetration/cyber HIL	Nominal/boundary verification of Firewall	DV	Open
DVP-VCU-158-B	VCU220-SYS3-B16-08	B16	Penetration/cyber HIL	Fault/degraded-case injection for Firewall	DV	Open
DVP-VCU-159-A	VCU220-SYS3-B16-09	B16	Penetration/cyber HIL	Nominal/boundary verification of TARA update	DV	Open
DVP-VCU-159-B	VCU220-SYS3-B16-09	B16	Penetration/cyber HIL	Fault/degraded-case injection for TARA update	DV	Open
DVP-VCU-160-A	VCU220-SYS3-B16-10	B16	Penetration/cyber HIL	Nominal/boundary verification of Safety-security coanalysis	DV	Open
DVP-VCU-160-B	VCU220-SYS3-B16-10	B16	Penetration/cyber HIL	Fault/degraded-case injection for Safety-security coanalysis	DV	Open
DVP-VCU-161-A	VCU220-SYS3-B17-01	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Safety supervisor	DV	Open
DVP-VCU-161-B	VCU220-SYS3-B17-01	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Safety supervisor	DV	Open
DVP-VCU-162-A	VCU220-SYS3-B17-02	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of FFI	DV	Open
DVP-VCU-162-B	VCU220-SYS3-B17-02	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for FFI	DV	Open
DVP-VCU-163-A	VCU220-SYS3-B17-03	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Reaction budget	DV	Open
DVP-VCU-163-B	VCU220-SYS3-B17-03	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Reaction budget	DV	Open
DVP-VCU-164-A	VCU220-SYS3-B17-04	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Safe-state selection	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-164-B	VCU220-SYS3-B17-04	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Safe-state selection	DV	Open
DVP-VCU-165-A	VCU220-SYS3-B17-05	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Watchdog diversity	DV	Open
DVP-VCU-165-B	VCU220-SYS3-B17-05	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Watchdog diversity	DV	Open
DVP-VCU-166-A	VCU220-SYS3-B17-06	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Calibration monitor	DV	Open
DVP-VCU-166-B	VCU220-SYS3-B17-06	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Calibration monitor	DV	Open
DVP-VCU-167-A	VCU220-SYS3-B17-07	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Network monitor	DV	Open
DVP-VCU-167-B	VCU220-SYS3-B17-07	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Network monitor	DV	Open
DVP-VCU-168-A	VCU220-SYS3-B17-08	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of External feedback	DV	Open
DVP-VCU-168-B	VCU220-SYS3-B17-08	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for External feedback	DV	Open
DVP-VCU-169-A	VCU220-SYS3-B17-09	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Latent diagnostics	DV	Open
DVP-VCU-169-B	VCU220-SYS3-B17-09	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Latent diagnostics	DV	Open
DVP-VCU-170-A	VCU220-SYS3-B17-10	B17	Safety HIL/WCET/fault injection	Nominal/boundary verification of Safety case evidence	DV	Open
DVP-VCU-170-B	VCU220-SYS3-B17-10	B17	Safety HIL/WCET/fault injection	Fault/degraded-case injection for Safety case evidence	DV	Open
DVP-VCU-171-A	VCU220-SYS3-B18-01	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Thermal resistance	DV	Open
DVP-VCU-171-B	VCU220-SYS3-B18-01	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Thermal resistance	DV	Open
DVP-VCU-172-A	VCU220-SYS3-B18-02	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Power integrity	DV	Open
DVP-VCU-172-B	VCU220-SYS3-B18-02	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Power integrity	DV	Open
DVP-VCU-173-A	VCU220-SYS3-B18-03	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of DDR SI	DV	Open
DVP-VCU-173-B	VCU220-SYS3-B18-03	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for DDR SI	DV	Open
DVP-VCU-174-A	VCU220-SYS3-B18-04	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of PCIe/storage SI	DV	Open
DVP-VCU-174-B	VCU220-SYS3-B18-04	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for PCIe/storage SI	DV	Open
DVP-VCU-175-A	VCU220-SYS3-B18-05	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Ethernet SI	DV	Open
DVP-VCU-175-B	VCU220-SYS3-B18-05	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Ethernet SI	DV	Open
DVP-VCU-176-A	VCU220-SYS3-B18-06	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of EMC emissions	DV	Open
DVP-VCU-176-B	VCU220-SYS3-B18-06	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for EMC emissions	DV	Open
DVP-VCU-177-A	VCU220-SYS3-B18-07	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of EMC immunity	DV	Open
DVP-VCU-177-B	VCU220-SYS3-B18-07	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for EMC immunity	DV	Open
DVP-VCU-178-A	VCU220-SYS3-B18-08	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Mechanical	DV	Open
DVP-VCU-178-B	VCU220-SYS3-B18-08	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Mechanical	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-179-A	VCU220-SYS3-B18-09	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Condensation/corrosion	DV	Open
DVP-VCU-179-B	VCU220-SYS3-B18-09	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Condensation/corrosion	DV	Open
DVP-VCU-180-A	VCU220-SYS3-B18-10	B18	SI/PI/EMC/thermal/ environmental	Nominal/boundary verification of Serviceability	DV	Open
DVP-VCU-180-B	VCU220-SYS3-B18-10	B18	SI/PI/EMC/thermal/ environmental	Fault/degraded-case injection for Serviceability	DV	Open
DVP-VCU-181-A	VCU220-SYS3-B19-01	B19	EOL/FAT/production audit	Nominal/boundary verification of Secure provisioning	DV	Open
DVP-VCU-181-B	VCU220-SYS3-B19-01	B19	EOL/FAT/production audit	Fault/degraded-case injection for Secure provisioning	DV	Open
DVP-VCU-182-A	VCU220-SYS3-B19-02	B19	EOL/FAT/production audit	Nominal/boundary verification of MCU test	DV	Open
DVP-VCU-182-B	VCU220-SYS3-B19-02	B19	EOL/FAT/production audit	Fault/degraded-case injection for MCU test	DV	Open
DVP-VCU-183-A	VCU220-SYS3-B19-03	B19	EOL/FAT/production audit	Nominal/boundary verification of SoC test	DV	Open
DVP-VCU-183-B	VCU220-SYS3-B19-03	B19	EOL/FAT/production audit	Fault/degraded-case injection for SoC test	DV	Open
DVP-VCU-184-A	VCU220-SYS3-B19-04	B19	EOL/FAT/production audit	Nominal/boundary verification of Power test	DV	Open
DVP-VCU-184-B	VCU220-SYS3-B19-04	B19	EOL/FAT/production audit	Fault/degraded-case injection for Power test	DV	Open
DVP-VCU-185-A	VCU220-SYS3-B19-05	B19	EOL/FAT/production audit	Nominal/boundary verification of Network identity	DV	Open
DVP-VCU-185-B	VCU220-SYS3-B19-05	B19	EOL/FAT/production audit	Fault/degraded-case injection for Network identity	DV	Open
DVP-VCU-186-A	VCU220-SYS3-B19-06	B19	EOL/FAT/production audit	Nominal/boundary verification of Calibration	DV	Open
DVP-VCU-186-B	VCU220-SYS3-B19-06	B19	EOL/FAT/production audit	Fault/degraded-case injection for Calibration	DV	Open
DVP-VCU-187-A	VCU220-SYS3-B19-07	B19	EOL/FAT/production audit	Nominal/boundary verification of Safety path	DV	Open
DVP-VCU-187-B	VCU220-SYS3-B19-07	B19	EOL/FAT/production audit	Fault/degraded-case injection for Safety path	DV	Open
DVP-VCU-188-A	VCU220-SYS3-B19-08	B19	EOL/FAT/production audit	Nominal/boundary verification of Configuration binding	DV	Open
DVP-VCU-188-B	VCU220-SYS3-B19-08	B19	EOL/FAT/production audit	Fault/degraded-case injection for Configuration binding	DV	Open
DVP-VCU-189-A	VCU220-SYS3-B19-09	B19	EOL/FAT/production audit	Nominal/boundary verification of Station quality	DV	Open
DVP-VCU-189-B	VCU220-SYS3-B19-09	B19	EOL/FAT/production audit	Fault/degraded-case injection for Station quality	DV	Open
DVP-VCU-190-A	VCU220-SYS3-B19-10	B19	EOL/FAT/production audit	Nominal/boundary verification of Genealogy	DV	Open
DVP-VCU-190-B	VCU220-SYS3-B19-10	B19	EOL/FAT/production audit	Fault/degraded-case injection for Genealogy	DV	Open
DVP-VCU-191-A	VCU220-SYS3-B20-01	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Mission life	DV	Open
DVP-VCU-191-B	VCU220-SYS3-B20-01	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Mission life	DV	Open
DVP-VCU-192-A	VCU220-SYS3-B20-02	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of PCN/EOL	DV	Open
DVP-VCU-192-B	VCU220-SYS3-B20-02	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for PCN/EOL	DV	Open
DVP-VCU-193-A	VCU220-SYS3-B20-03	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of FRACAS	DV	Open
DVP-VCU-193-B	VCU220-SYS3-B20-03	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for FRACAS	DV	Open
DVP-VCU-194-A	VCU220-SYS3-B20-04	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Telemetry	DV	Open
DVP-VCU-194-B	VCU220-SYS3-B20-04	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Telemetry	DV	Open
DVP-VCU-195-A	VCU220-SYS3-B20-05	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Storage endurance	DV	Open

DVP ID	SYS3 Parent	Block	Method	Scenario / Stimulus	Gate	Status
DVP-VCU-195-B	VCU220-SYS3-B20-05	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Storage endurance	DV	Open
DVP-VCU-196-A	VCU220-SYS3-B20-06	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Thermal aging	DV	Open
DVP-VCU-196-B	VCU220-SYS3-B20-06	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Thermal aging	DV	Open
DVP-VCU-197-A	VCU220-SYS3-B20-07	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Calibration governance	DV	Open
DVP-VCU-197-B	VCU220-SYS3-B20-07	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Calibration governance	DV	Open
DVP-VCU-198-A	VCU220-SYS3-B20-08	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Software compatibility	DV	Open
DVP-VCU-198-B	VCU220-SYS3-B20-08	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Software compatibility	DV	Open
DVP-VCU-199-A	VCU220-SYS3-B20-09	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Spares/tool support	DV	Open
DVP-VCU-199-B	VCU220-SYS3-B20-09	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Spares/tool support	DV	Open
DVP-VCU-200-A	VCU220-SYS3-B20-10	B20	Reliability/lifecycle/service audit	Nominal/boundary verification of Decommissioning	DV	Open
DVP-VCU-200-B	VCU220-SYS3-B20-10	B20	Reliability/lifecycle/service audit	Fault/degraded-case injection for Decommissioning	DV	Open

10. System Validation Cases

VAL ID	Area	Validation Objective	Method	Acceptance	Status
VAL-VCU-B01-01	B01	Validate LV Power Input, PMIC, Wake/Sleep & Power-State Control in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B01-02	B01	Validate single-fault/degraded/recovery behavior of LV Power Input, PMIC, Wake/Sleep & Power-State Control including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B02-01	B02	Validate Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B02-02	B02	Validate single-fault/degraded/recovery behavior of Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B03-01	B03	Validate Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B03-02	B03	Validate single-fault/degraded/recovery behavior of Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization including power-cycle and	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open

VAL ID	Area	Validation Objective	Method	Acceptance	Status
		communication faults.			
VAL-VCU-B04-01	B04	Validate Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B04-02	B04	Validate single-fault/degraded/recovery behavior of Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B05-01	B05	Validate Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B05-02	B05	Validate single-fault/degraded/recovery behavior of Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B06-01	B06	Validate Regenerative Braking, Deceleration Coordination & Brake-Energy Manager in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B06-02	B06	Validate single-fault/degraded/recovery behavior of Regenerative Braking, Deceleration Coordination & Brake-Energy Manager including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B07-01	B07	Validate Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B07-02	B07	Validate single-fault/degraded/recovery behavior of Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B08-01	B08	Validate Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B08-02	B08	Validate single-fault/degraded/recovery behavior of Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B09-01	B09	Validate HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open

VAL ID	Area	Validation Objective	Method	Acceptance	Status
VAL-VCU-B09-02	B09	Validate single-fault/degraded/recovery behavior of HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B10-01	B10	Validate Energy, Thermal, Performance Limit & Derating Supervisor in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B10-02	B10	Validate single-fault/degraded/recovery behavior of Energy, Thermal, Performance Limit & Derating Supervisor including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B11-01	B11	Validate CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B11-02	B11	Validate single-fault/degraded/recovery behavior of CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B12-01	B12	Validate Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B12-02	B12	Validate single-fault/degraded/recovery behavior of Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B13-01	B13	Validate Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B13-02	B13	Validate single-fault/degraded/recovery behavior of Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B14-01	B14	Validate Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B14-02	B14	Validate single-fault/degraded/recovery behavior of Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B15-01	B15	Validate Boot, OTA Software Update, Configuration,	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/networ	Open

VAL ID	Area	Validation Objective	Method	Acceptance	Status
		Variant & Calibration Management in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.		k acceptance criteria satisfied	
VAL-VCU-B15-02	B15	Validate single-fault/degraded/recovery behavior of Boot, OTA Software Update, Configuration, Variant & Calibration Management including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B16-01	B16	Validate Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B16-02	B16	Validate single-fault/degraded/recovery behavior of Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B17-01	B17	Validate Functional Safety Supervisor, Freedom-from-Interference & Health Management in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B17-02	B17	Validate single-fault/degraded/recovery behavior of Functional Safety Supervisor, Freedom-from-Interference & Health Management including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B18-01	B18	Validate Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B18-02	B18	Validate single-fault/degraded/recovery behavior of Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B19-01	B19	Validate Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B19-02	B19	Validate single-fault/degraded/recovery behavior of Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning including power-cycle and communication faults.	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority; diagnostic and recovery evidence complete	Open
VAL-VCU-B20-01	B20	Validate Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS in an integrated vehicle/vehicle-HIL environment across nominal and boundary states.	Vehicle / HIL / bench as applicable	All linked SYS.3/safety/latency/network acceptance criteria satisfied	Open
VAL-VCU-B20-02	B20	Validate single-fault/degraded/recove	Fault injection / vehicle / HIL	No unsafe increase in torque/HV/chassis authority;	Open

VAL ID	Area	Validation Objective	Method	Acceptance	Status
		ry behavior of Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS including power-cycle and communication faults.		diagnostic and recovery evidence complete	

11. Traceability Index

The full 200-row RTM with RA/SYS.1/SYS.2/SYS.3/SG/FSR/HSR/SWSR/ICD/DVP/VAL references is controlled in the master workbook. The compact index below supports document review without duplicating every wide traceability column.

SYS3	Block	SG	ICD	DVP	VAL	Status
VCU220-SYS3-B01-01	B01	SG-11	ICD-VCU-001	DVP-VCU-001-A;DVP-VCU-001-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-02	B01	SG-11	ICD-VCU-001	DVP-VCU-002-A;DVP-VCU-002-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-03	B01	SG-11	ICD-VCU-001	DVP-VCU-003-A;DVP-VCU-003-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-04	B01	SG-11	ICD-VCU-001	DVP-VCU-004-A;DVP-VCU-004-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-05	B01	SG-11	ICD-VCU-001	DVP-VCU-005-A;DVP-VCU-005-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-06	B01	SG-11	ICD-VCU-001	DVP-VCU-006-A;DVP-VCU-006-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-07	B01	SG-11	ICD-VCU-001	DVP-VCU-007-A;DVP-VCU-007-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-08	B01	SG-11	ICD-VCU-001	DVP-VCU-008-A;DVP-VCU-008-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-09	B01	SG-11	ICD-VCU-001	DVP-VCU-009-A;DVP-VCU-009-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B01-10	B01	SG-11	ICD-VCU-001	DVP-VCU-010-A;DVP-VCU-010-B	VAL-VCU-B01-01	Complete
VCU220-SYS3-B02-01	B02	SG-01	ICD-VCU-003	DVP-VCU-011-A;DVP-VCU-011-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-02	B02	SG-01	ICD-VCU-003	DVP-VCU-012-A;DVP-VCU-012-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-03	B02	SG-01	ICD-VCU-003	DVP-VCU-013-A;DVP-VCU-013-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-04	B02	SG-01	ICD-VCU-003	DVP-VCU-014-A;DVP-VCU-014-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-05	B02	SG-01	ICD-VCU-003	DVP-VCU-015-A;DVP-VCU-015-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-06	B02	SG-01	ICD-VCU-003	DVP-VCU-016-A;DVP-VCU-016-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-07	B02	SG-01	ICD-VCU-003	DVP-VCU-017-A;DVP-VCU-017-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-08	B02	SG-01	ICD-VCU-003	DVP-VCU-018-A;DVP-VCU-018-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-09	B02	SG-01	ICD-VCU-003	DVP-VCU-019-A;DVP-VCU-019-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B02-10	B02	SG-01	ICD-VCU-003	DVP-VCU-020-A;DVP-VCU-020-B	VAL-VCU-B02-01	Complete
VCU220-SYS3-B03-01	B03	SG-06	ICD-VCU-005	DVP-VCU-021-A;DVP-VCU-021-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-02	B03	SG-06	ICD-VCU-005	DVP-VCU-022-A;DVP-VCU-022-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-03	B03	SG-06	ICD-VCU-005	DVP-VCU-023-A;DVP-VCU-023-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-04	B03	SG-06	ICD-VCU-005	DVP-VCU-024-A;DVP-VCU-024-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-05	B03	SG-06	ICD-VCU-005	DVP-VCU-025-A;DVP-VCU-025-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-06	B03	SG-06	ICD-VCU-005	DVP-VCU-026-A;DVP-VCU-026-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-07	B03	SG-06	ICD-VCU-005	DVP-VCU-027-A;DVP-VCU-027-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-08	B03	SG-06	ICD-VCU-005	DVP-VCU-028-A;DVP-VCU-028-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B03-09	B03	SG-06	ICD-VCU-005	DVP-VCU-029-A;DVP-	VAL-VCU-B03-01	Complete

SYS3	Block	SG	ICD	DVP	VAL	Status
				VCU-029-B		
VCU220-SYS3-B03-10	B03	SG-06	ICD-VCU-005	DVP-VCU-030-A;DVP-VCU-030-B	VAL-VCU-B03-01	Complete
VCU220-SYS3-B04-01	B04	SG-01	ICD-VCU-007	DVP-VCU-031-A;DVP-VCU-031-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-02	B04	SG-01	ICD-VCU-007	DVP-VCU-032-A;DVP-VCU-032-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-03	B04	SG-01	ICD-VCU-007	DVP-VCU-033-A;DVP-VCU-033-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-04	B04	SG-01	ICD-VCU-007	DVP-VCU-034-A;DVP-VCU-034-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-05	B04	SG-01	ICD-VCU-007	DVP-VCU-035-A;DVP-VCU-035-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-06	B04	SG-01	ICD-VCU-007	DVP-VCU-036-A;DVP-VCU-036-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-07	B04	SG-01	ICD-VCU-007	DVP-VCU-037-A;DVP-VCU-037-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-08	B04	SG-01	ICD-VCU-007	DVP-VCU-038-A;DVP-VCU-038-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-09	B04	SG-01	ICD-VCU-007	DVP-VCU-039-A;DVP-VCU-039-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B04-10	B04	SG-01	ICD-VCU-007	DVP-VCU-040-A;DVP-VCU-040-B	VAL-VCU-B04-01	Complete
VCU220-SYS3-B05-01	B05	SG-01	ICD-VCU-009	DVP-VCU-041-A;DVP-VCU-041-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-02	B05	SG-01	ICD-VCU-009	DVP-VCU-042-A;DVP-VCU-042-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-03	B05	SG-01	ICD-VCU-009	DVP-VCU-043-A;DVP-VCU-043-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-04	B05	SG-01	ICD-VCU-009	DVP-VCU-044-A;DVP-VCU-044-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-05	B05	SG-01	ICD-VCU-009	DVP-VCU-045-A;DVP-VCU-045-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-06	B05	SG-01	ICD-VCU-009	DVP-VCU-046-A;DVP-VCU-046-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-07	B05	SG-01	ICD-VCU-009	DVP-VCU-047-A;DVP-VCU-047-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-08	B05	SG-01	ICD-VCU-009	DVP-VCU-048-A;DVP-VCU-048-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-09	B05	SG-01	ICD-VCU-009	DVP-VCU-049-A;DVP-VCU-049-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B05-10	B05	SG-01	ICD-VCU-009	DVP-VCU-050-A;DVP-VCU-050-B	VAL-VCU-B05-01	Complete
VCU220-SYS3-B06-01	B06	SG-03	ICD-VCU-011	DVP-VCU-051-A;DVP-VCU-051-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-02	B06	SG-03	ICD-VCU-011	DVP-VCU-052-A;DVP-VCU-052-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-03	B06	SG-03	ICD-VCU-011	DVP-VCU-053-A;DVP-VCU-053-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-04	B06	SG-03	ICD-VCU-011	DVP-VCU-054-A;DVP-VCU-054-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-05	B06	SG-03	ICD-VCU-011	DVP-VCU-055-A;DVP-VCU-055-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-06	B06	SG-03	ICD-VCU-011	DVP-VCU-056-A;DVP-VCU-056-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-07	B06	SG-03	ICD-VCU-011	DVP-VCU-057-A;DVP-VCU-057-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-08	B06	SG-03	ICD-VCU-011	DVP-VCU-058-A;DVP-VCU-058-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-09	B06	SG-03	ICD-VCU-011	DVP-VCU-059-A;DVP-VCU-059-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B06-10	B06	SG-03	ICD-VCU-011	DVP-VCU-060-A;DVP-VCU-060-B	VAL-VCU-B06-01	Complete
VCU220-SYS3-B07-01	B07	SG-05	ICD-VCU-013	DVP-VCU-061-A;DVP-VCU-061-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-02	B07	SG-05	ICD-VCU-013	DVP-VCU-062-A;DVP-VCU-062-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-03	B07	SG-05	ICD-VCU-013	DVP-VCU-063-A;DVP-VCU-063-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-04	B07	SG-05	ICD-VCU-013	DVP-VCU-064-A;DVP-	VAL-VCU-B07-01	Complete

SYS3	Block	SG	ICD	DVP	VAL	Status
				VCU-064-B		
VCU220-SYS3-B07-05	B07	SG-05	ICD-VCU-013	DVP-VCU-065-A;DVP-VCU-065-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-06	B07	SG-05	ICD-VCU-013	DVP-VCU-066-A;DVP-VCU-066-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-07	B07	SG-05	ICD-VCU-013	DVP-VCU-067-A;DVP-VCU-067-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-08	B07	SG-05	ICD-VCU-013	DVP-VCU-068-A;DVP-VCU-068-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-09	B07	SG-05	ICD-VCU-013	DVP-VCU-069-A;DVP-VCU-069-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B07-10	B07	SG-05	ICD-VCU-013	DVP-VCU-070-A;DVP-VCU-070-B	VAL-VCU-B07-01	Complete
VCU220-SYS3-B08-01	B08	SG-03	ICD-VCU-015	DVP-VCU-071-A;DVP-VCU-071-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-02	B08	SG-03	ICD-VCU-015	DVP-VCU-072-A;DVP-VCU-072-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-03	B08	SG-03	ICD-VCU-015	DVP-VCU-073-A;DVP-VCU-073-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-04	B08	SG-03	ICD-VCU-015	DVP-VCU-074-A;DVP-VCU-074-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-05	B08	SG-03	ICD-VCU-015	DVP-VCU-075-A;DVP-VCU-075-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-06	B08	SG-03	ICD-VCU-015	DVP-VCU-076-A;DVP-VCU-076-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-07	B08	SG-03	ICD-VCU-015	DVP-VCU-077-A;DVP-VCU-077-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-08	B08	SG-03	ICD-VCU-015	DVP-VCU-078-A;DVP-VCU-078-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-09	B08	SG-03	ICD-VCU-015	DVP-VCU-079-A;DVP-VCU-079-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B08-10	B08	SG-03	ICD-VCU-015	DVP-VCU-080-A;DVP-VCU-080-B	VAL-VCU-B08-01	Complete
VCU220-SYS3-B09-01	B09	SG-04	ICD-VCU-017	DVP-VCU-081-A;DVP-VCU-081-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-02	B09	SG-04	ICD-VCU-017	DVP-VCU-082-A;DVP-VCU-082-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-03	B09	SG-04	ICD-VCU-017	DVP-VCU-083-A;DVP-VCU-083-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-04	B09	SG-04	ICD-VCU-017	DVP-VCU-084-A;DVP-VCU-084-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-05	B09	SG-04	ICD-VCU-017	DVP-VCU-085-A;DVP-VCU-085-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-06	B09	SG-04	ICD-VCU-017	DVP-VCU-086-A;DVP-VCU-086-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-07	B09	SG-04	ICD-VCU-017	DVP-VCU-087-A;DVP-VCU-087-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-08	B09	SG-04	ICD-VCU-017	DVP-VCU-088-A;DVP-VCU-088-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-09	B09	SG-04	ICD-VCU-017	DVP-VCU-089-A;DVP-VCU-089-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B09-10	B09	SG-04	ICD-VCU-017	DVP-VCU-090-A;DVP-VCU-090-B	VAL-VCU-B09-01	Complete
VCU220-SYS3-B10-01	B10	SG-08	ICD-VCU-019	DVP-VCU-091-A;DVP-VCU-091-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-02	B10	SG-08	ICD-VCU-019	DVP-VCU-092-A;DVP-VCU-092-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-03	B10	SG-08	ICD-VCU-019	DVP-VCU-093-A;DVP-VCU-093-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-04	B10	SG-08	ICD-VCU-019	DVP-VCU-094-A;DVP-VCU-094-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-05	B10	SG-08	ICD-VCU-019	DVP-VCU-095-A;DVP-VCU-095-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-06	B10	SG-08	ICD-VCU-019	DVP-VCU-096-A;DVP-VCU-096-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-07	B10	SG-08	ICD-VCU-019	DVP-VCU-097-A;DVP-VCU-097-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-08	B10	SG-08	ICD-VCU-019	DVP-VCU-098-A;DVP-VCU-098-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B10-09	B10	SG-08	ICD-VCU-019	DVP-VCU-099-A;DVP-	VAL-VCU-B10-01	Complete

SYS3	Block	SG	ICD	DVP	VAL	Status
				VCU-099-B		
VCU220-SYS3-B10-10	B10	SG-08	ICD-VCU-019	DVP-VCU-100-A;DVP-VCU-100-B	VAL-VCU-B10-01	Complete
VCU220-SYS3-B11-01	B11	SG-09	ICD-VCU-021	DVP-VCU-101-A;DVP-VCU-101-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-02	B11	SG-09	ICD-VCU-021	DVP-VCU-102-A;DVP-VCU-102-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-03	B11	SG-09	ICD-VCU-021	DVP-VCU-103-A;DVP-VCU-103-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-04	B11	SG-09	ICD-VCU-021	DVP-VCU-104-A;DVP-VCU-104-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-05	B11	SG-09	ICD-VCU-021	DVP-VCU-105-A;DVP-VCU-105-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-06	B11	SG-09	ICD-VCU-021	DVP-VCU-106-A;DVP-VCU-106-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-07	B11	SG-09	ICD-VCU-021	DVP-VCU-107-A;DVP-VCU-107-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-08	B11	SG-09	ICD-VCU-021	DVP-VCU-108-A;DVP-VCU-108-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-09	B11	SG-09	ICD-VCU-021	DVP-VCU-109-A;DVP-VCU-109-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B11-10	B11	SG-09	ICD-VCU-021	DVP-VCU-110-A;DVP-VCU-110-B	VAL-VCU-B11-01	Complete
VCU220-SYS3-B12-01	B12	SG-09	ICD-VCU-023	DVP-VCU-111-A;DVP-VCU-111-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-02	B12	SG-09	ICD-VCU-023	DVP-VCU-112-A;DVP-VCU-112-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-03	B12	SG-09	ICD-VCU-023	DVP-VCU-113-A;DVP-VCU-113-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-04	B12	SG-09	ICD-VCU-023	DVP-VCU-114-A;DVP-VCU-114-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-05	B12	SG-09	ICD-VCU-023	DVP-VCU-115-A;DVP-VCU-115-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-06	B12	SG-09	ICD-VCU-023	DVP-VCU-116-A;DVP-VCU-116-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-07	B12	SG-09	ICD-VCU-023	DVP-VCU-117-A;DVP-VCU-117-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-08	B12	SG-09	ICD-VCU-023	DVP-VCU-118-A;DVP-VCU-118-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-09	B12	SG-09	ICD-VCU-023	DVP-VCU-119-A;DVP-VCU-119-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B12-10	B12	SG-09	ICD-VCU-023	DVP-VCU-120-A;DVP-VCU-120-B	VAL-VCU-B12-01	Complete
VCU220-SYS3-B13-01	B13	SG-07	ICD-VCU-025	DVP-VCU-121-A;DVP-VCU-121-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-02	B13	SG-07	ICD-VCU-025	DVP-VCU-122-A;DVP-VCU-122-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-03	B13	SG-07	ICD-VCU-025	DVP-VCU-123-A;DVP-VCU-123-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-04	B13	SG-07	ICD-VCU-025	DVP-VCU-124-A;DVP-VCU-124-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-05	B13	SG-07	ICD-VCU-025	DVP-VCU-125-A;DVP-VCU-125-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-06	B13	SG-07	ICD-VCU-025	DVP-VCU-126-A;DVP-VCU-126-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-07	B13	SG-07	ICD-VCU-025	DVP-VCU-127-A;DVP-VCU-127-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-08	B13	SG-07	ICD-VCU-025	DVP-VCU-128-A;DVP-VCU-128-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-09	B13	SG-07	ICD-VCU-025	DVP-VCU-129-A;DVP-VCU-129-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B13-10	B13	SG-07	ICD-VCU-025	DVP-VCU-130-A;DVP-VCU-130-B	VAL-VCU-B13-01	Complete
VCU220-SYS3-B14-01	B14	SG-12	ICD-VCU-027	DVP-VCU-131-A;DVP-VCU-131-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-02	B14	SG-12	ICD-VCU-027	DVP-VCU-132-A;DVP-VCU-132-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-03	B14	SG-12	ICD-VCU-027	DVP-VCU-133-A;DVP-VCU-133-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-04	B14	SG-12	ICD-VCU-027	DVP-VCU-134-A;DVP-	VAL-VCU-B14-01	Complete

SYS3	Block	SG	ICD	DVP	VAL	Status
				VCU-134-B		
VCU220-SYS3-B14-05	B14	SG-12	ICD-VCU-027	DVP-VCU-135-A;DVP-VCU-135-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-06	B14	SG-12	ICD-VCU-027	DVP-VCU-136-A;DVP-VCU-136-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-07	B14	SG-12	ICD-VCU-027	DVP-VCU-137-A;DVP-VCU-137-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-08	B14	SG-12	ICD-VCU-027	DVP-VCU-138-A;DVP-VCU-138-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-09	B14	SG-12	ICD-VCU-027	DVP-VCU-139-A;DVP-VCU-139-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B14-10	B14	SG-12	ICD-VCU-027	DVP-VCU-140-A;DVP-VCU-140-B	VAL-VCU-B14-01	Complete
VCU220-SYS3-B15-01	B15	SG-10	ICD-VCU-029	DVP-VCU-141-A;DVP-VCU-141-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-02	B15	SG-10	ICD-VCU-029	DVP-VCU-142-A;DVP-VCU-142-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-03	B15	SG-10	ICD-VCU-029	DVP-VCU-143-A;DVP-VCU-143-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-04	B15	SG-10	ICD-VCU-029	DVP-VCU-144-A;DVP-VCU-144-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-05	B15	SG-10	ICD-VCU-029	DVP-VCU-145-A;DVP-VCU-145-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-06	B15	SG-10	ICD-VCU-029	DVP-VCU-146-A;DVP-VCU-146-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-07	B15	SG-10	ICD-VCU-029	DVP-VCU-147-A;DVP-VCU-147-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-08	B15	SG-10	ICD-VCU-029	DVP-VCU-148-A;DVP-VCU-148-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-09	B15	SG-10	ICD-VCU-029	DVP-VCU-149-A;DVP-VCU-149-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B15-10	B15	SG-10	ICD-VCU-029	DVP-VCU-150-A;DVP-VCU-150-B	VAL-VCU-B15-01	Complete
VCU220-SYS3-B16-01	B16	SG-10	ICD-VCU-001	DVP-VCU-151-A;DVP-VCU-151-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-02	B16	SG-10	ICD-VCU-001	DVP-VCU-152-A;DVP-VCU-152-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-03	B16	SG-10	ICD-VCU-001	DVP-VCU-153-A;DVP-VCU-153-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-04	B16	SG-10	ICD-VCU-001	DVP-VCU-154-A;DVP-VCU-154-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-05	B16	SG-10	ICD-VCU-001	DVP-VCU-155-A;DVP-VCU-155-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-06	B16	SG-10	ICD-VCU-001	DVP-VCU-156-A;DVP-VCU-156-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-07	B16	SG-10	ICD-VCU-001	DVP-VCU-157-A;DVP-VCU-157-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-08	B16	SG-10	ICD-VCU-001	DVP-VCU-158-A;DVP-VCU-158-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-09	B16	SG-10	ICD-VCU-001	DVP-VCU-159-A;DVP-VCU-159-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B16-10	B16	SG-10	ICD-VCU-001	DVP-VCU-160-A;DVP-VCU-160-B	VAL-VCU-B16-01	Complete
VCU220-SYS3-B17-01	B17	SG-01	ICD-VCU-003	DVP-VCU-161-A;DVP-VCU-161-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-02	B17	SG-01	ICD-VCU-003	DVP-VCU-162-A;DVP-VCU-162-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-03	B17	SG-01	ICD-VCU-003	DVP-VCU-163-A;DVP-VCU-163-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-04	B17	SG-01	ICD-VCU-003	DVP-VCU-164-A;DVP-VCU-164-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-05	B17	SG-01	ICD-VCU-003	DVP-VCU-165-A;DVP-VCU-165-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-06	B17	SG-01	ICD-VCU-003	DVP-VCU-166-A;DVP-VCU-166-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-07	B17	SG-01	ICD-VCU-003	DVP-VCU-167-A;DVP-VCU-167-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-08	B17	SG-01	ICD-VCU-003	DVP-VCU-168-A;DVP-VCU-168-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B17-09	B17	SG-01	ICD-VCU-003	DVP-VCU-169-A;DVP-	VAL-VCU-B17-01	Complete

SYS3	Block	SG	ICD	DVP	VAL	Status
				VCU-169-B		
VCU220-SYS3-B17-10	B17	SG-01	ICD-VCU-003	DVP-VCU-170-A;DVP-VCU-170-B	VAL-VCU-B17-01	Complete
VCU220-SYS3-B18-01	B18	SG-11	ICD-VCU-005	DVP-VCU-171-A;DVP-VCU-171-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-02	B18	SG-11	ICD-VCU-005	DVP-VCU-172-A;DVP-VCU-172-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-03	B18	SG-11	ICD-VCU-005	DVP-VCU-173-A;DVP-VCU-173-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-04	B18	SG-11	ICD-VCU-005	DVP-VCU-174-A;DVP-VCU-174-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-05	B18	SG-11	ICD-VCU-005	DVP-VCU-175-A;DVP-VCU-175-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-06	B18	SG-11	ICD-VCU-005	DVP-VCU-176-A;DVP-VCU-176-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-07	B18	SG-11	ICD-VCU-005	DVP-VCU-177-A;DVP-VCU-177-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-08	B18	SG-11	ICD-VCU-005	DVP-VCU-178-A;DVP-VCU-178-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-09	B18	SG-11	ICD-VCU-005	DVP-VCU-179-A;DVP-VCU-179-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B18-10	B18	SG-11	ICD-VCU-005	DVP-VCU-180-A;DVP-VCU-180-B	VAL-VCU-B18-01	Complete
VCU220-SYS3-B19-01	B19	SG-12	ICD-VCU-007	DVP-VCU-181-A;DVP-VCU-181-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-02	B19	SG-12	ICD-VCU-007	DVP-VCU-182-A;DVP-VCU-182-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-03	B19	SG-12	ICD-VCU-007	DVP-VCU-183-A;DVP-VCU-183-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-04	B19	SG-12	ICD-VCU-007	DVP-VCU-184-A;DVP-VCU-184-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-05	B19	SG-12	ICD-VCU-007	DVP-VCU-185-A;DVP-VCU-185-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-06	B19	SG-12	ICD-VCU-007	DVP-VCU-186-A;DVP-VCU-186-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-07	B19	SG-12	ICD-VCU-007	DVP-VCU-187-A;DVP-VCU-187-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-08	B19	SG-12	ICD-VCU-007	DVP-VCU-188-A;DVP-VCU-188-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-09	B19	SG-12	ICD-VCU-007	DVP-VCU-189-A;DVP-VCU-189-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B19-10	B19	SG-12	ICD-VCU-007	DVP-VCU-190-A;DVP-VCU-190-B	VAL-VCU-B19-01	Complete
VCU220-SYS3-B20-01	B20	SG-12	ICD-VCU-009	DVP-VCU-191-A;DVP-VCU-191-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-02	B20	SG-12	ICD-VCU-009	DVP-VCU-192-A;DVP-VCU-192-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-03	B20	SG-12	ICD-VCU-009	DVP-VCU-193-A;DVP-VCU-193-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-04	B20	SG-12	ICD-VCU-009	DVP-VCU-194-A;DVP-VCU-194-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-05	B20	SG-12	ICD-VCU-009	DVP-VCU-195-A;DVP-VCU-195-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-06	B20	SG-12	ICD-VCU-009	DVP-VCU-196-A;DVP-VCU-196-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-07	B20	SG-12	ICD-VCU-009	DVP-VCU-197-A;DVP-VCU-197-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-08	B20	SG-12	ICD-VCU-009	DVP-VCU-198-A;DVP-VCU-198-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-09	B20	SG-12	ICD-VCU-009	DVP-VCU-199-A;DVP-VCU-199-B	VAL-VCU-B20-01	Complete
VCU220-SYS3-B20-10	B20	SG-12	ICD-VCU-009	DVP-VCU-200-A;DVP-VCU-200-B	VAL-VCU-B20-01	Complete

12. Manufacturing / EOL / Service Evidence Handoff

- Verify ECU current consumption, rail sequencing, PMIC watchdog and brownout safe-state using calibrated production fixtures.

- Bind central ECU serial, safety MCU/HPA firmware, HSM identity, vehicle VIN/variant, network configuration and safety calibration before READY release.
- Run I/O, CAN FD, Ethernet, memory/storage, thermal-sensor, wake/sleep and independent safety-chain proof tests.
- Protect production/service routines behind authenticated roles and vehicle-state interlocks; plant mode must not bypass torque/HV safety.
- Retain station ID, software/calibration hashes, key/certificate provisioning status, measurement traceability and pass/fail genealogy.

13. Reference Register

Ref ID	Category	Document / Product	Source	Link	Use
REF-VCU-001	Process	Automotive SPICE 4.0 PRM/PAM	VDA QMC	Official source	SYS.1/SYS.2/SYS.3 process and assessment baseline
REF-VCU-002	Functional Safety	ISO 26262:2018 Road vehicles - Functional safety	ISO	Official source	Item/HARA/FSC/TSC/ system/HW/SW safety lifecycle
REF-VCU-003	Cybersecurity	ISO/SAE 21434:2021 Road vehicles - Cybersecurity engineering	ISO	Official source	TARA and cybersecurity lifecycle
REF-VCU-004	Software Update	ISO 24089:2023 + Amd 1:2024	ISO	Official source	Software update engineering
REF-VCU-005	Regulation	UN Regulation No.155 - Cyber security / CSMS	UNECE	Official source	Market cybersecurity type-approval comparator
REF-VCU-006	Regulation	UN Regulation No.156 - Software update / SUMS	UNECE	Official source	Market software-update type-approval comparator
REF-VCU-007	Diagnostics	ISO 14229-1:2020 UDS (Jan-2026 baseline)	ISO	Official source	UDS diagnostic service architecture
REF-VCU-008	Diagnostics	ISO 13400-2:2025 DoIP (Jan-2026 baseline)	ISO	Official source	IP diagnostic transport
REF-VCU-009	CAN	ISO 11898-1:2024 / ISO 11898-2:2024 Jan-2026 baseline	ISO	Official source	CAN/CAN FD data-link and high-speed physical layer
REF-VCU-010	Electrical	ISO 16750-2:2023 Electrical loads	ISO	Official source	LV electrical environmental loads
REF-VCU-011	Electrical	ISO 7637-2:2011 Conducted transients	ISO	Official source	12/24 V transient bench test baseline
REF-VCU-012	EMC	CISPR 25:2021	IEC	Official source	Component conducted/radiated emissions
REF-VCU-013	EMC	ISO 11452-2:2019 / ISO 11452-4:2020	ISO	Official source	Component radiated/BCI immunity
REF-VCU-014	AUTOSAR	AUTOSAR E2E Protocol Specification	AUTOSAR	Official source	Communication safety integrity profiles
REF-VCU-015	AUTOSAR	AUTOSAR Secure Onboard Communication (SecOC)	AUTOSAR	Official source	PDU authenticity and freshness
REF-VCU-016	Safety MCU	AURIX TC4x family	Infineon	Official source	Safety MCU / eMobility / vehicle motion control reference
REF-VCU-017	Safety MCU PMIC	OPTIREG TLF35585	Infineon	Official source	Safety PMIC/watchdog/rail/safe-state reference
REF-VCU-018	Central real-time compute	S32N55 / S32N family	NXP	Official source	Central real-time vehicle control / hardware isolation comparator
REF-VCU-019	HPA SoC	R-Car X5H / Gen5 central compute	Renesas	Official source	Multi-domain HPA / IVI / virtualization comparator
REF-VCU-020	Cockpit SoC	Snapdragon Cockpit Elite	Qualcomm	Official source	Premium virtualized cockpit/central compute comparator
REF-VCU-021	Hypervisor	QNX Hypervisor for Safety	QNX	Official source	Safety-qualified virtualization comparator
REF-VCU-022	Ethernet Switch	Brightlane Q622x central automotive Ethernet switch	Marvell	Official source	60/90-Gbps TSN/high-speed central switching comparator
REF-VCU-023	Ethernet PHY	Brightlane 88Q2110/88Q2112	Infineon/Marvell technology line	Official source	100/1000BASE-T1 PHY / TC10 / 802.1AS comparator

Ref ID	Category	Document / Product	Source	Link	Use
REF-VCU-024	CAN PHY	TCAN1043A/HG-Q1 CAN FD transceiver	Texas Instruments	Official source	CAN FD / wake / bus fault comparator
REF-VCU-025	Cockpit	Snapdragon Cockpit Platform	Qualcomm	Official source	Cockpit virtualized processing / UX ecosystem
REF-VCU-026	Central Compute	R-Car X5H architecture overview	Renesas	Official source	Centralized multi-domain SoC capabilities
REF-VCU-027	Software	AUTOSAR Standards	AUTOSAR	Official source	Classic/Adaptive architecture baseline

Appendix A - Block Ownership Map

Block	Controlled Architecture Scope
B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control
B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway
B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization
B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs
B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager
B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager
B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control
B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager
B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks
B10	Energy, Thermal, Performance Limit & Derating Supervisor
B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management
B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness
B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary
B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging
B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management
B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services
B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management
B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design
B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning
B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS

Appendix B - Evidence Chain

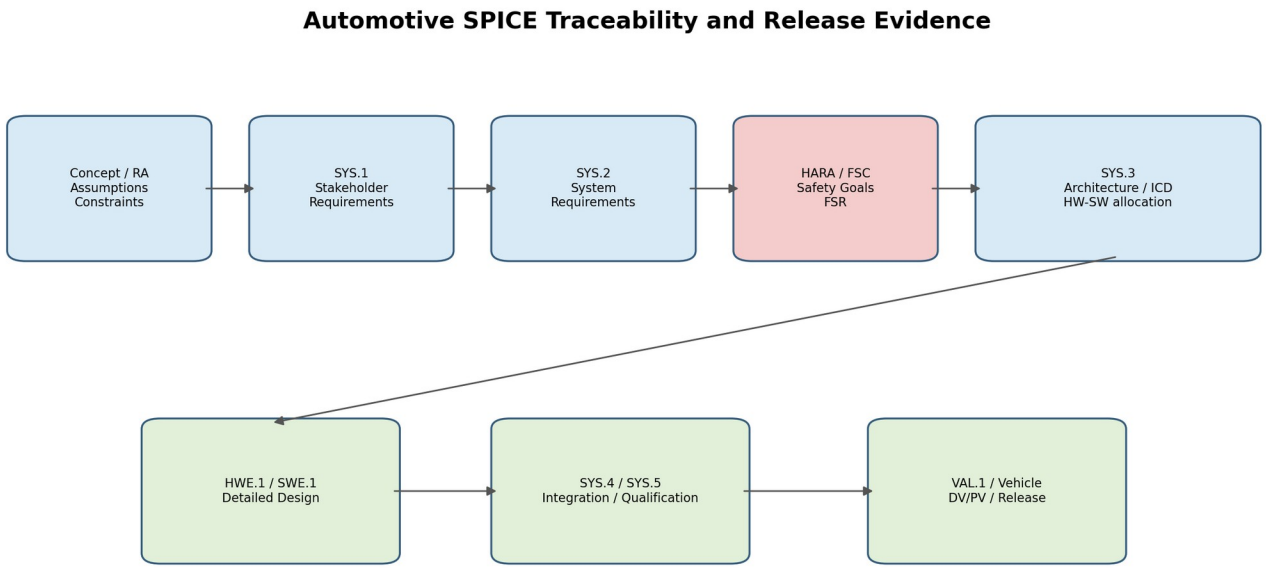


Figure 7 - SYS.3 to detailed-design and validation handoff

14. Rev A2 Detailed Architecture Hardening

This section is the production-intent addendum to the B01-B20 architecture. It closes the audit findings without changing the central design principle: authoritative motion/HV control remains on the independent safety domain, while the HPA/cockpit domain is consolidated for user experience and non-authoritative services.

14.1 B01/B18 - 130 W central-compute power, cold crank and thermal hierarchy

Load Domain	Nominal W	Peak W	Design Rule	Qualification
Safety MCU + PMIC + safety CAN	12	20	Independent rail; torque/HV supervision survives HPA reset	ASIL-D safety path reference
HPA SoC CPU/GPU/NPU	35	75	Depends on nominated silicon and display/AI load	Throttle/shedding first
LPDDR5X memory	8	14	Multi-channel memory + ECC where supported	Budget bandwidth and thermal separately
UFS/NVMe storage	2	5	Update/log burst	No safety authority
Ethernet switch/PHYs	5	10	1G baseline; Multi-G future branch	Safety traffic isolated
Display/SerDes interface silicon	3	8	Pixel links excluded from vehicle Ethernet backbone	External panels excluded
Misc I/O / sensors / housekeeping	5	8	ADC/discretes/wake/secure elements	Includes engineering margin
Central ECU total	70	130	Sustained design target; 150 W short input allocation	External backlights/audio amplifier excluded

Rev A2 uses 70 W nominal / 130 W sustained central ECU and a 150 W short electrical input envelope. The safety MCU rail is power/reset independent from the HPA. Cold crank is supported by a wide-input/buck-boost front end; only the safety domain receives explicitly credited hold-up energy.

Hold-Up Design Input	Rev A2 Value
Safety-domain load	20 W reference
Hold-up interval	100 ms reference
Storage voltage	12 V initial to 6 V minimum usable
Converter efficiency	90%
Ideal capacitance	~41.2 mF
Design-start with 1.5 margin	~61.7 mF
Whole 130 W HPA hold-up	Not credited; controlled reset is allowed

Thermal State	Trigger / Principle	HPA Action	Vehicle / Safety Action
T0 Normal	< supplier-calibrated threshold	Full released workloads	All modes subject to SOP
T1 Compute Shed	Approaching thermal limit	Throttle Android background / GPU / NPU / noncritical animations	No motion degradation
T2 Cockpit Degrade	Higher temperature persistence	Reduce display complexity, recorder/OTA/background services	Sport/Track may be inhibited
T3 HPA Controlled Shutdown	Near HPA protective limit	Shutdown/restart HPA if required	Safety MCU continues torque/HV supervision
T4 Safety-Domain Limit	Safety MCU/PMIC own thermal threshold	Controlled torque/HV safe response per safety concept	Vehicle function degraded/stopped

14.2 B02/B05/B17 - ASIL-D torque safety chain, decomposition and safe-state selection

Safety Function	Target	Architecture	Boundary
SG-01 Unintended positive/excess torque	ASIL D target	Safety MCU authoritative torque arbiter + independent torque monitor + hard inhibit; inverter local torque/SC protection remains independent	No HPA direct actuator authority
Nominal torque function	ASIL D or decomposed implementation	Primary command and independent monitor can use ISO26262-9 decomposition only after dependent-failure analysis	Decomposition is not automatic
Pedal acquisition	ASIL D path	Dual independent pedal channels, range/rationality/rate plausibility, brake override	Fault cannot increase torque
Delivered torque plausibility	ASIL D path	Compare command vs inverter actual torque/current/speed and acceleration plausibility	Mismatch -> inhibit/controlled degradation
HV READY authorization	ASIL C/D by item	VCU orchestrates; BMS owns pack contactors/precharge; inverter enable depends on valid HV state	No VCU bypass of BMS protection
Safety reaction	<=10 ms internal capability	DTI + qualification + inhibit-request	Not a mechanical torque decay guarantee

Safety Function	Target	Architecture	Boundary
		budget; vehicle FTTI remains HARA/TSC-derived	

Fault / Condition	VCU Response	Timing	Independent Layer	Recovery Rule
Unintended positive torque / command corruption	Immediate positive-torque inhibit; brake/ESC retain authority	<=10 ms internal inhibit request	Inverter hard limiter + local protection	No limp torque until fault cleared
Pedal plausibility failure	Zero driver-demand torque; controlled propulsion fallback if vehicle-level safety concept allows	<=10 ms qualification/inhibit	Brake override active	Driver warning
Loss of one CAN safety bus	Continue on healthy redundant path if integrity valid	Within message timeout	No authority increase	Service DTC
Loss of both torque communication paths	Controlled torque removal / local inverter safe state	TSC-defined	Local inverter watchdog	Context-dependent coast/stop
HPA/Android crash	Safety MCU continues torque/HV supervision; cockpit degraded	No motion interruption required	Independent safety power/reset	Restart HPA guest
Thermal overload HPA	Shed GPU/NPU/background, then controlled HPA shutdown	Seconds-level thermal state	Safety MCU unaffected	Preserve legal/safety warning path
BMS SOP invalid	Reduce torque/regen to conservative limits or zero as allocated	<=SOP timeout	No increase in torque/charge	Driver warning
ESC/ABS active	Revoke/reduce regen and propulsion as requested by ESC	<=10 ms control loop	Brake ECU final authority	No driver-mode override
Crash input valid	Torque zero + HV de-energization request per vehicle safety concept	Vehicle crash FTTI	BMS/inverter local crash paths remain	No software re-enable

The internal <=10 ms value is an electronic detection/qualification/inhibit-request capability. It does not assert that the mechanical vehicle torque has decayed within 10 ms. Final FTTI and vehicle-level safe response remain FSC/TSC results.

14.3 B03/B12 - Hypervisor, memory, DMA, IPC and freedom-from-interference

Resource	Rev A2 Budget	Design Rule	Verification
Safety MCU CPU	<=70% worst-case utilization	Leave >=30% WCET/interrupt margin at max CAN/fault load	WCET + stress HIL
Safety MCU RAM/Flash	<=75% allocated	Keep diagnostic/update margin; no runtime heap in critical loops	Link map + runtime monitor
HPA CPU/GPU/NPU	<=75% sustained released workload	Reserve for UI spikes/update/background isolation	Target workload stress
HPA memory bandwidth	<=65% sustained released workload	Reserve >=35% to avoid DDR contention under multi-display/network bursts	SoC performance counters
HPA RAM	<=80% normal high-watermark	Bound each VM/cgroup; safety-support services preallocate bounded pools	Long-duration stress
DMA / IOMMU	Explicit allow-list	QM guest DMA cannot reach safety memory/MMIO	IOMMU fault injection
Cache / interconnect QoS	Priority/way/bandwidth partitioning where silicon supports	QM workload cannot starve safety-support/control IPC	Contention benchmark
Interrupt ownership	Static routing	Safety interrupts cannot be masked by Android guest	Fault injection
Power / reset	Independent roots	HPA reset/brownout does not reset safety MCU/PMIC/CAN	Brownout/reset test
Clock	Independent safety clock supervision	HPA PLL/clock fault cannot remove safety monitor timing	Clock fault injection

The HPA hypervisor guest environment is treated as QM unless a safety-qualified hypervisor/OS combination is deliberately nominated and used according to its safety manual. Rev A2 does not take such credit for motion authority.

State	Function	Reference Timing	Safety Rule
OFF	Safety MCU off; always-on wake monitor active	Sleep current OEM target	No torque/HV authority
WAKE	Safety PMIC/MCU powers first; self-test/watchdog/HSM init	<=50 ms start reference	HPA still off
SAFETY_READY	CAN A/B, pedal/discrete, torque monitor and HV-state supervisor available	<=150 ms reference from valid wake	Android/cockpit boot not required
HPA_BOOT	SoC/hypervisor/guests start under independent power/reset supervision	<=2 s service target, platform-specific	Safety functions continue
DRIVE_READY	Valid BMS/ESC/inverter/HV state + no critical DTC + safety checks complete	Vehicle-state dependent	Mode manager enabled
HPA_RESTART	Local guest/SoC reset after failure	Safety domain remains active	Torque/HV authority retained by safety MCU
SHUTDOWN	Torque zero/HV state safe; logs flush;	Controlled	Safety MCU last-off

State	Function	Reference Timing	Safety Rule
	HPA off before safety domain if possible		

14.4 B07 - Driver-mode eligibility and deterministic blending

Mode	Vehicle State	Eligibility	Power Cap	Regen	Jerk Target	Release Boundary
Eco	Any drive-ready	SOP valid; no safety inhibit	110 kW reference or lower	High regen	<=1.2 m/s3 design target	User selectable
Normal	Any drive-ready	SOP valid	Up to 220 kW	Medium	<=2.0 m/s3 target	Default after fault/update unless OEM policy differs
Sport	Forward / suitable speed	BMS/inverter thermal/SOP margin; ESC ready	Up to 220 kW	Low-medium	<=3.5 m/s3 target	Reject on thermal/traction degradation
Track	Explicit eligible state	Market/variant permit, explicit driver confirmation, ESC/local ECU support, tyre/chassis state, no critical DTC	Supplier-approved boost only	Configurable	<=5.0 m/s3 track calibration target	Geofence optional support, not sole safety/legal control
Valet	Drive-ready	Signed owner/fleet policy	80 kW default or lower	Standard	<=1.5 m/s3 target	Cannot raise cap without authorization
Mode Transition	Any accepted request	Safety MCU validates speed/gear/SOP/thermal/brake/ESC	Cross-fade 200-500 ms	Cross-fade	Jerk bounded by active mode/vehicle stability	Ack <=50 ms separate from map blend

Implementation rule: mode selection is a two-phase commit. The HMI request is validated first; only after all prerequisites pass does the safety-domain mode manager publish an accepted mode generation and apply a synchronized calibration cross-fade. Rejection leaves the previous complete mode active.

14.5 B11/B12 - CAN FD, E2E/SecOC, Ethernet and time synchronization

Flow / Network	Protection	Rev A2 Rule	Rationale
Torque command 10 ms CAN FD	E2E mandatory; SecOC if TARA requires	HSM-accelerated auth/freshness; TARA-selected truncation; deadline/bus-load analysis	Do not blanket remove SecOC
Inverter status 10 ms CAN FD	E2E mandatory; authenticity per TARA	Age/counter/CRC; delivered torque/current plausibility	Safety MCU consumes directly
BMS SOP 10-20 ms CAN FD	E2E mandatory; authenticity per TARA	Stale/invalid can only reduce authority	No HPA bypass
ESC/brake 10 ms CAN FD	E2E mandatory; authenticity per TARA	ESC/ABS override has priority	Dual-bus where safety allocation requires
Ethernet service backbone	VLAN/ACL/rate limit + MACsec where TARA allocates	1GBASE-T1 baseline; 2.5/10G upgrade branch	Not sole torque-safety path
HPA->Safety MCU IPC	Whitelisted mailbox / safe IPC	Schema, range, counter, rate limit, freshness; no raw pointer/MMIO sharing	All HPA requests untrusted
Time sync	IEEE 802.1AS-2025 program reference for Ethernet time	Time quality + monotonicity; torque loop not dependent on Ethernet time	Flight recorder/service correlation
CAN roadmap	ISO11898-1:2024 compatible strategy	CAN FD baseline; CAN XL future branch assessed at next platform freeze	Do not mix without gateway policy

Planning target: <=40% sustained utilization on the safety CAN FD bus unless formal schedulability proof permits another value. The analysis must include arbitration, bit stuffing, retransmissions, E2E fields, SecOC freshness/authenticator data, gateway latency and worst-case fault traffic.

Ethernet time uses IEEE 802.1AS-2025 as the new-design reference. Safety torque control remains independent of Ethernet time; gPTP is used for service-oriented data correlation, logging and other time-sensitive Ethernet functions.

14.6 B13/B14 - Safety telltales, diagnostics and event recorder

Topic	Rev A2 Requirement	Implementation	Boundary
Capture window	30 s pre + 30 s post trigger reference	RAM circular buffer; triggered flush to NVM	Vehicle event / safety / crash / watchdog / torque mismatch
Core channels	Pedal, brake, mode, torque request/limit reason, BMS SOP, inverter actual torque, ESC state, gear, speed, HV state, rail voltages, bus utilization	Sample rates tiered by signal criticality	Avoid unnecessary personal IVI content
Time	IEEE 802.1AS-2025 for Ethernet correlation plus local monotonic safety MCU time	Store time quality / sync state	CAN safety logic not dependent on gPTP
Integrity	CRC/signature/hash chain for critical event package	Write-complete marker and power-loss-safe commit	Forensic trust
Privacy	Purpose-limited safety/service data; configurable retention and access policy	Market legal review	Separate personal cockpit logs
Endurance	RAM-first buffering, write-rate budget,	Storage endurance model	No continuous high-rate NVM writes

Topic	Rev A2 Requirement	Implementation	Boundary
	health counter		

A QM Android display failure shall not silently remove a safety/regulatory-critical warning. The final display architecture must nominate a safety-qualified render path, monitored safety overlay, independent telltale or other vehicle-level fallback according to the warning criticality.

14.7 B15/B16 - OTA, SBOM, HSM, SecOC, privacy and production debug

Function	Requirement	Implementation	Boundary
Secure boot	HSM roots on Safety MCU and HPA; measured/verified chain	Anti-rollback, signed boot/config	No unsigned recovery image
OTA	A/B or equivalent atomic strategy with signed dependency manifest	Update only in allowed vehicle state; power-loss recovery	ISO24089 / R156 process
SBOM	Machine-readable ECU software/component inventory	Link version -> vulnerability status -> affected VIN/variant	Supply-chain governance
Crypto agility	Algorithm/key-length abstraction and certificate rotation	Avoid hardcoded lifecycle crypto dead-end	OEM policy controls algorithms
Debug	Production debug disabled or authenticated	Physical service policy and tamper logging	No generic unlock
SecOC	TARA-driven on safety/security relevant PDUs	HSM acceleration; schedule verified	Independent from E2E
Ethernet	VLAN/ACL/rate limiting; MACsec where threat analysis allocates	HPA/IVI not permitted direct safety-network injection	Gateway default deny
IDS	CAN/Ethernet anomaly monitoring without blocking safety timing	Event forwarding to backend when available	Safety reaction takes priority
Data Domain	**Principle**	**Control**	**Boundary**
Driver profile / mode preference	Store minimum attributes required for feature	Encrypted at rest; owner/profile access controls	Market privacy/consent rules
Valet/geofence policy	Treat location and owner constraints as personal/security data	Purpose limitation and retention policy	Geofence not sole safety control
Flight recorder	Separate safety/event channels from entertainment history	Role-based service access; retention schedule	Legal/insurance rules vary
Cloud telemetry	Opt-in/contractual lawful basis where required	Data minimization, pseudonymization, backend ACL	No cloud dependency for safety
Voice/assistant/cockpit analytics	Outside safety authority	Dedicated privacy consent and delete/export workflow	No access to raw torque controls

14.8 B19/B20 - EOL isolation screening, lifecycle and standards watch

Item	Status	Program Action	Owner
ISO 26262 Ed.3	Under development in 2026	Run delta review before SOP if edition changes before compliance freeze	Safety Lead
UN R155/R156 amendments	Active interpretation/amendment activity in 2025-2026	Freeze applicable regulation series at homologation gate	Cyber/Homologation
UN R10 07-series evolution	Current series/amendments evolving	Use market-specific applicable series at EMC homologation freeze	EMC/Homologation
AUTOSAR R25-11	Current release in 2026; program may remain on controlled R24-11 baseline	Perform delta assessment; do not upgrade stack mid-program without impact analysis	SW Architect
IEEE 802.1AS-2025	Supersedes 802.1AS-2020	Use 2025 baseline for new design unless supplier stack is frozen older with compatibility evidence	Network Architect
ISO 6469-3 next edition	Under development	Track changes affecting HV operational/interface safety	HV Safety

EOL must verify not only basic boot and I/O but also architectural independence: HPA reset, malformed IPC, unauthorized Ethernet route and wrong mode/calibration variant shall be injected or covered by a validated production control plan. Long-term support additionally tracks SBOM vulnerabilities, certificate expiry, hypervisor/OS lifecycle and critical semiconductor PCN/EOL.

15. Rev A2 Detailed SYS.3 Requirement Addendum

The following 46 requirements are added to the original 200-row SYS.3 baseline. They are assigned Rev A2 IDs without renumbering existing requirements, preserving traceability for already-reviewed Rev A1 content.

SYS3 ID	Block	Requirement	Statement	Acceptance	Allocation	Safety
VCU220-SYS3-B01-11	B01	Cold-crank front-end	The central ECU input power stage shall use a buck-boost or equivalent front end able to maintain the safety domain through the released OEM cold-crank profile rather than relying on bulk capacitance alone.	Safety MCU/CAN/HV supervision remain alive or transition deterministically through the released LV profile.	HW	Safety-related
VCU220-SYS3-B01-12	B01	Safety hold-up energy	The independent safety-domain supply shall provide energy hold-up sized from its actual safety load and reset thresholds; Rev A2 design reference is 20 W for 100 ms, equivalent to about 41 mF ideal for 12-to-6 V at 90% conversion before design margin.	Selected capacitance includes ESR, tolerance, aging, temperature and inrush limits; HPA hold-up is not credited.	HW	Safety-related
VCU220-SYS3-B01-13	B01	Input transient power	The LV front end shall be electrically sized for 150 W short-duration central ECU demand while thermal design uses 130 W sustained peak reference unless nominated silicon evidence changes the budget.	No PMIC/input protection current limit or thermal shutdown occurs during released cockpit/compute peak.	HW	Safety-related
VCU220-SYS3-B01-14	B01	Independent safety rail	Safety MCU, safety PMIC/watchdog and at least one safety CAN path shall use an independently supervised rail/reset path not controlled by the HPA SoC.	HPA reset/power fault does not reset or depower authoritative motion supervision.	HW	Safety-related
VCU220-SYS3-B02-11	B02	ASIL-D torque authority	The authoritative positive/negative propulsion torque arbitration and independent torque monitor shall support the SG-01 ASIL-D safety goal directly or through a formally approved ISO 26262-9 decomposition.	No contradiction remains between item HARA ASIL and allocated safety requirement integrity.	HW/SW	ASIL D target
VCU220-SYS3-B02-12	B02	Independent torque inhibit	The safety MCU/PMIC shall provide a hardware-enforced torque/inverter inhibit output or equivalent independent path that cannot be blocked by Android/HPA software.	Valid severe-torque fault causes inhibit request inside the <=10 ms internal electronic capability budget.	HW	ASIL D target
VCU220-SYS3-B02-13	B02	Safety boot independence	Safety MCU self-test, watchdog, CAN and torque/HV supervision shall become available independently of Android/cockpit boot.	Reference SAFETY_READY <=150 ms after valid wake; final timing from platform/OEM startup requirements.	HW/SW	Safety-related
VCU220-SYS3-B02-14	B02	Dual-source torque plausibility	Safety torque arbitration shall compare driver demand, brake/gear/state limits, BMS/inverter capability and delivered torque feedback so that a single corrupted demand source cannot increase authority.	Fault/stale source can only reduce authority; delivered torque mismatch is diagnosed and contained.	SW/System	ASIL D target
VCU220-SYS3-B03-11	B03	HPA resource reserve	Released HPA workloads shall preserve >=25% CPU/GPU reserve and >=35% sustained memory-bandwidth reserve under maximum cockpit, Ethernet and logging load unless nominated SoC evidence justifies another controlled margin.	No vehicle-service deadline violation under synthetic worst-case resource contention.	HW/SW	Safety-supporting
VCU220-SYS3-B03-12	B03	DMA containment	IOMMU/SMMU and MMIO ownership shall prevent QM guest or display/network DMA from accessing safety memory/peripherals or corrupting safety-supporting inter-processor	Unauthorized DMA generates contained fault without safety MCU impact.	HW/SW	Safety-supporting

SYS3 ID	Block	Requirement	Statement	Acceptance	Allocation	Safety
			buffers.			
VCU220-SYS3-B03-13	B03	Bounded safety-support memory	Any safety-relevant HPA helper service shall use bounded/preallocated memory and explicit high-watermark/OOM handling; Android/QM dynamic allocation shall be quota controlled.	Memory pressure cannot trigger platform-wide failure that removes safety MCU control or required gateway service.	SW	Safety-supporting
VCU220-SYS3-B03-14	B03	Guest fault containment	A guest kernel panic or VM termination shall be locally recoverable without forcing safety MCU reset or inadvertent torque/HV state change.	HPA guest restart fault injection passes; safety domain remains deterministic.	SW/System	Safety-related
VCU220-SYS3-B05-11	B05	Context-dependent safe state	Torque fault handling shall distinguish unintended-positive-torque faults requiring immediate inhibit from failures where controlled torque ramp or limited limp propulsion is safer, subject to the vehicle FSC/TSC.	Safe-state matrix is validated per operational situation; torque zero for every fault is not assumed.	SW/System	ASIL D/C
VCU220-SYS3-B05-12	B05	Torque gradient control	Nominal driver/mode torque changes shall be rate/jerk limited independently from emergency safety interventions.	Mode transition never produces an uncontrolled torque step; safety override may exceed comfort jerk targets.	SW	Safety-related
VCU220-SYS3-B05-13	B05	Charge/service drive inhibit	Propulsion enable shall require valid park/gear, charging connector/charge state, HV READY, BMS/inverter readiness and service-mode interlocks.	No drive torque while incompatible charging/service state is active.	HW/SW	Safety-related
VCU220-SYS3-B07-11	B07	Mode eligibility matrix	Every driver mode shall have explicit vehicle-state, thermal/SOP, chassis/ESC, fault, market and user-authorization prerequisites evaluated by the safety domain.	Rejected mode publishes reason and does not partially apply calibration.	SW/System	Safety-related
VCU220-SYS3-B07-12	B07	Track-mode governance	Track mode shall use market/variant enable, explicit driver confirmation and local ECU safety envelopes; geofence may support policy but shall not be the sole safety/legal control.	Track cannot disable ABS/ESC hard protection or bypass BMS/inverter capability.	SW/System	Safety-related
VCU220-SYS3-B07-13	B07	Boost thermal debt	Any overboost/track power extension shall use a bounded timer/energy or thermal-debt model and shall fall back bumplessly when SOP/thermal allowance expires.	No repeated mode cycling bypasses thermal/energy protection.	SW	Safety-related
VCU220-SYS3-B07-14	B07	Mode rollback	Invalid calibration, update mismatch, watchdog reset or critical DTC shall revert to a controlled safe/default road profile before torque authority is increased.	No stale Track/Sport calibration persists after integrity failure.	SW/System	Safety-related
VCU220-SYS3-B11-11	B11	Safety CAN utilization reserve	Safety CAN FD networks shall be scheduled with released utilization/deadline reserve including arbitration, stuffing, retransmission, SecOC/E2E overhead and fault traffic.	Reference design target <=40% sustained safety-bus utilization unless formal schedulability proof accepts a different value.	HW/SW	Safety-related
VCU220-SYS3-B11-12	B11	E2E and SecOC allocation	High-rate motion PDUs shall use AUTOSAR E2E for communication-safety integrity; SecOC shall be enabled where TARA requires authenticity/freshness and shall be included in CPU/bus timing analysis.	Security allocation cannot violate the 10 ms motion deadline or be removed solely for performance convenience.	SW/System	Safety/Cyber
VCU220-SYS3-B11-13	B11	HPA firewall boundary	HPA/IVI Ethernet or IPC traffic shall not route directly to powertrain actuation PDUs without safety-gateway whitelist, range, rate, freshness and state qualification.	Attempted unauthorized route is blocked and logged.	HW/SW	Safety/Cyber
VCU220-SYS3-B11-14	B11	Ethernet roadmap	1GBASE-T1 remains the Rev A2 vehicle IP baseline while switch/PCB/harness	Upgrade does not change safety authority or bypass gateway policy.	HW/System	Functional

SYS3 ID	Block	Requirement	Statement	Acceptance	Allocation	Safety
			architecture shall retain a controlled 2.5/10G or CAN XL upgrade path where future cockpit/service traffic requires it.			
VCU220-SYS3-B12-11	B12	Current time-sync baseline	Ethernet time synchronization shall use IEEE 802.1AS-2025 or an approved compatible profile for new design, with time quality/holdover status propagated to event logging and time-sensitive services.	Torque safety loop remains safe without Ethernet time synchronization.	HW/SW	Safety-supporting
VCU220-SYS3-B12-12	B12	Safe inter-processor mailbox	Safety MCU to/from HPA IPC shall use bounded schemas, counters/CRC, age, rate limit and explicit ownership; raw shared pointers or uncontrolled MMIO access are prohibited across the safety boundary.	Malformed/stale HPA request cannot increase torque/HV authority.	HW/SW	Safety-related
VCU220-SYS3-B13-11	B13	Safety telltale availability	Safety/regulatory-critical telltales shall have an independent source-of-truth and bounded presentation path. If the HPA renderer is QM, an independent safety overlay/fallback indicator shall cover allocated critical warnings.	Android/HPA crash does not silently remove mandatory safety warning.	HW/SW	Safety-related
VCU220-SYS3-B13-12	B13	Driver-mode HMI authority	HMI shall only request modes and display accepted state/reason; it shall not own safety calibration or torque limit memory.	Compromised HMI cannot directly activate Track/Sport or raise Valet cap.	SW/System	Safety/Cyber
VCU220-SYS3-B14-11	B14	Flight recorder	The safety/event recorder shall maintain a circular RAM capture with a 30 s pre-event and 30 s post-event reference window for critical torque/HV/network/power signals, followed by power-loss-safe NVM commit on trigger.	Recorder contains reason codes, data validity and time quality and remains within storage endurance budget.	SW/System	Safety-supporting
VCU220-SYS3-B14-12	B14	Diagnostic protocol baseline	New DoIP designs shall align with ISO 13400-2:2025 while UDS application services align with ISO 14229-1:2020 or approved OEM profile.	Diagnostic update cannot monopolize safety network or bypass authorization.	SW/System	Functional/Cyber
VCU220-SYS3-B15-11	B15	SBOM and dependency control	Every ECU software release shall include a machine-readable SBOM/dependency manifest linked to supplier/version/vulnerability and vehicle-variant applicability.	Affected component/vulnerability can be traced to released ECU/VIN population.	SW/System	Cyber
VCU220-SYS3-B15-12	B15	Atomic update and rollback	Software/calibration update shall use atomic A/B or equivalent recovery strategy, signed dependency validation and rollback protection, with activation limited to an approved vehicle state.	Power loss or failed guest update cannot corrupt the safety MCU production image or strand the vehicle in undefined state.	HW/SW	Cyber/Safety
VCU220-SYS3-B15-13	B15	R156/RXSWIN evidence	Where UN R156 applies, software update and RXSWIN-related configuration evidence shall be version-controlled and traceable to the approved vehicle SUMS process.	Update approval/change record is retrievable for homologation audit.	SW/System	Cyber/Regulatory
VCU220-SYS3-B16-11	B16	Crypto agility	Security architecture shall support controlled algorithm/key/certificate rotation without redesign of safety message semantics or ECU hardware where feasible.	No hard-coded lifecycle dependency prevents mandated cryptographic migration.	HW/SW	Cyber
VCU220-SYS3-B16-12	B16	Production debug control	JTAG/SWD/engineering debug shall be disabled or strongly authenticated in production with auditable service unlock and no generic master bypass.	Unauthorized debug attempt cannot expose safety keys or modify torque software.	HW/SW	Cyber/Safety
VCU220-SYS3-B16-13	B16	Privacy boundary	Driver profile,	Safety operation does not	SW/System	Cyber/Privacy

SYS3 ID	Block	Requirement	Statement	Acceptance	Allocation	Safety
			geofence/valet, cockpit analytics and cloud telemetry shall be treated as personal/security-sensitive data with market-specific retention, access and consent controls.	depend on cloud/personal-data availability.		
VCU220-SYS3-B17-11	B17	ASIL decomposition governance	Any ASIL decomposition used for SG-01 shall include dependent-failure analysis, independence evidence and explicit allocation to safety MCU/inverter mechanisms before credit is taken.	No decomposition claim based only on two software tasks on the same failure root.	HW/SW/System	ASIL D target
VCU220-SYS3-B17-12	B17	Common-cause central ECU analysis	Safety analysis shall include common-cause loss from ECU power connector, enclosure thermal event, water ingress, common oscillator/reset, EMC upset and shared network/switch failure.	Local inverter/BMS/ESC safety layers and safety-domain independence close credible common causes.	HW/System	Safety-related
VCU220-SYS3-B17-13	B17	Safe-state scenario table	Safety supervisor shall select response from an approved operational safe-state matrix rather than a single global fallback.	Fault injection demonstrates correct immediate inhibit, controlled degradation, limp mode or shutdown per scenario.	SW/System	Safety-related
VCU220-SYS3-B18-11	B18	Revised ECU power budget	The central ECU hardware and cooling shall be designed for a 70 W nominal / 130 W sustained peak reference with 150 W short input envelope unless selected silicon/package evidence releases another controlled budget.	Thermal/PI tests close worst HPA+DDR+Ethernet+SerDes load without safety rail violation.	HW/System	Safety-supporting
VCU220-SYS3-B18-12	B18	Thermal shedding hierarchy	Thermal control shall shed noncritical HPA GPU/NPU/background/animation workloads before reducing any authoritative safety MCU vehicle-control capability.	HPA may degrade/shutdown while safety MCU and critical warning path remain operational.	HW/SW	Safety-related
VCU220-SYS3-B18-13	B18	Updated immunity baseline	EMC immunity planning shall use current ISO 11452 general principles and relevant test parts plus OEM requirements; vehicle homologation shall use the applicable UN R10 series.	No safety output or torque authority corruption during released immunity tests.	HW/System	Safety-related
VCU220-SYS3-B18-14	B18	LV electrical baseline	Power-supply design shall align with ISO 16750-2:2023 and ISO 7637-2:2011 where applicable, supplemented by the OEM vehicle-specific crank/load-dump/reverse-battery profile.	No generic 6 V/12 V assumption substitutes for released vehicle electrical profile.	HW/System	Safety-related
VCU220-SYS3-B19-11	B19	EOL partition isolation test	EOL shall verify that HPA reset, guest crash, malformed IPC and Ethernet injection cannot change safety MCU torque/HV outputs beyond the defined safe response.	100% production screen or validated sampling/control-plan allocation is documented.	HW/SW	Safety/Cyber
VCU220-SYS3-B19-12	B19	Mode/calibration binding	Driver-mode, torque, regen and thermal calibration packages shall be signed and cryptographically bound to approved vehicle/powertrain/market variants.	Wrong/unsigned Track/Sport calibration blocks DRIVE_READY or mode availability.	SW/System	Safety/Cyber
VCU220-SYS3-B20-11	B20	Standards/regulation watchlist	The program shall maintain a controlled standards/regulation watchlist through SOP and service life, including ISO 26262 revision, R155/R156, R10, AUTOSAR and diagnostic/network standards.	Relevant post-freeze change triggers formal impact assessment instead of silent adoption.	System	Functional
VCU220-SYS3-B20-12	B20	Software supply-chain lifecycle	SBOM, vulnerability status, supplier support/EOL, certificate expiry and OS/hypervisor lifecycle shall be included in long-term service	No unsupported safety/cyber dependency remains unmanaged during contracted support life.	SW/System	Cyber/Functional

SYS3 ID	Block	Requirement	Statement	Acceptance	Allocation	Safety
			planning.			

16. Rev A2 Hardware and Software Safety Requirement Addendum

The HSR/SWSR additions are intentionally focused on independence, safety-domain power, ASIL-D torque inhibit, resource interference, critical warnings, secure communication and update recovery.

16.1 Hardware Safety Requirements

HSR ID	Parent	SG	Hardware Safety Requirement	Timing	Allocation	Verification
HSR-VCU-061	VCU220-SYS3-B01-12	SG-01	Safety-domain energy storage and PMIC shall retain safety MCU/CAN/torque-monitor function for the allocated hold-up interval.	100 ms reference	B01/B02 power	Brownout/load-step
HSR-VCU-062	VCU220-SYS3-B01-14	SG-01	Safety MCU rail/reset shall be independent from HPA power/reset control.	Continuous	Safety PMIC/MCU	Power/reset fault injection
HSR-VCU-063	VCU220-SYS3-B02-12	SG-01	Hardware torque/inverter inhibit shall be independently controllable by safety path.	<=10 ms electronic request	Safety MCU/PMIC/inverter interface	Fault injection
HSR-VCU-064	VCU220-SYS3-B02-14	SG-01	Safety hardware shall support independent accelerator/brake/vehicle-state plausibility acquisition or receive safety-rated protected inputs as allocated.	1-10 ms loops	Safety MCU I/O	Sensor fault injection
HSR-VCU-065	VCU220-SYS3-B03-12	SG-06	IOMMU/SMMU/MPU configuration shall prevent HPA QM DMA/MMIO access to safety assets.	Continuous	HPA SoC / interconnect	DMA attack/fault injection
HSR-VCU-066	VCU220-SYS3-B11-11	SG-02	Safety CAN transceivers/controllers shall support the released redundant topology and deadlines.	10 ms motion loops	CAN A/B	Bus-off/fault injection
HSR-VCU-067	VCU220-SYS3-B13-11	SG-07	Critical warning path shall include independent display/telltale fallback or safety-qualified renderer as allocated.	Fault duration	HMI safety path	Guest/display fault
HSR-VCU-068	VCU220-SYS3-B18-11	SG-01	ECU power/thermal hardware shall support 130 W sustained peak reference without safety PMIC brownout or thermal propagation.	Peak workload	B18 thermal/PDN	CFD/chamber/PI
HSR-VCU-069	VCU220-SYS3-B17-12	SG-01	Safety common-cause analysis shall include shared connector, enclosure, clock/reset and EMC roots and allocate independent mitigation.	Lifecycle	B17/B18	DFA/CCF review
HSR-VCU-070	VCU220-SYS3-B19-11	SG-06	EOL shall verify partition isolation and safety path independence using controlled HPA reset/IPC/network injections.	EOL	Manufacturing test	EOL fault injection

16.2 Software Safety Requirements

SWSR ID	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
SWSR-VCU-061	VCU220-SYS3-B02-11	SG-01	Safety software shall implement ASIL-D torque demand calculation/monitoring or approved decomposed channels and shall reject any HPA torque request outside hard safety bounds.	1-10 ms	Safety MCU torque manager	SIL/HIL/fault injection
SWSR-VCU-062	VCU220-SYS3-B05-11	SG-01	Safety supervisor shall select fault response from an operational safe-state matrix rather than applying unconditional torque zero.	<=10 ms decision for severe torque fault	Safety supervisor	Scenario HIL
SWSR-VCU-063	VCU220-SYS3-B07-11	SG-05	Mode manager shall evaluate eligibility atomically before applying	<=50 ms request result	Mode manager	Mode HIL

SWSR ID	Parent	SG	Software Safety Requirement	Timing	Allocation	Verification
			any calibration element.			
SWSR-VCU-064	VCU220-SYS3-B07-13	SG-05	Overboost/Track software shall accumulate thermal/energy debt and shall not reset the allowance by rapid mode cycling.	10-100 ms update	Mode/Thermal manager	Abuse tests
SWSR-VCU-065	VCU220-SYS3-B11-12	SG-02	Communication stack shall keep E2E and SecOC responsibilities separate and schedule both according to the approved PDU/TARA profile.	10 ms torque loop	Safety comm stack	CAN timing/cyber HIL
SWSR-VCU-066	VCU220-SYS3-B12-12	SG-06	HPA-to-safety IPC shall validate schema, length, range, age, counter and rate before data can affect safety calculations.	Per message	IPC gateway	Malformed IPC fuzzing
SWSR-VCU-067	VCU220-SYS3-B13-11	SG-07	HMI safety service shall detect loss of critical telltale rendering and command fallback indication.	<= allocated warning FTTI	HMI safety service	Display fault injection
SWSR-VCU-068	VCU220-SYS3-B14-11	SG-10	Flight recorder shall capture defined pre/post-event safety channels with integrity markers and bounded NVM write behavior.	Event trigger	Diagnostics/logger	Power-fail/log stress
SWSR-VCU-069	VCU220-SYS3-B15-12	SG-10	Update manager shall inhibit unsafe activation states, verify signed dependencies and recover atomically after power interruption.	Before activation	OTA manager	Update fault injection
SWSR-VCU-070	VCU220-SYS3-B16-13	SG-10	Privacy/security policy shall isolate personal cockpit data from safety logging and shall enforce retention/access controls.	Lifecycle	Data governance	Privacy/security review
SWSR-VCU-071	VCU220-SYS3-B18-12	SG-11	Thermal manager shall shed QM workloads before degrading safety MCU control resources.	Thermal state transitions	Thermal manager	Thermal stress
SWSR-VCU-072	VCU220-SYS3-B20-11	SG-10	Standards/compliance watchlist changes shall generate formal impact tasks and shall not silently change released behavior.	Lifecycle	Compliance/change management	Audit

17. Rev A2 DVP Expansion - 92 Additional Verification Cases

Each new SYS.3 hardening requirement receives one nominal/boundary test and one degraded/fault-injection test. This keeps the design changes objectively verifiable rather than leaving them as narrative recommendations.

DVP ID	SYS3 Parent	Method	Scenario	Acceptance
DVP-VCU-201-A	VCU220-SYS3-B01-11	LV transient / load-step / brownout bench	Nominal/boundary: Cold-crank front-end	Safety MCU/CAN/HV supervision remain alive or transition deterministically through the released LV profile.
DVP-VCU-201-B	VCU220-SYS3-B01-11	LV transient / load-step / brownout bench	Fault/degraded: Cold-crank front-end	Safety MCU/CAN/HV supervision remain alive or transition deterministically through the released LV profile.
DVP-VCU-202-A	VCU220-SYS3-B01-12	LV transient / load-step / brownout bench	Nominal/boundary: Safety hold-up energy	Selected capacitance includes ESR, tolerance, aging, temperature and inrush limits; HPA hold-up is not credited.
DVP-VCU-202-B	VCU220-SYS3-B01-12	LV transient / load-step / brownout bench	Fault/degraded: Safety hold-up energy	Selected capacitance includes ESR, tolerance, aging, temperature and inrush limits; HPA hold-up is not credited.
DVP-VCU-203-A	VCU220-SYS3-B01-13	LV transient / load-step / brownout bench	Nominal/boundary: Input transient power	No PMIC/input protection current limit or thermal shutdown occurs during released cockpit/compute peak.
DVP-VCU-203-B	VCU220-SYS3-B01-13	LV transient / load-step / brownout bench	Fault/degraded: Input transient power	No PMIC/input protection current limit or thermal shutdown occurs during released cockpit/compute peak.
DVP-VCU-204-A	VCU220-SYS3-B01-14	LV transient / load-step / brownout bench	Nominal/boundary: Independent safety rail	HPA reset/power fault does not reset or depower authoritative motion supervision.
DVP-VCU-204-B	VCU220-SYS3-B01-14	LV transient / load-step / brownout bench	Fault/degraded: Independent safety rail	HPA reset/power fault does not reset or depower authoritative motion supervision.
DVP-VCU-205-A	VCU220-SYS3-B02-11	Safety HIL / fault injection / timing	Nominal/boundary: ASIL-D torque authority	No contradiction remains between item HARA ASIL and allocated safety requirement integrity.
DVP-VCU-205-B	VCU220-SYS3-B02-11	Safety HIL / fault injection / timing	Fault/degraded: ASIL-D torque authority	No contradiction remains between item HARA ASIL and allocated safety requirement integrity.
DVP-VCU-206-A	VCU220-SYS3-B02-12	Safety HIL / fault injection / timing	Nominal/boundary: Independent torque inhibit	Valid severe-torque fault causes inhibit request inside the <=10 ms internal electronic capability budget.
DVP-VCU-206-B	VCU220-SYS3-B02-12	Safety HIL / fault injection / timing	Fault/degraded: Independent torque inhibit	Valid severe-torque fault causes inhibit request inside the <=10 ms internal electronic capability budget.
DVP-VCU-207-A	VCU220-SYS3-B02-13	Safety HIL / fault injection / timing	Nominal/boundary: Safety boot independence	Reference SAFETY_READY <=150 ms after valid wake; final timing from platform/OEM startup requirements.
DVP-VCU-207-B	VCU220-SYS3-B02-13	Safety HIL / fault injection / timing	Fault/degraded: Safety boot independence	Reference SAFETY_READY <=150 ms after valid wake; final timing from platform/OEM startup requirements.
DVP-VCU-208-A	VCU220-SYS3-B02-14	Safety HIL / fault injection / timing	Nominal/boundary: Dual-source torque plausibility	Fault/stale source can only reduce authority; delivered torque mismatch is diagnosed and contained.
DVP-VCU-208-B	VCU220-SYS3-B02-14	Safety HIL / fault injection / timing	Fault/degraded: Dual-source torque plausibility	Fault/stale source can only reduce authority; delivered torque mismatch is diagnosed and contained.
DVP-VCU-209-A	VCU220-SYS3-B03-11	Target stress / IOMMU / VM fault injection	Nominal/boundary: HPA resource reserve	No vehicle-service deadline violation under synthetic worst-case resource contention.
DVP-VCU-209-B	VCU220-SYS3-B03-11	Target stress / IOMMU / VM fault injection	Fault/degraded: HPA resource reserve	No vehicle-service deadline violation under synthetic worst-case resource contention.
DVP-VCU-210-A	VCU220-SYS3-B03-12	Target stress / IOMMU / VM fault injection	Nominal/boundary: DMA containment	Unauthorized DMA generates contained fault without safety MCU impact.
DVP-VCU-210-B	VCU220-SYS3-B03-12	Target stress / IOMMU / VM fault injection	Fault/degraded: DMA containment	Unauthorized DMA generates contained fault without safety MCU impact.
DVP-VCU-211-A	VCU220-SYS3-B03-13	Target stress / IOMMU / VM fault injection	Nominal/boundary: Bounded safety-support memory	Memory pressure cannot trigger platform-wide failure that removes safety MCU control or required gateway service.
DVP-VCU-211-B	VCU220-SYS3-B03-13	Target stress / IOMMU / VM fault injection	Fault/degraded: Bounded safety-support memory	Memory pressure cannot trigger platform-wide failure that removes safety MCU control or required gateway service.
DVP-VCU-212-A	VCU220-SYS3-B03-14	Target stress / IOMMU / VM fault injection	Nominal/boundary: Guest fault containment	HPA guest restart fault injection passes; safety domain remains deterministic.
DVP-VCU-212-B	VCU220-SYS3-B03-14	Target stress / IOMMU / VM fault injection	Fault/degraded: Guest fault containment	HPA guest restart fault injection passes; safety domain remains deterministic.
DVP-VCU-213-A	VCU220-SYS3-B05-11	Powertrain HIL / dyno / fault injection	Nominal/boundary: Context-dependent safe state	Safe-state matrix is validated per operational situation; torque zero for

DVP ID	SYS3 Parent	Method	Scenario	Acceptance
				every fault is not assumed.
DVP-VCU-213-B	VCU220-SYS3-B05-11	Powertrain HIL / dyno / fault injection	Fault/degraded: Context-dependent safe state	Safe-state matrix is validated per operational situation; torque zero for every fault is not assumed.
DVP-VCU-214-A	VCU220-SYS3-B05-12	Powertrain HIL / dyno / fault injection	Nominal/boundary: Torque gradient control	Mode transition never produces an uncontrolled torque step; safety override may exceed comfort jerk targets.
DVP-VCU-214-B	VCU220-SYS3-B05-12	Powertrain HIL / dyno / fault injection	Fault/degraded: Torque gradient control	Mode transition never produces an uncontrolled torque step; safety override may exceed comfort jerk targets.
DVP-VCU-215-A	VCU220-SYS3-B05-13	Powertrain HIL / dyno / fault injection	Nominal/boundary: Charge/service drive inhibit	No drive torque while incompatible charging/service state is active.
DVP-VCU-215-B	VCU220-SYS3-B05-13	Powertrain HIL / dyno / fault injection	Fault/degraded: Charge/service drive inhibit	No drive torque while incompatible charging/service state is active.
DVP-VCU-216-A	VCU220-SYS3-B07-11	Mode HIL / vehicle dynamics	Nominal/boundary: Mode eligibility matrix	Rejected mode publishes reason and does not partially apply calibration.
DVP-VCU-216-B	VCU220-SYS3-B07-11	Mode HIL / vehicle dynamics	Fault/degraded: Mode eligibility matrix	Rejected mode publishes reason and does not partially apply calibration.
DVP-VCU-217-A	VCU220-SYS3-B07-12	Mode HIL / vehicle dynamics	Nominal/boundary: Track-mode governance	Track cannot disable ABS/ESC hard protection or bypass BMS/inverter capability.
DVP-VCU-217-B	VCU220-SYS3-B07-12	Mode HIL / vehicle dynamics	Fault/degraded: Track-mode governance	Track cannot disable ABS/ESC hard protection or bypass BMS/inverter capability.
DVP-VCU-218-A	VCU220-SYS3-B07-13	Mode HIL / vehicle dynamics	Nominal/boundary: Boost thermal debt	No repeated mode cycling bypasses thermal/energy protection.
DVP-VCU-218-B	VCU220-SYS3-B07-13	Mode HIL / vehicle dynamics	Fault/degraded: Boost thermal debt	No repeated mode cycling bypasses thermal/energy protection.
DVP-VCU-219-A	VCU220-SYS3-B07-14	Mode HIL / vehicle dynamics	Nominal/boundary: Mode rollback	No stale Track/Sport calibration persists after integrity failure.
DVP-VCU-219-B	VCU220-SYS3-B07-14	Mode HIL / vehicle dynamics	Fault/degraded: Mode rollback	No stale Track/Sport calibration persists after integrity failure.
DVP-VCU-220-A	VCU220-SYS3-B11-11	CANoe/Ethernet network + cyber HIL	Nominal/boundary: Safety CAN utilization reserve	Reference design target <=40% sustained safety-bus utilization unless formal schedulability proof accepts a different value.
DVP-VCU-220-B	VCU220-SYS3-B11-11	CANoe/Ethernet network + cyber HIL	Fault/degraded: Safety CAN utilization reserve	Reference design target <=40% sustained safety-bus utilization unless formal schedulability proof accepts a different value.
DVP-VCU-221-A	VCU220-SYS3-B11-12	CANoe/Ethernet network + cyber HIL	Nominal/boundary: E2E and SecOC allocation	Security allocation cannot violate the 10 ms motion deadline or be removed solely for performance convenience.
DVP-VCU-221-B	VCU220-SYS3-B11-12	CANoe/Ethernet network + cyber HIL	Fault/degraded: E2E and SecOC allocation	Security allocation cannot violate the 10 ms motion deadline or be removed solely for performance convenience.
DVP-VCU-222-A	VCU220-SYS3-B11-13	CANoe/Ethernet network + cyber HIL	Nominal/boundary: HPA firewall boundary	Attempted unauthorized route is blocked and logged.
DVP-VCU-222-B	VCU220-SYS3-B11-13	CANoe/Ethernet network + cyber HIL	Fault/degraded: HPA firewall boundary	Attempted unauthorized route is blocked and logged.
DVP-VCU-223-A	VCU220-SYS3-B11-14	CANoe/Ethernet network + cyber HIL	Nominal/boundary: Ethernet roadmap	Upgrade does not change safety authority or bypass gateway policy.
DVP-VCU-223-B	VCU220-SYS3-B11-14	CANoe/Ethernet network + cyber HIL	Fault/degraded: Ethernet roadmap	Upgrade does not change safety authority or bypass gateway policy.
DVP-VCU-224-A	VCU220-SYS3-B12-11	IPC/time fault injection	Nominal/boundary: Current time-sync baseline	Torque safety loop remains safe without Ethernet time synchronization.
DVP-VCU-224-B	VCU220-SYS3-B12-11	IPC/time fault injection	Fault/degraded: Current time-sync baseline	Torque safety loop remains safe without Ethernet time synchronization.
DVP-VCU-225-A	VCU220-SYS3-B12-12	IPC/time fault injection	Nominal/boundary: Safe inter-processor mailbox	Malformed/stale HPA request cannot increase torque/HV authority.
DVP-VCU-225-B	VCU220-SYS3-B12-12	IPC/time fault injection	Fault/degraded: Safe inter-processor mailbox	Malformed/stale HPA request cannot increase torque/HV authority.
DVP-VCU-226-A	VCU220-SYS3-B13-11	HMI/display fault injection	Nominal/boundary: Safety telltale availability	Android/HPA crash does not silently remove mandatory safety warning.
DVP-VCU-226-B	VCU220-SYS3-B13-11	HMI/display fault injection	Fault/degraded: Safety telltale availability	Android/HPA crash does not silently remove mandatory safety warning.
DVP-VCU-227-A	VCU220-SYS3-B13-12	HMI/display fault injection	Nominal/boundary: Driver-mode HMI authority	Compromised HMI cannot directly activate Track/Sport or raise Valet cap.
DVP-VCU-227-B	VCU220-SYS3-B13-12	HMI/display fault injection	Fault/degraded: Driver-mode HMI authority	Compromised HMI cannot directly activate Track/Sport or raise Valet cap.
DVP-VCU-228-A	VCU220-SYS3-B14-11	Diagnostics/log power-fail test	Nominal/boundary: Flight recorder	Recorder contains reason codes, data validity and time quality and remains within storage endurance budget.
DVP-VCU-228-B	VCU220-SYS3-B14-11	Diagnostics/log power-fail test	Fault/degraded: Flight recorder	Recorder contains reason codes, data validity and time quality and remains within storage endurance budget.
DVP-VCU-229-A	VCU220-SYS3-B14-12	Diagnostics/log power-fail test	Nominal/boundary: Diagnostic protocol baseline	Diagnostic update cannot monopolize safety network or bypass authorization.
DVP-VCU-229-B	VCU220-SYS3-B14-12	Diagnostics/log power-fail test	Fault/degraded: Diagnostic protocol baseline	Diagnostic update cannot monopolize safety network or bypass authorization.

DVP ID	SYS3 Parent	Method	Scenario	Acceptance
DVP-VCU-230-A	VCU220-SYS3-B15-11	OTA/update fault injection	Nominal/boundary: SBOM and dependency control	Affected component/vulnerability can be traced to released ECU/VIN population.
DVP-VCU-230-B	VCU220-SYS3-B15-11	OTA/update fault injection	Fault/degraded: SBOM and dependency control	Affected component/vulnerability can be traced to released ECU/VIN population.
DVP-VCU-231-A	VCU220-SYS3-B15-12	OTA/update fault injection	Nominal/boundary: Atomic update and rollback	Power loss or failed guest update cannot corrupt the safety MCU production image or strand the vehicle in undefined state.
DVP-VCU-231-B	VCU220-SYS3-B15-12	OTA/update fault injection	Fault/degraded: Atomic update and rollback	Power loss or failed guest update cannot corrupt the safety MCU production image or strand the vehicle in undefined state.
DVP-VCU-232-A	VCU220-SYS3-B15-13	OTA/update fault injection	Nominal/boundary: R156/RXSWIN evidence	Update approval/change record is retrievable for homologation audit.
DVP-VCU-232-B	VCU220-SYS3-B15-13	OTA/update fault injection	Fault/degraded: R156/RXSWIN evidence	Update approval/change record is retrievable for homologation audit.
DVP-VCU-233-A	VCU220-SYS3-B16-11	Penetration / HSM / configuration test	Nominal/boundary: Crypto agility	No hard-coded lifecycle dependency prevents mandated cryptographic migration.
DVP-VCU-233-B	VCU220-SYS3-B16-11	Penetration / HSM / configuration test	Fault/degraded: Crypto agility	No hard-coded lifecycle dependency prevents mandated cryptographic migration.
DVP-VCU-234-A	VCU220-SYS3-B16-12	Penetration / HSM / configuration test	Nominal/boundary: Production debug control	Unauthorized debug attempt cannot expose safety keys or modify torque software.
DVP-VCU-234-B	VCU220-SYS3-B16-12	Penetration / HSM / configuration test	Fault/degraded: Production debug control	Unauthorized debug attempt cannot expose safety keys or modify torque software.
DVP-VCU-235-A	VCU220-SYS3-B16-13	Penetration / HSM / configuration test	Nominal/boundary: Privacy boundary	Safety operation does not depend on cloud/personal-data availability.
DVP-VCU-235-B	VCU220-SYS3-B16-13	Penetration / HSM / configuration test	Fault/degraded: Privacy boundary	Safety operation does not depend on cloud/personal-data availability.
DVP-VCU-236-A	VCU220-SYS3-B17-11	Safety CCF/fault-injection HIL	Nominal/boundary: ASIL decomposition governance	No decomposition claim based only on two software tasks on the same failure root.
DVP-VCU-236-B	VCU220-SYS3-B17-11	Safety CCF/fault-injection HIL	Fault/degraded: ASIL decomposition governance	No decomposition claim based only on two software tasks on the same failure root.
DVP-VCU-237-A	VCU220-SYS3-B17-12	Safety CCF/fault-injection HIL	Nominal/boundary: Common-cause central ECU analysis	Local inverter/BMS/ESC safety layers and safety-domain independence close credible common causes.
DVP-VCU-237-B	VCU220-SYS3-B17-12	Safety CCF/fault-injection HIL	Fault/degraded: Common-cause central ECU analysis	Local inverter/BMS/ESC safety layers and safety-domain independence close credible common causes.
DVP-VCU-238-A	VCU220-SYS3-B17-13	Safety CCF/fault-injection HIL	Nominal/boundary: Safe-state scenario table	Fault injection demonstrates correct immediate inhibit, controlled degradation, limp mode or shutdown per scenario.
DVP-VCU-238-B	VCU220-SYS3-B17-13	Safety CCF/fault-injection HIL	Fault/degraded: Safe-state scenario table	Fault injection demonstrates correct immediate inhibit, controlled degradation, limp mode or shutdown per scenario.
DVP-VCU-239-A	VCU220-SYS3-B18-11	Power/thermal/EMC/SI-PI lab	Nominal/boundary: Revised ECU power budget	Thermal/PI tests close worst HPA+DDR+Ethernet+SerDes load without safety rail violation.
DVP-VCU-239-B	VCU220-SYS3-B18-11	Power/thermal/EMC/SI-PI lab	Fault/degraded: Revised ECU power budget	Thermal/PI tests close worst HPA+DDR+Ethernet+SerDes load without safety rail violation.
DVP-VCU-240-A	VCU220-SYS3-B18-12	Power/thermal/EMC/SI-PI lab	Nominal/boundary: Thermal shedding hierarchy	HPA may degrade/shutdown while safety MCU and critical warning path remain operational.
DVP-VCU-240-B	VCU220-SYS3-B18-12	Power/thermal/EMC/SI-PI lab	Fault/degraded: Thermal shedding hierarchy	HPA may degrade/shutdown while safety MCU and critical warning path remain operational.
DVP-VCU-241-A	VCU220-SYS3-B18-13	Power/thermal/EMC/SI-PI lab	Nominal/boundary: Updated immunity baseline	No safety output or torque authority corruption during released immunity tests.
DVP-VCU-241-B	VCU220-SYS3-B18-13	Power/thermal/EMC/SI-PI lab	Fault/degraded: Updated immunity baseline	No safety output or torque authority corruption during released immunity tests.
DVP-VCU-242-A	VCU220-SYS3-B18-14	Power/thermal/EMC/SI-PI lab	Nominal/boundary: LV electrical baseline	No generic 6 V/12 V assumption substitutes for released vehicle electrical profile.
DVP-VCU-242-B	VCU220-SYS3-B18-14	Power/thermal/EMC/SI-PI lab	Fault/degraded: LV electrical baseline	No generic 6 V/12 V assumption substitutes for released vehicle electrical profile.
DVP-VCU-243-A	VCU220-SYS3-B19-11	EOL/provisioning/station validation	Nominal/boundary: EOL partition isolation test	100% production screen or validated sampling/control-plan allocation is documented.
DVP-VCU-243-B	VCU220-SYS3-B19-11	EOL/provisioning/station validation	Fault/degraded: EOL partition isolation test	100% production screen or validated sampling/control-plan allocation is documented.
DVP-VCU-244-A	VCU220-SYS3-B19-12	EOL/provisioning/station validation	Nominal/boundary: Mode/calibration	Wrong/unsigned Track/Sport

DVP ID	SYS3 Parent	Method	Scenario	Acceptance
			binding	calibration blocks DRIVE_READY or mode availability.
DVP-VCU-244-B	VCU220-SYS3-B19-12	EOL/provisioning/station validation	Fault/degraded: Mode/calibration binding	Wrong/unsigned Track/Sport calibration blocks DRIVE_READY or mode availability.
DVP-VCU-245-A	VCU220-SYS3-B20-11	Lifecycle/process audit	Nominal/boundary: Standards/regulation watchlist	Relevant post-freeze change triggers formal impact assessment instead of silent adoption.
DVP-VCU-245-B	VCU220-SYS3-B20-11	Lifecycle/process audit	Fault/degraded: Standards/regulation watchlist	Relevant post-freeze change triggers formal impact assessment instead of silent adoption.
DVP-VCU-246-A	VCU220-SYS3-B20-12	Lifecycle/process audit	Nominal/boundary: Software supply-chain lifecycle	No unsupported safety/cyber dependency remains unmanaged during contracted support life.
DVP-VCU-246-B	VCU220-SYS3-B20-12	Lifecycle/process audit	Fault/degraded: Software supply-chain lifecycle	No unsupported safety/cyber dependency remains unmanaged during contracted support life.

18. Rev A2 System Validation Addendum

The following system-level validation cases close cross-domain behavior that cannot be proven by a component bench alone.

VAL ID	Area	Validation Objective	Method	Acceptance	Status
VAL-VCU-041	Power/Thermal	Validate 130 W sustained central ECU load, 150 W electrical transient, thermal shedding and independent safety rail.	Thermal chamber + power bench + target workload	No safety-domain reset; HPA shedding follows state machine.	Open
VAL-VCU-042	Brownout	Validate OEM cold crank, safety-domain hold-up and HPA controlled reset/recovery.	LV transient bench / vehicle	Safety supervision remains available or degrades deterministically.	Open
VAL-VCU-043	ASIL-D torque	Validate independent torque monitor/inhibit/decomposition assumptions across sensor/CPU/network/inverter faults.	Closed-loop HIL + dyno	SG-01 safety path timing/independence evidence closes.	Open
VAL-VCU-044	FFI/Platform	Validate Android/HPA crash, DMA violation, memory/DDR saturation, reset and guest restart containment.	Target stress + HIL	No unauthorized torque/HV effect; safety MCU continues.	Open
VAL-VCU-045	Network/Cyber	Validate E2E/SecOC schedulability, CAN utilization, firewall and HPA-to-safety route policy.	CANoe + HSM benchmark + penetration	All 10 ms deadlines and security controls close.	Open
VAL-VCU-046	Driver Modes	Validate Eco/Normal/Sport/Track/Valet eligibility, blend, boost debt and rollback.	Dyno/vehicle/HIL	No torque step or safety-limit bypass.	Open
VAL-VCU-047	Cockpit Safety	Validate critical warning/telltale availability during HPA guest/display failure.	HMI/display fault injection	Allocated warnings remain visible/audible through fallback.	Open
VAL-VCU-048	OTA/Lifecycle	Validate atomic update, SBOM dependency, power-loss recovery, rollback, certificate/crypto lifecycle.	OTA rig + cyber lab	No incompatible or unsigned activation; recoverable failure.	Open
VAL-VCU-049	Compliance	Validate current regulation/standards applicability and vehicle-level interface evidence before homologation.	Compliance audit / vehicle witness	Applicable R10/R100/R13-H/R79/R155/R156 evidence mapped.	Open
VAL-VCU-050	Flight Recorder	Validate 30s pre/post capture, time quality, NVM power-fail integrity, retention/privacy policy.	Bench/vehicle/service tool	Event bundle complete, integrity-protected and endurance compliant.	Open

19. Rev A2 Traceability, Compliance and Release-Control Summary

Rev A2 does not overwrite the JAN-2026 technical baseline history. It adds explicit change-controlled traceability from audit observation to engineering decision, upstream requirements, SYS.3 allocation, safety requirements, DVP and system validation.

Traceability Layer	Rev A1	Rev A2	Release Meaning
RA	60	68	Expanded stakeholder/design authority
SYS.1	50	58	New stakeholder needs for power, FFI, compliance and lifecycle
SYS.2	100	115	New system-level hardening requirements
HARA	40	48	Central-compute-specific hazards added
SYS.3	200	246	46 detailed hardening requirements added
HSR	60	70	Power/independence/HMI/EOL hardware safety additions
SW Safety	60	72	Mode, IPC, network, thermal, OTA and governance additions
DFMEA	100	120	New physical/software failure modes
TARA	30	40	New cross-domain and supply-chain threats
ICD	30	38	Safety IPC, telltale, inhibit, charge, recorder, OTA and timing interfaces
DVP	400	492	Two objective tests per new SYS.3 hardening requirement
Validation	40	50	Cross-domain vehicle/system validation
RTM	200	246	One trace row per SYS.3 requirement

19.1 Current standards / regulatory controls

Domain	Standard / Regulation	2026 Status / Note	VCU220 Evidence
Functional safety	ISO 26262:2018	Published baseline; third edition under development in 2026	Safety lifecycle/HARA/FSC/TSC/HW/SW safety case
Cybersecurity	ISO/SAE 21434:2021	Published	TARA, lifecycle cyber risk management
Software update	ISO 24089:2023	Published	Update engineering, dependency/rollback evidence
ASPICE	Automotive SPICE 4.0	Program process baseline	SYS.1/SYS.2/SYS.3 + HWE/SWE/SYS.4/5 handoff
Electrical loads	ISO 16750-2:2023	Current published edition	LV operating/transient test profile plus OEM requirements
Conducted transients	ISO 7637-2:2011	Published/confirmed; 12/24V	Bench transient immunity; OEM pulse profile still controls
EMC immunity	ISO 11452-1:2025 + relevant parts	Current general principles edition	Component immunity plan
Emissions	CISPR 25:2021	Published	On-board receiver protection / component emissions
CAN	ISO 11898-1:2024	Published; includes CAN Classic/FD/XL framework	Network architecture/future branch
DoIP	ISO 13400-2:2025	Current published transport/network edition	Diagnostics over IP
UDS	ISO 14229-1:2020	Published	Diagnostic services
EV electrical safety	ISO 6469-3:2021	Published; revision underway	HV state/interface vehicle safety evidence
Ethernet timing	IEEE 802.1AS-2025	Active; supersedes 2020	gPTP/time-sync architecture
Ethernet sleep/wake	OPEN Alliance TC10	Automotive Ethernet sleep/wake specifications	Partial networking/wake design
Cyber type approval	UN R155	Apply current series/amendments at market homologation	CSMS/type approval evidence
Software update type approval	UN R156	Apply current series/amendments at market homologation	SUMS/RXSWIN/update compliance
Vehicle EMC	UN R10	Apply applicable current series at homologation	Vehicle/component EMC type approval
EV powertrain safety	UN R100	Vehicle-level applicability	Rechargeable energy storage/electric powertrain interfaces
Braking interface	UN R13-H / applicable market braking regulation	Vehicle-level interface	Regen/friction blend must not compromise braking compliance
Steering interface	UN R79 / applicable market steering regulation	Vehicle-level interface	Driver-mode steering request remains inside EPS/regulatory boundary